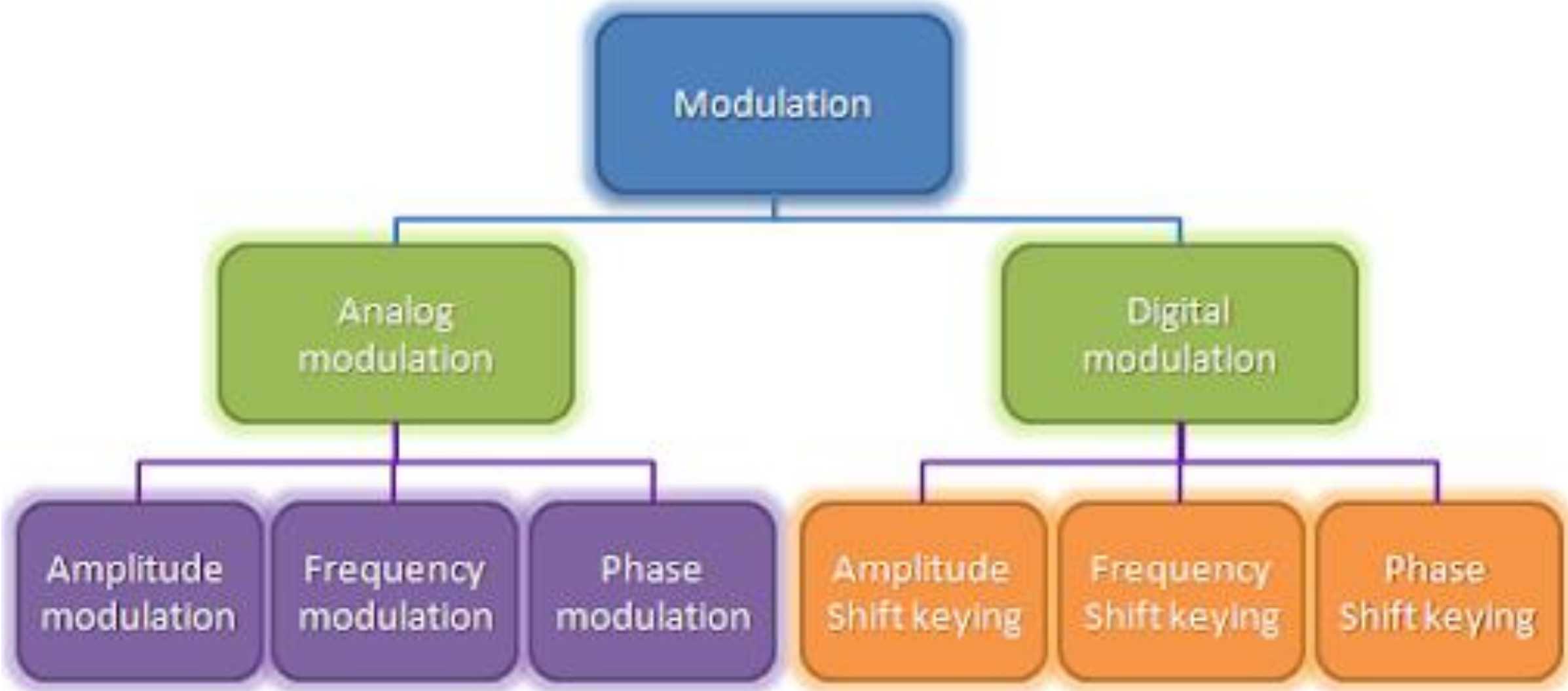
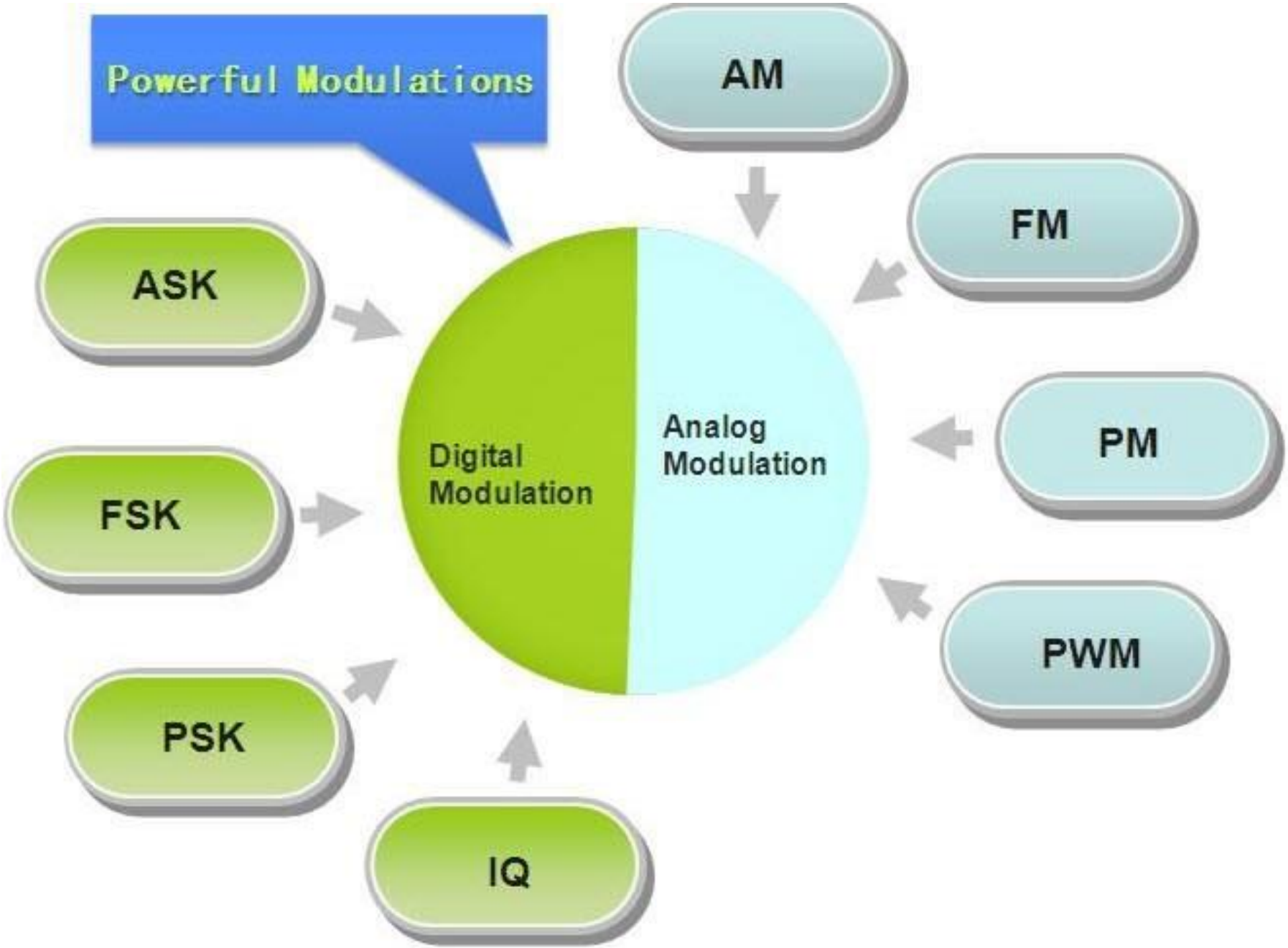
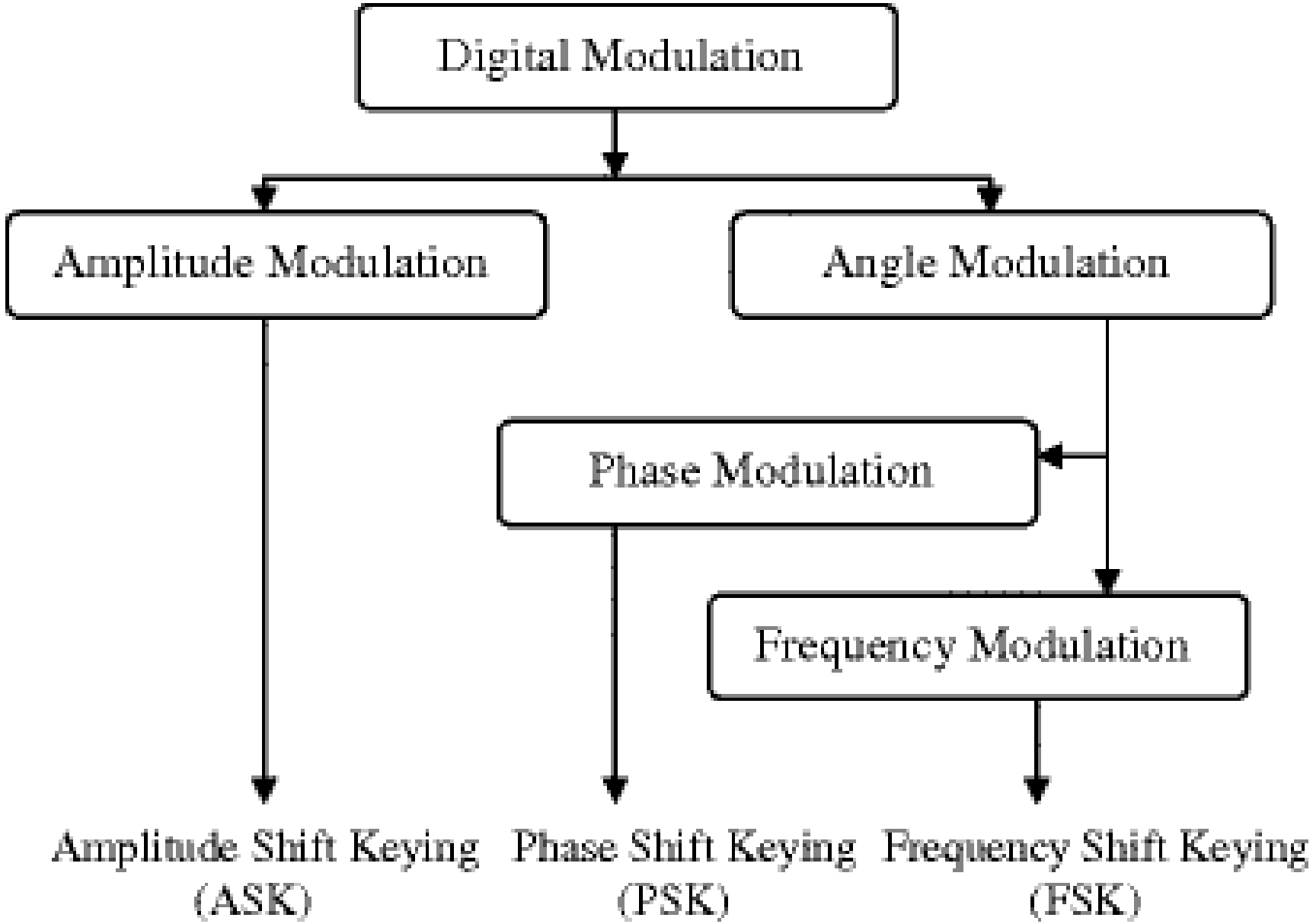
UNIT–IV

Modulation Techniques: Constant Envelop Modulation, **MSK**, **GMSK**, Combined Linear and Constant Envelope Modulation Technique, **MPSK**, **QAM**, **OFDM**, Introduction to **Spread Spectrum**, PN Sequence, **DS-SS**, **FH-SS**, Performance of **DS-SS** and **FH-SS**.


Modulation
Analog modulation
Digital modulation
Amplitude modulation
Frequency modulation
Phase modulation
Amplitude Shift keying
Frequency Shift keying
Phase Shift keying

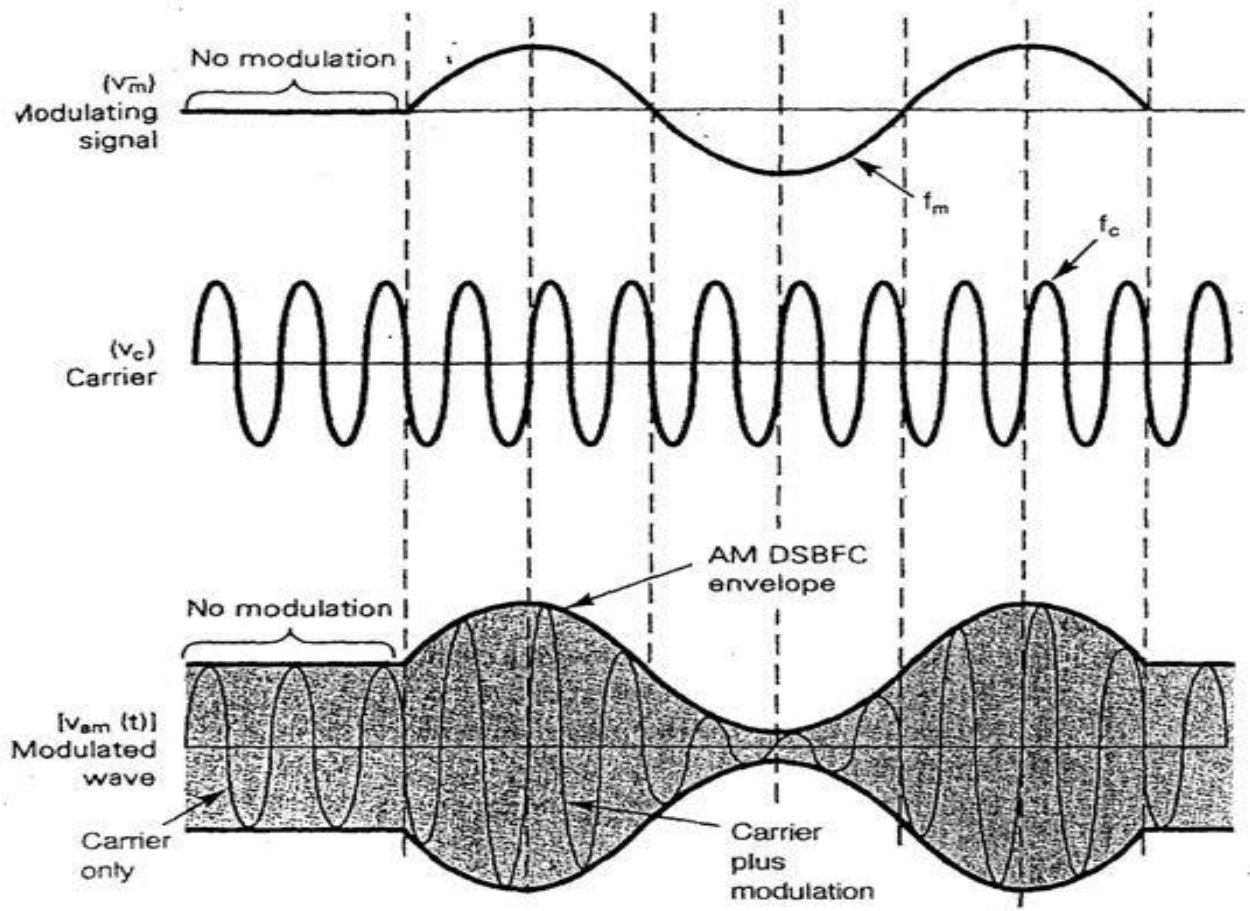

Powerful Modulations
AM
FM
PM
PWM
ASK
FSK
PSK
IQ
Digital Modulation
Analog Modulation


Digital Modulation
Amplitude Modulation
Angle Modulation
Phase Modulation
Frequency Modulation
Amplitude Shift Keying (ASK)
Phase Shift Keying (PSK)
Frequency Shift Keying (FSK)

Amplitude Modulation Envelope

- Most commonly used is AM *double-sideband full carrier* **(DSBFC)**, or sometimes called conventional AM or AM.

NB :
- Repetition rate of the envelope ≈ frequency of the modulating signal.
- Shape of the envelope ≈ shape of the modulating signal



AM Envelope Generation

Linear Modulation Techniques:

Digital modulation can be broadly classified as:

1. Linear (change Amplitude or phase)
2. Non linear modulation techniques (change frequency).

Linear Modulation Techniques:

- The amplitude /phase of the transmitted signal s(t), varies linearly with the modulating digital signal, m(t).
- These are bandwidth efficient (because it doesn't change frequency) and hence are very attractive for use in wireless communication systems where there is an increasing demand to accommodate more and more users within a limited spectrum.

Digital Modulation

- Two **general** classifications of digital modulation methods:

  1) **Linear** : amplitude of Tx signal varies **linearly** with modulating information signal, $m(t)$.

  2) **Constant Envelope** : *non-linear* methods where amplitude of Tx signal held **constant** regardless of the variation in the modulating information signal, $m(t)$.

Classification

Linear modulation

- Amplitude of the transmitted signal varies linearly with message signal
- Bandwidth efficient
- QPSK, OQPSK

Non linear (constant envelope) modulation

- Amplitude of the transmitted signal does not vary with the amplitude of the message signal
- Power efficient class C amplifiers can be used
- Low out of band radiation
- Limiter-discriminator can be used for demodulation
- FSK,MSK, GMSK

CONSTANT ENVELOPE

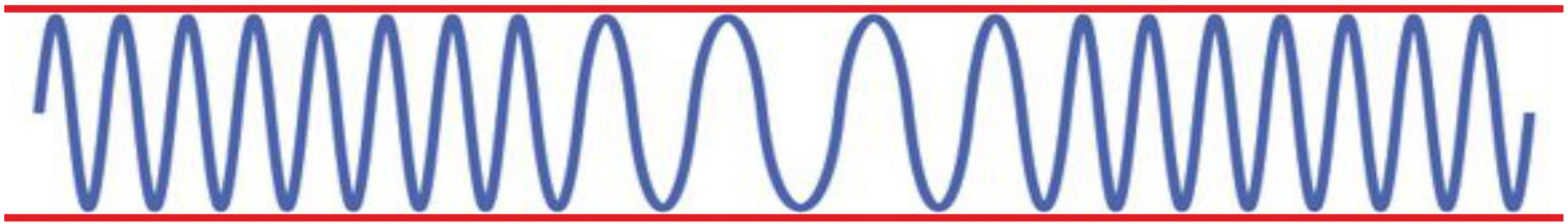
Non-Linear Modulation in this Amplitude of Modulated wave is constant

Constant Envelope is achieved when a sinusoidal waveform reaches equilibrium in a specific system. This happens when negative feedback in a control system, such as in radio automatic gain control or when an amplifier reaches steady state. Steady state, as defined in electrical engineering, occurs after a system becomes settled. To be more specific, control systems are unstable until they reach a steady state. Constant envelope needs to occur for the system to be stable, where there is the least amount of noise and feedback gain has rendered the system steady.

Constant Envelope Modulation

- Amplitude of the carrier is constant, regardless of the variation in the modulating signal
  - Better immunity to fluctuations due to fading.
  - Better random noise immunity
  - Power efficient
- They occupy larger bandwidth

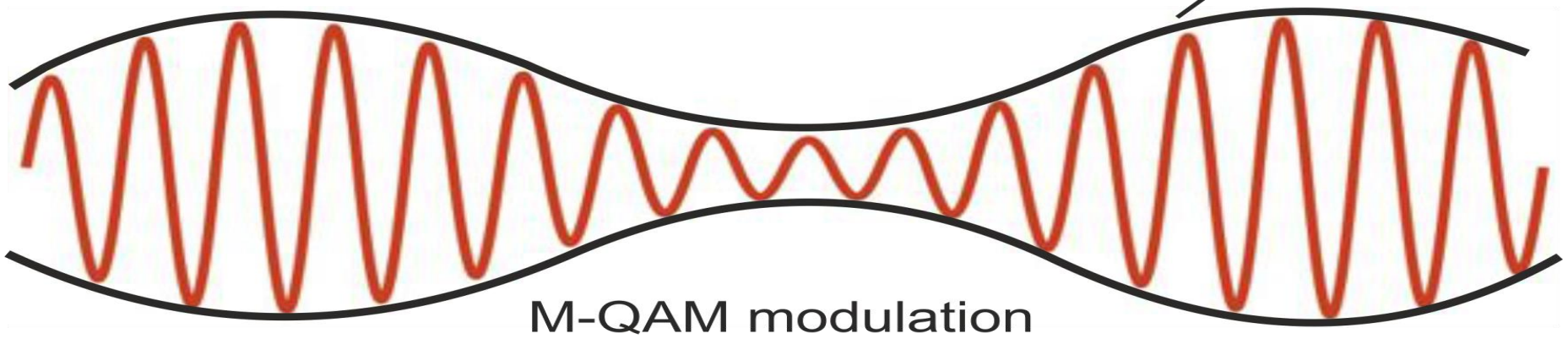
Constant envelope modulation example



M-CPFSK modulation --> Time

PEP [dBm] = RMS [dBm]

Non-constant envelope modulation example



M-QAM modulation

PEP [dBm] = RMS [dBm] + PAPR [dB]

Frequency Modulation

- Complex envelope for phase modulation (PM) and frequency modulation (FM) is: $g(t) = A_c e^{j\theta(t)}$
  - The real envelope $R(t) = |g(t)| = A_c \rightarrow$ constant value
  - PM and FM are **constant envelope** modulation methods
    - » AM & DSB-SC are linear since $R(t)$ is linearly $\propto$ to $m(t)$
- Bandpass signal is: $s(t) = A_c \cos[2\pi f_c t + \theta(t)]$
- Relationship between $\theta(t)$ & $m(t)$ determines type of modulation $\rightarrow$ PM or FM
  - For PM we have $\theta(t) = D_p\, m(t)$
    - » Linear relationship between phase and modulating signal
    - » Still **nonlinear** modulation since: $s(t) = A_c \,\mathrm{Re}\{e^{j2\pi f_c t} e^{jD_p m(t)}\}$

(-ve)Feedback is used to create a feedback signal to control gain, reduce distortion, control output voltage, improve stability or create instability, as in an oscillator. Some examples of constant envelope modulations are as FSK, GFSK, MSK, GMSK and Feher's IJF - All constant envelope modulations allow power amplifiers to operate at or near saturation levels. Although, the power spectrum efficiency of a non-constant amplitude envelope is always higher than that of a constant envelope modulation.[downside of constant envelope]

What is constant envelope modulation?

We call those modulations **constant envelope** modulations; that is, no signal is **modulated** on the amplitude. ... **Constant envelope modulation** can be a linear or a nonlinear **modulation** in digital mobile systems. In general, **constant envelope modulation** is used for the nonlinear **modulation** (FM) in analog mobile systems.

Is QPSK constant envelope?

Similar to BPSK, **QPSK** is a **constant envelope** waveform with discontinuous phases at the symbol boundaries.

The Constant Envelope Modulation Power Advantage

Constant Envelope Modulation provides significant advantages over the standard PSK forms of modulation used with Direct Sequence Spread Spectrum Systems.

Constant envelope modulations such as FSK, GFSK, MSK, GMSK and Feher's IJF **[intersymbol-interference and jitter-free (IJF)]** - nonlinearly filtered ISI-free modulations offer not only enhanced spectral efficiency in DS systems, they also provide an inherent transmitted power advantage. All constant envelope modulations allow a DS transmiter's power amplifiers to operate at or near saturation levels. On the other hand, standard BPSK, QPSK, QAM and even π/4-DCQPSK modulations contain AM components in the modulated envelope, which require from 3 to 6 dB of back off (from saturation) in the output power amplifier to reduce or eliminate spectrum splatter of sideband components that might cause Adjacent Channel Interference **(ACI).**

Most non-constant envelope modulations actually require fully linear power amplification and thus for similar power outputs (to saturated RF PA stages) require RF power amplifiers that are up to 4 times more capable (to handle the PEP required by the waveform). These larger power amplifiers are less efficient, consume more primary power, generate more heat and are more expensive than their Class C amplifier counterparts.

MSK, GMSK, Combined Linear and Constant Envelope Modulation Technique, MPSK, QAM, OFDM

M S K

Minimum Shift Keying

Minimum-Shift Keying (MSK) and GMSK

- **Definition**

  MSK is continuous 2FSK with minimum modulation index h=0.5

- **Properties of MSK**

  1)continuous binary FSK ;

  2)modulation index being minimum

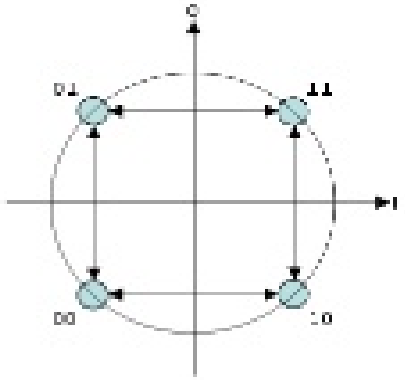
  3)two signals are orthogonal over the bit interval ;

  4)being identical with OQPSK when the pulse shape is sinusoidal type

INTRODUCTION

In digital modulation, **minimum-shift keying**(**MSK**) is a type of continuous-phase frequency-**shift keying** that was developed in the late 1950s and 1960s.

Similar to OQPSK(Offset quadrature phase-shift keying), **MSK** is encoded with bits alternating between quadrature components, with the Q component delayed by half the symbol period.



What is meant by MSK? **[in QPSK BW required is more so we go for MSK]**

Minimum shift keying, **MSK**, is a form frequency modulation based on a system called continuous-phase frequency-shift keying.

In digital modulation, **minimum-shift keying** (**MSK**) is a type of continuous-phase frequency-shift keying that was **developed in the late 1950s by Collins Radio employees Melvin L.** ... Similar to OQPSK, **MSK** is encoded with bits alternating between quadrature components, with the Q component delayed by half the symbol period.

Why MSK is called shaped QPSK?

The **MSK** modulation is a constant envelope signal with continuous phase that results from modulating the instantaneous frequency with rectangular pulses. ... **MSK** is considered to be a special case of **Offset QPSK** (**OQPSK**) with half sinusoidal pulse weighting rather than rectangular

What is the difference between MSK and GMSK?

An **MSK** signal with I-Q components is formed by passing the modulation signal through a half-sine shaper before modulation. **GMSK**/GFSK modulation uses the same technique as **MSK** but instead of the half-sine pulse shape, input bits have the shape of the Gaussian bell curve.

What is the difference between FSK and MSK?

Minimum frequency-shift keying or minimum-shift keying (**MSK**) is a particular spectrally efficient form of coherent **FSK**. In **MSK**, the **difference between** the higher and lower frequency is identical to half the bit rate. Consequently, the waveforms that represent a 0 and a 1 bit differ by exactly half a carrier period.

Which modulation technique is used in 3g?

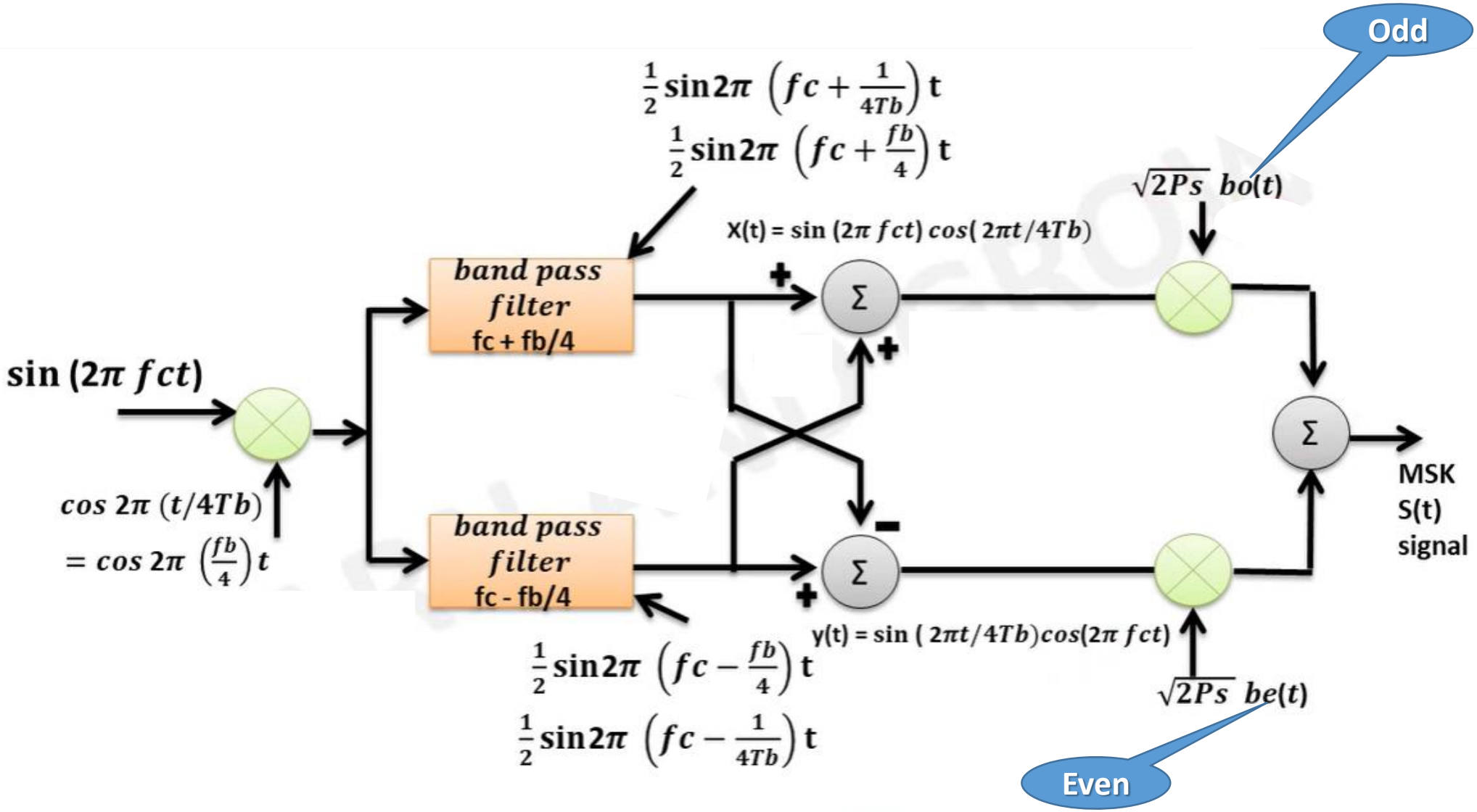
In addition, adaptive modulation allows the system to overcome fading and other interference. Both **QAM** and **QPSK** are modulation techniques used in IEEE 802.11 (Wi-Fi*), IEEE 802.16 (WiMAX*) and 3G (**WCDMA**/HSDPA) wireless technologies.

Which modulation technique is used in 4g?

LTE devices use **QPSK**, **16QAM** and **64QAM** to modulate data and control information. The eNode B supports all of these modulation techniques for the Down Link direction. However, **64QAM** is optional in the Uplink direction. A modulation technique is selected based on the measured Signal to Noise Ratio (SNR).

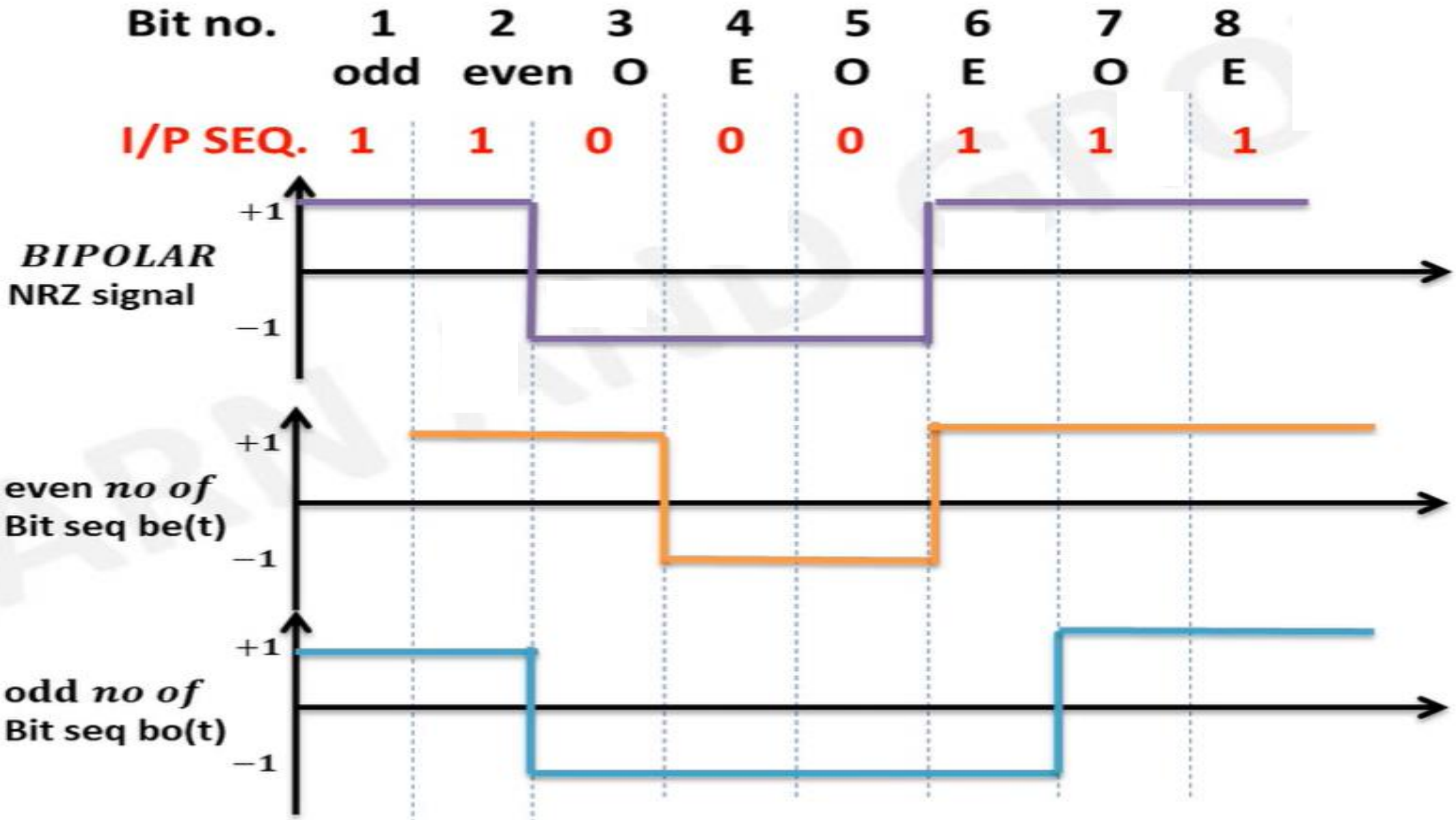
Which modulation technique is used in 5g?

Phase shift keying: 5G technology implements **quadrature phase shift keying**, **QPSK** as the lowest order modulation format. Although this will provide the slowest data throughput it will also provide the most robust link and as such it can be used when signal levels are low or when interference is high.

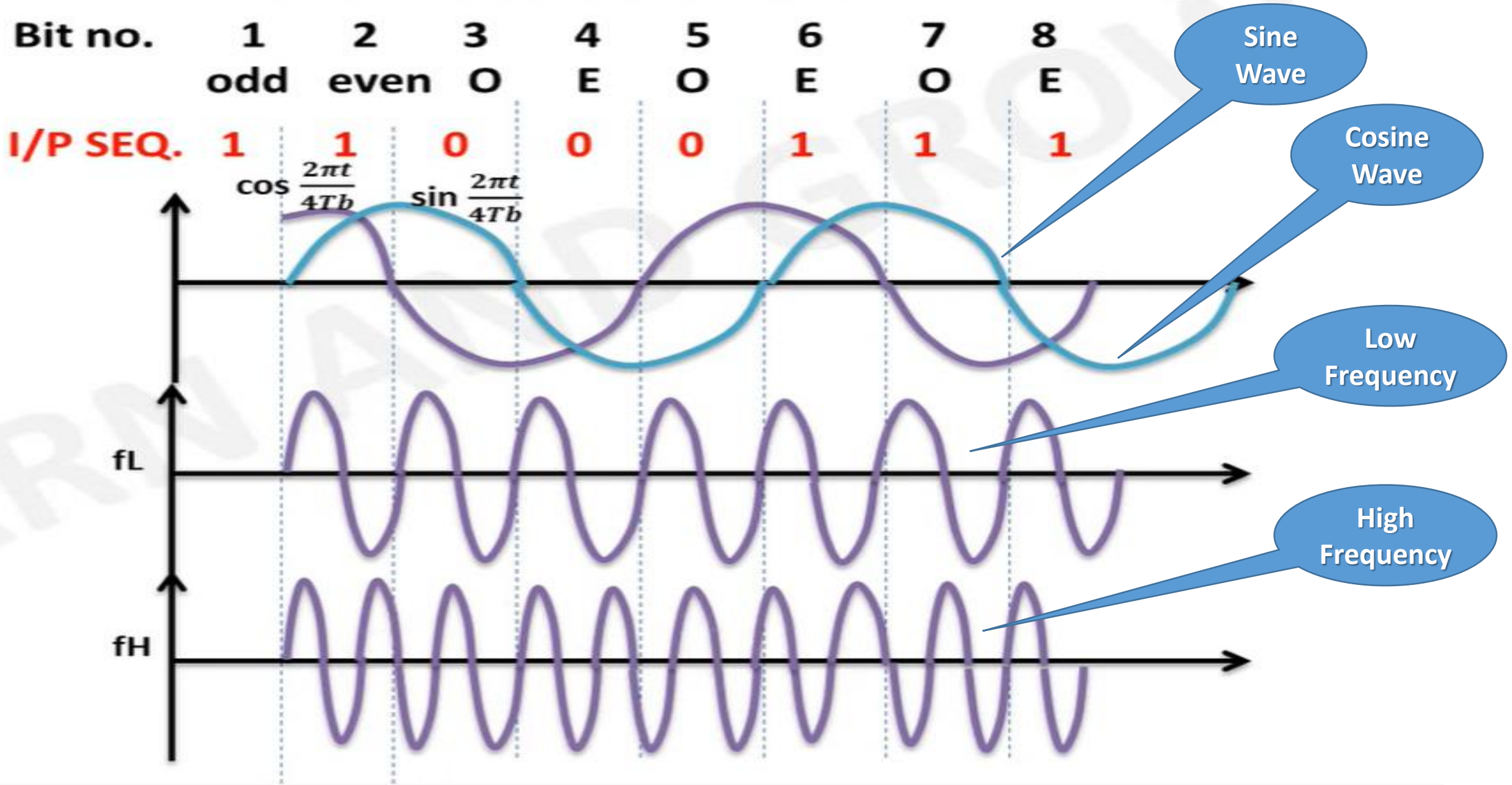


$$S(t) = \sqrt{2Ps}\ [be(t)\sin(2\pi t/4Tb)]\cos(2\pi fct) + \sqrt{2Ps}\ [bo(t)\cos(2\pi t/4Tb)]\sin(2\pi fct)$$

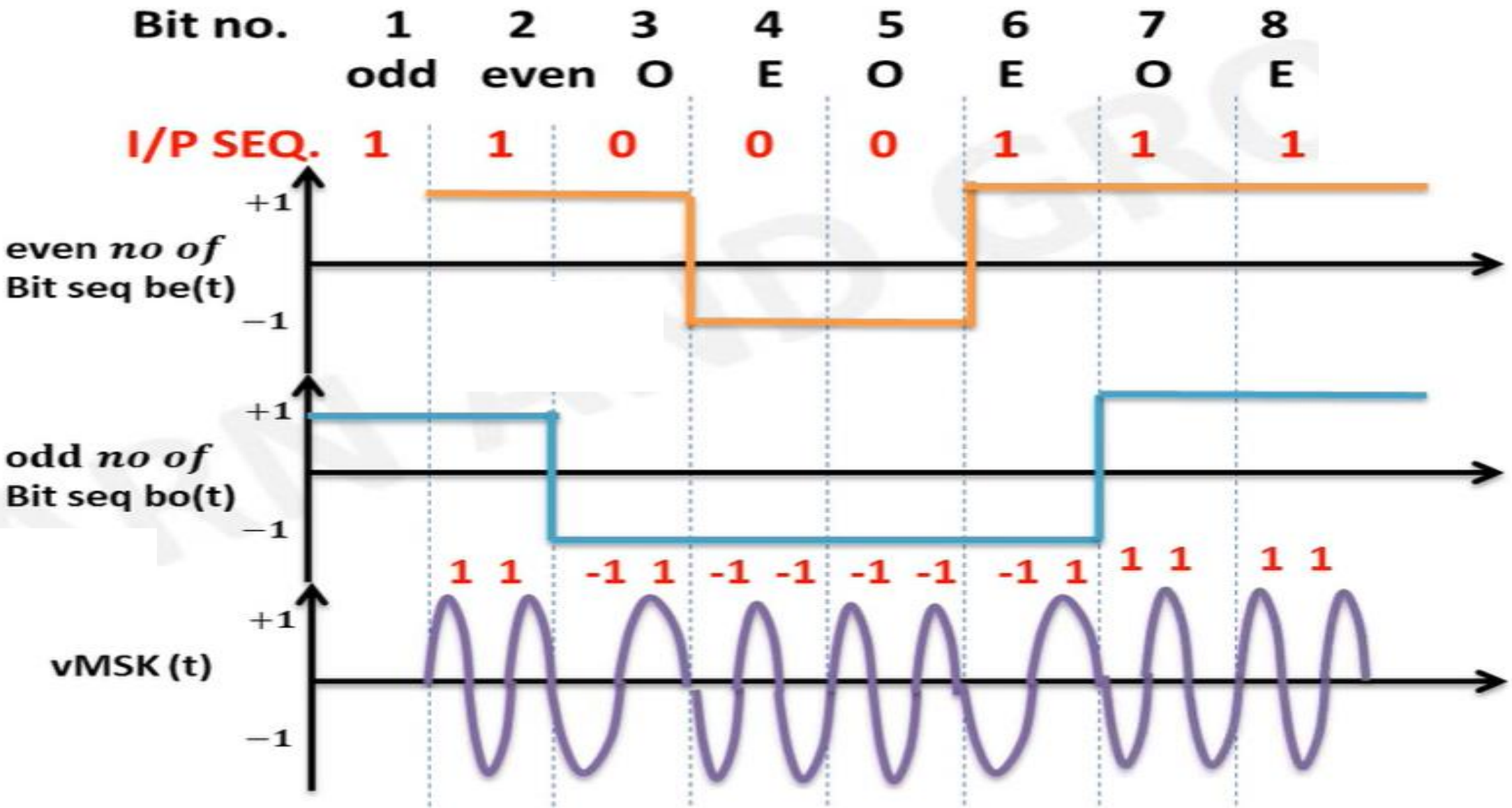
WAVE FORMS OF MSK SIGNAL



WAVE FORMS OF MSK SIGNAL



WAVE FORMS OF MSK SIGNAL



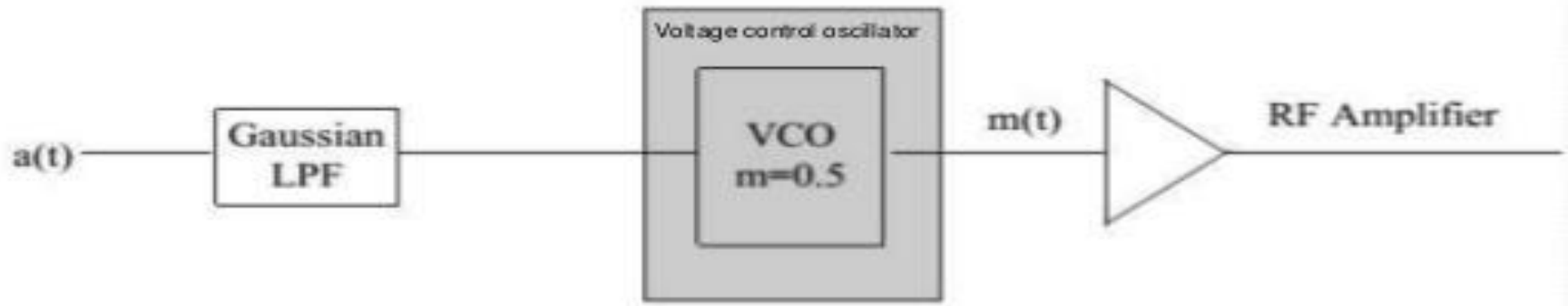
GMSK

Gaussian Minimum Shift Keying

GMSK:

- Gaussian Minimum Shift Keying.
- Continuous phase digital frequency modulation
- Modulation index h=1/2
- Gaussian Frequency Shaping Filter
- GMSK = **MSK** + **Gaussian filter**
- Characterized by the value of BT
  - **T** = bit duration
  - **B** = 3dB Bandwidth of the shaping filter
  - **BT** = 0.3 for GSM **B**andwidth **T**ime
  - **BT** = 0.5 for DECT (Digital Enhanced Cordless Telecommunications)
  - Data rate: 270.8 kbps

GMSK generator



Generating GMSK using a Gaussian filter and VCO

What is GMSK Modulation ?

Gaussian Minimum Shift Keying

Gaussian Minimum Shift Keying, ***GMSK is a form of modulation based on frequency shift keying that has no phase discontinuities and provides efficient use of spectrum as well as enabling high efficiency radio power amplifiers.***

Gaussian Minimum Shift Keying, or to give it its full title Gaussian filtered Minimum Shift Keying, GMSK, is a form of frequency modulation that is used in radio communications systems.

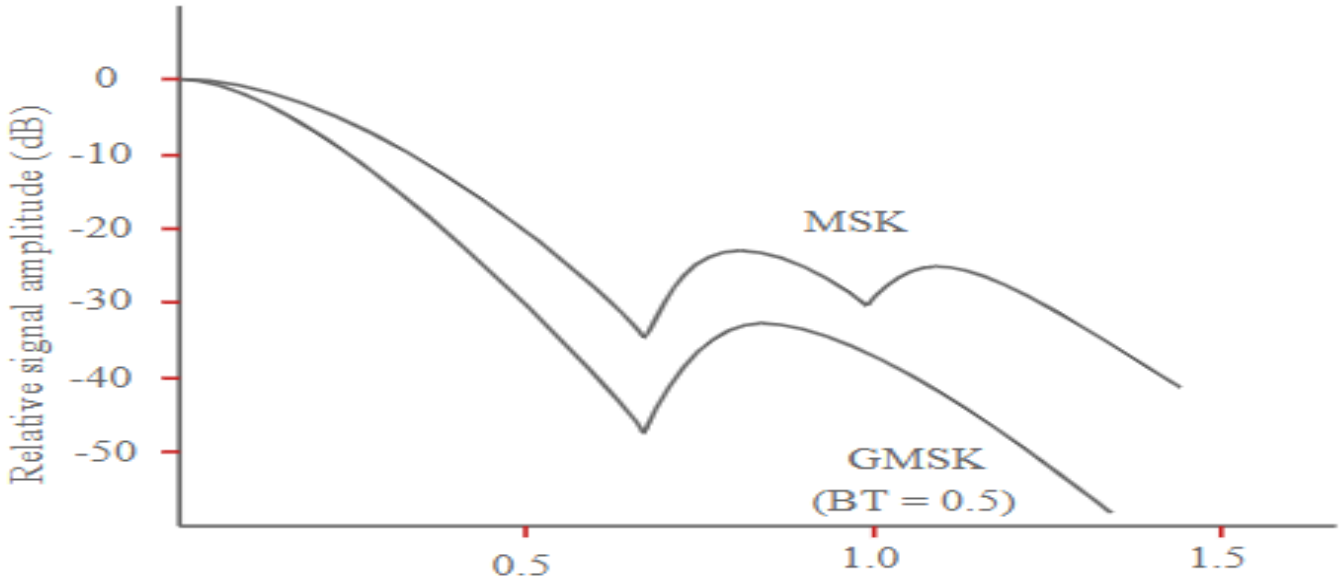
GMSK is probably most widely associated wit the 2G GSM mobile communications system where it proved to be an effective form of modulation. It was one of the reasons that GSM cellphones had a long battery life in view of the high efficiency that could be obtained from the RF power amplifiers.

GMSK basics

GMSK modulation is based on MSK, which is itself a form of continuous-phase frequency-shift keying, CPFSK. One of the problems with standard forms of PSK is that sidebands extend out from the carrier. To overcome this, MSK and its derivative GMSK can be used.

MSK and also GMSK modulation are what is known as a continuous phase scheme. Here there are no phase discontinuities because the frequency changes occur at the carrier zero crossing points. This arises as a result of the unique factor of MSK that the frequency difference between the logical one and logical zero states is always equal to half the data rate. This can be expressed in terms of the modulation index, and it is always equal to 0.5.

A plot of the spectrum of an MSK signal shows sidebands extending well beyond a bandwidth equal to the data rate. This can be reduced by passing the modulating signal through a low pass filter prior to applying it to the carrier. The requirements for the filter are that it should have a sharp cut-off, narrow bandwidth and its impulse response should show no overshoot. The ideal filter is known as a Gaussian filter which has a Gaussian shaped response to an impulse and no ringing. In this way the basic MSK signal is converted to GMSK modulation.

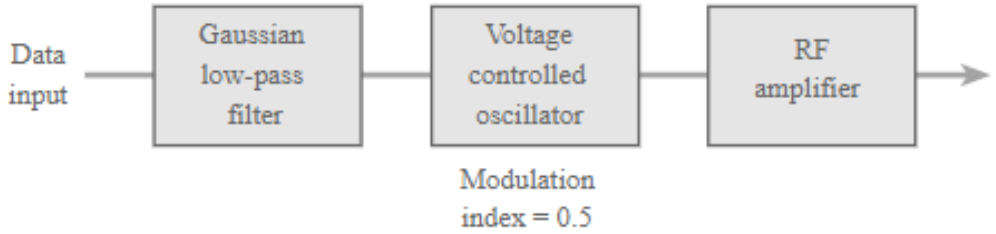


GMSK spectral density with offset from carrier

One of the advantages of MSK modulation in general and in this specific case, GMSK is that it. The MSK modulation makes the phase change linear and limited to ± (π/2) over a bit interval T. This enables MSK to provide a significant improvement over QPSK. Because of the effect of the linear phase change, the power spectral density has low side lobes that help to control adjacent-channel interference. However the main lobe becomes wider than the quadrature shift keying.

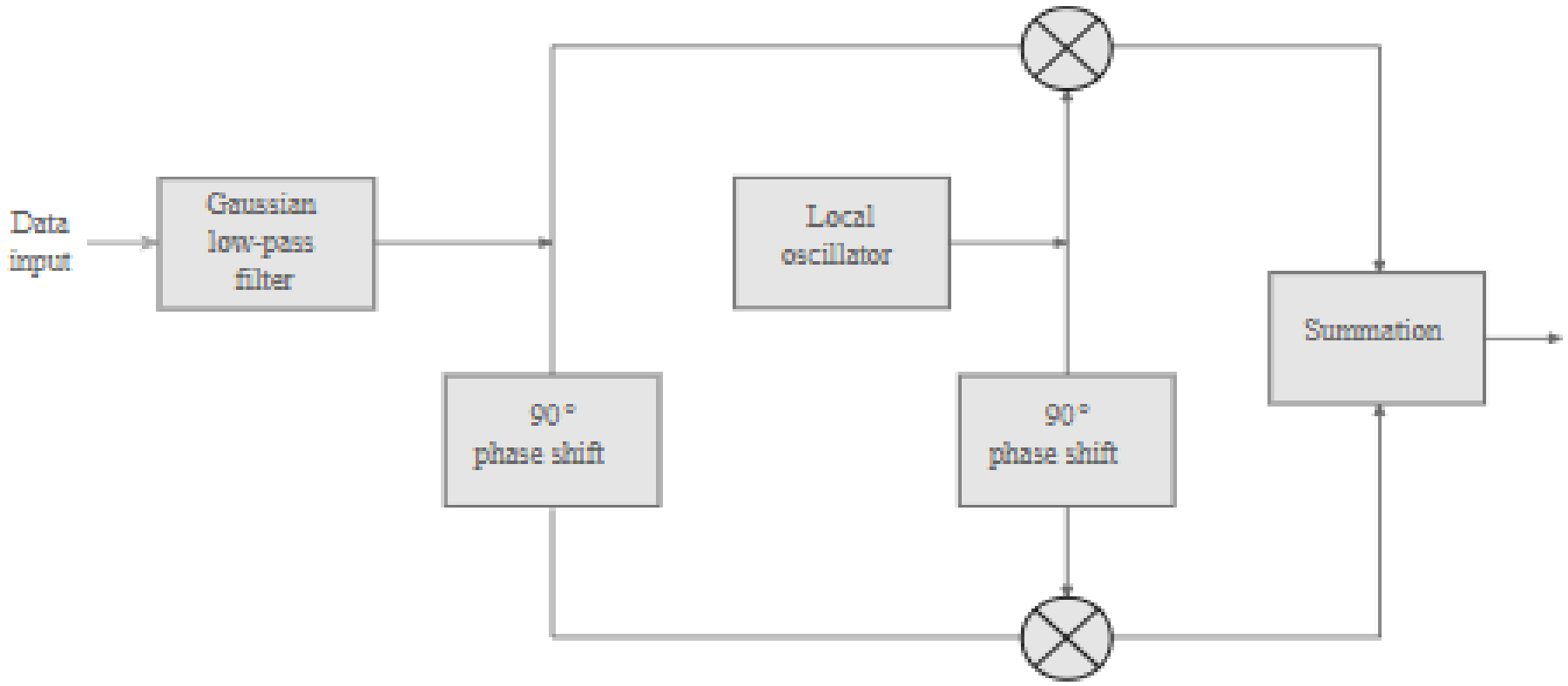
Generating GMSK modulation

There are two main ways in which GMSK modulation can be generated. The most obvious way is to filter the modulating signal using a Gaussian filter and then apply this to a frequency modulator where the modulation index is set to 0.5. This method is very simple and straightforward but it has the drawback that the modulation index **must exactly equal 0.5**. *In practice this analogue method is not suitable because component tolerances drift and cannot be set exactly.*



GMSK modulator using VCO

A second method is more widely used. Here what is known as a quadrature modulator is used. The term quadrature means that the phase of a signal is in quadrature or 90 degrees to another one. The quadrature modulator uses one signal that is said to be in-phase and another that is in quadrature to this. In view of the in-phase and quadrature elements this type of modulator is often said to be an I-Q modulator. Using this type of modulator the modulation index can be maintained at exactly 0.5 without the need for any settings or adjustments. This makes it much easier to use, and capable of providing the required level of performance without the need for adjustments. For demodulation the technique can be used in reverse.



GMSK modulator using an I-Q modulator

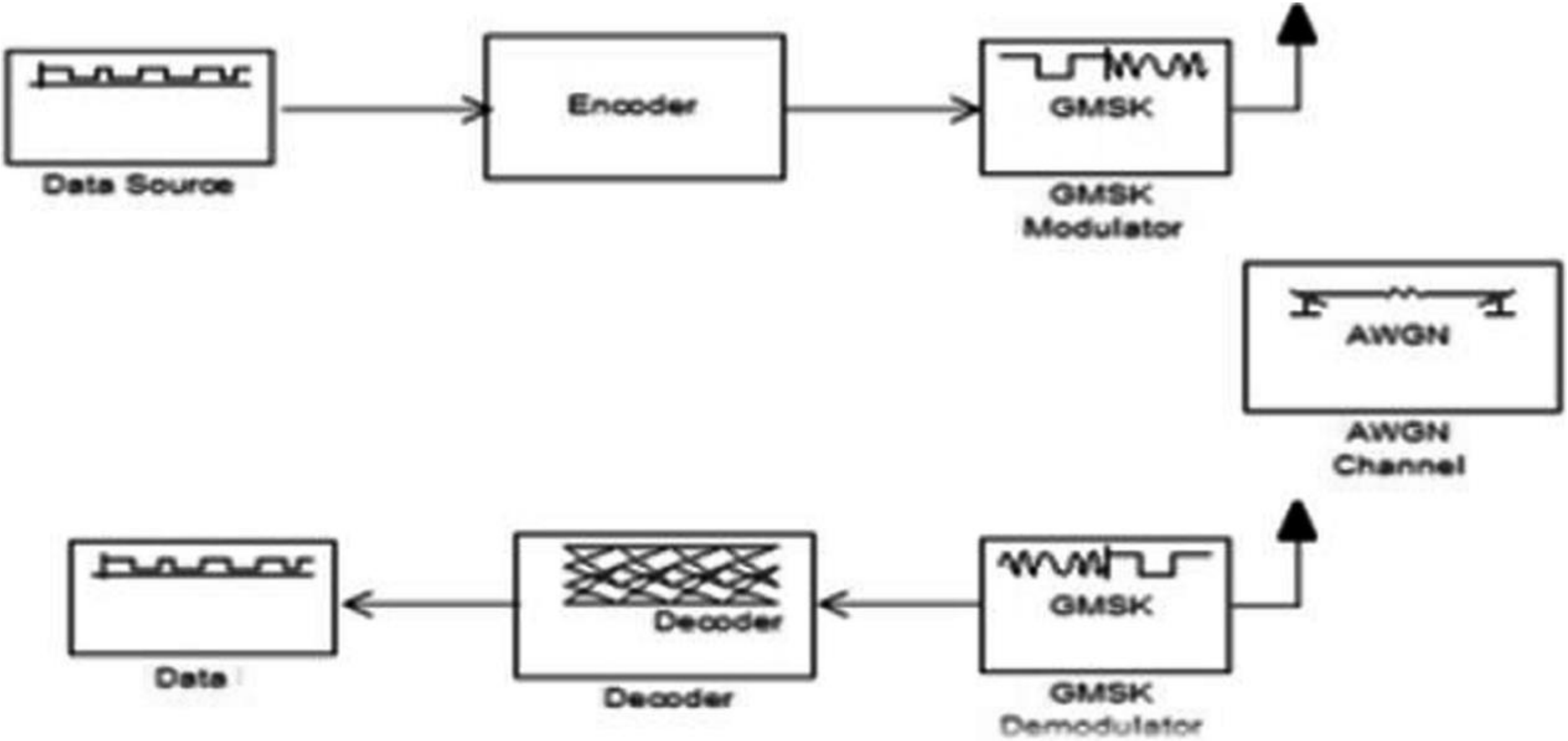
GMSK stands for **Gaussian Minimum Shift Keying**. This is a modulation scheme in. which the phase of the carrier is instantaneously varied by the \modulating" signal (i.e. the information to transmit).

Why is GMSK used in GSM?

Due to its efficient use of the spectrum, **GMSK** modulation has been **used** in a number of radio communications applications. The most notable one being in **GSM** Cellular Networks which has billions of subscribers worldwide. **GMSK** modulation is derived from MSK, which is a type of Continuous-Phase Frequency-Shift Keying.

GMSK - Gaussian Minimum Shift Keying

- GMSK is a form of continuous-phase FSK, in which the phase is changed between symbols to provide a constant envelope.
- The RF bandwidth is controlled by the Gaussian low-pass filter bandwidth.
- The degree of filtering is expressed by multiplying the filter 3dB bandwidth by the bit period of the transmission, ie. by BT.
- As bandwidth of this filter is lowered the amount of *intersymbol-interference* introduced increases.
- GMSK allows efficient class C non-linear amplifiers to be used, however even with a low BT value its bandwidth efficiency is less than filtered QPSK.
    - GMSK generally achieves a bandwidth efficiency less than 0.7 bits per second per Hz (QPSK can be as high as 1.6 bits per second per Hz).


Data Source
Encoder
GMSK
GMSK
Modulator
AWGN
AWGN
Channel
Data
Decoder
Decoder
GMSK
GMSK
Demodulator

Gaussian Minimum Shift Keying (GMSK)

- GMSK is a simple binary modulation scheme which may be viewed as a derivative of MSK.
- In GMSK, the sidelobe levels of the spectrum are further reduced by passing the modulating NRZ data waveform through a premodulation Gaussian pulse-shaping filter [7].
- In practice, GMSK is most attractive for its excellent power efficiency (due to the constant envelope).

Advantages of GMSK modulation

There are several advantages to the use of GMSK modulation for a radio communications system. One is obviously the improved spectral efficiency when compared to other phase shift keyed modes.

A further advantage of GMSK is that it can be amplified by a non-linear amplifier and remain undistorted This is because there are no elements of the signal that are carried as amplitude variations. This advantage is of particular importance when using small portable transmitters, such as those required by cellular technology. Non-linear amplifiers are more efficient in terms of the DC power input from the power rails that they convert into a radio frequency signal. This means that the power consumption for a given output is much less, and this results in lower levels of battery consumption; a very important factor for cell phones.

A further advantage of GMSK modulation again arises from the fact that none of the information is carried as amplitude variations. This means that is immune to amplitude variations and therefore more resilient to noise, than some other forms of modulation, because most noise is mainly amplitude based.

GMSK, Gaussian Minimum Shift Keying proved to be a particularly successful form of modulation for GSM and some other radio communications systems. Its combination of spectral efficiency and capability to enable efficient power amplifier operation enabled it to be an excellent choice for radio communications systems of the time.

M-ary

P S K

M-ary phase-shift keying

- What is M-ary PSK Modulation?
- M-ary Phase Shift Keying - or MPSK - is a modulation where data bits select one of M phase shifted versions of the carrier to transmit the data.
- The M possible waveforms all have the same amplitude and frequency but different phases.
- The signal constellations consist of M equally spaced points on a circle.

Note: $M = 2^n$ where n is integr No. 1,2,3,.....



In any communication system, what are the two major resources that are used efficiently ?

Answer:

Channel bandwidth and Transmit Power are two primary resources used efficiently.

Bandwidth efficiency depends upon bit rate per unit bandwidth.

Power utilization efficiency depends upon the S/N ratio required to attain certain bit error probability.

M-ary (Multi level) Signaling Scheme

- BASK, BFSK, BPSK and DPSK are binary modulation techniques.
- They are represented using only two symbols consisting of single bit either '1' or '0'.
- Two amplitudes, two frequencies or two phases are used to represent two symbols respectively.
- The maximum bandwidth of BASK is 2fb, BFSK is 4fb, BPSK is $2f_b$ and DPSK is f_b.

- To reduce the bandwidth requirement, two are more bits are combined to form one symbol.
- Thus the number of symbols is always greater than two, given by,

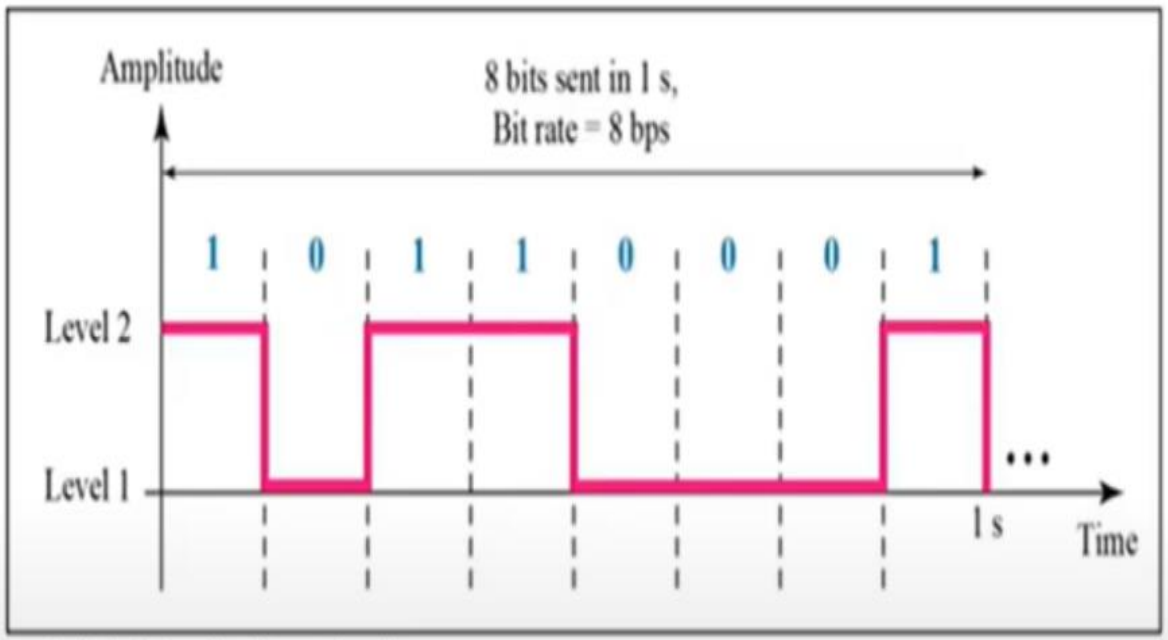
$$M = 2^n$$

where,

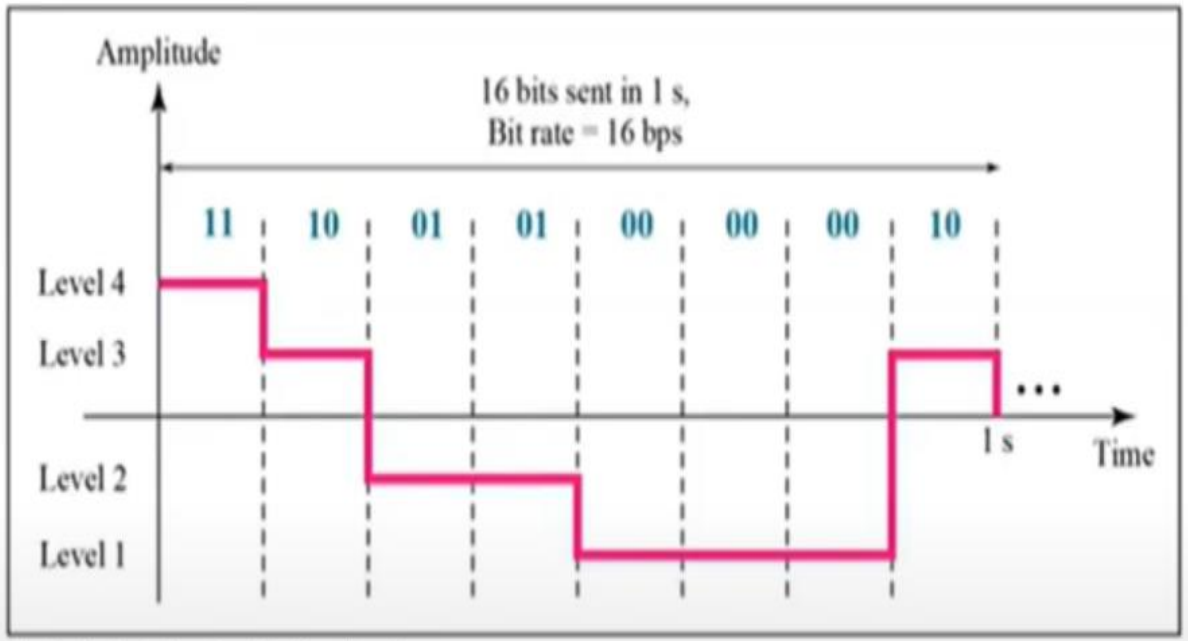
n = number of bits combined (n=2,3,4, ...)

M = Total number of symbols or levels.

- The signal representation which uses the combination of two or more bits to represent a single symbol is known as M-ary (Multi- level) Signaling Scheme.



a. A digital signal with two levels



b. A digital signal with four levels

Binary : Levels = M= 2

M-ary : Levels = M = 4

- M-level signal is used to modulate the carrier, then this technique is known as M-ary Digital Modulation Technique.
- In this, one of the M possible symbol is transmitted during each symbol duration, T_s, given by,

$$T_s = T_b \log_2 M = nT_b, \qquad \textit{Where } T_b = \textit{one bit duration}$$

- Based on the type of modulation used, they are classified as
  - *M-ary ASK*
  - *M-ary FSK*
  - *M-ary PSK*
  - *M-ary Amplitude and Phase Shift Keying (QAM)*

M-ary ASK Modulation Technique

- In M-ary Amplitude Shift Keying (M-ary ASK), there are M different amplitude levels of the carrier.
- The signal is represented by:

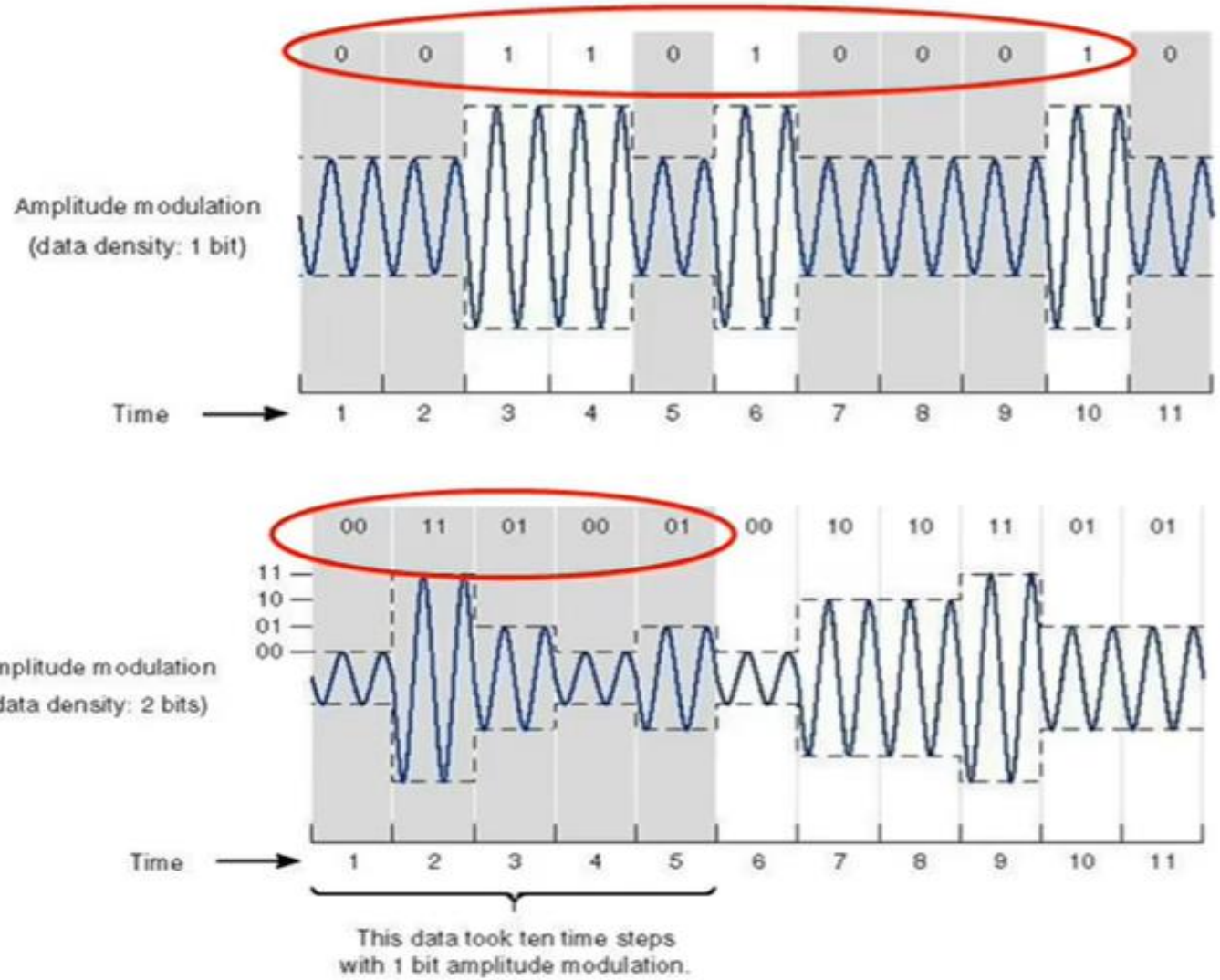
$$s_i(t) = A_i \cos(2\pi f_c t), \quad 0 \le t \le T_s \quad \text{for } i = 1,2,\ldots,M$$

$A_i = (2i - 1 - M)d$, where $2d$ is the difference between two consecutive signal amplitudes.

- Let $M = 4$ and $d = 1$, the four signal amplitudes will be -3, -1, 1 and 3 V.

The M-ASK signals will be: for $0 \le t \le T_s$

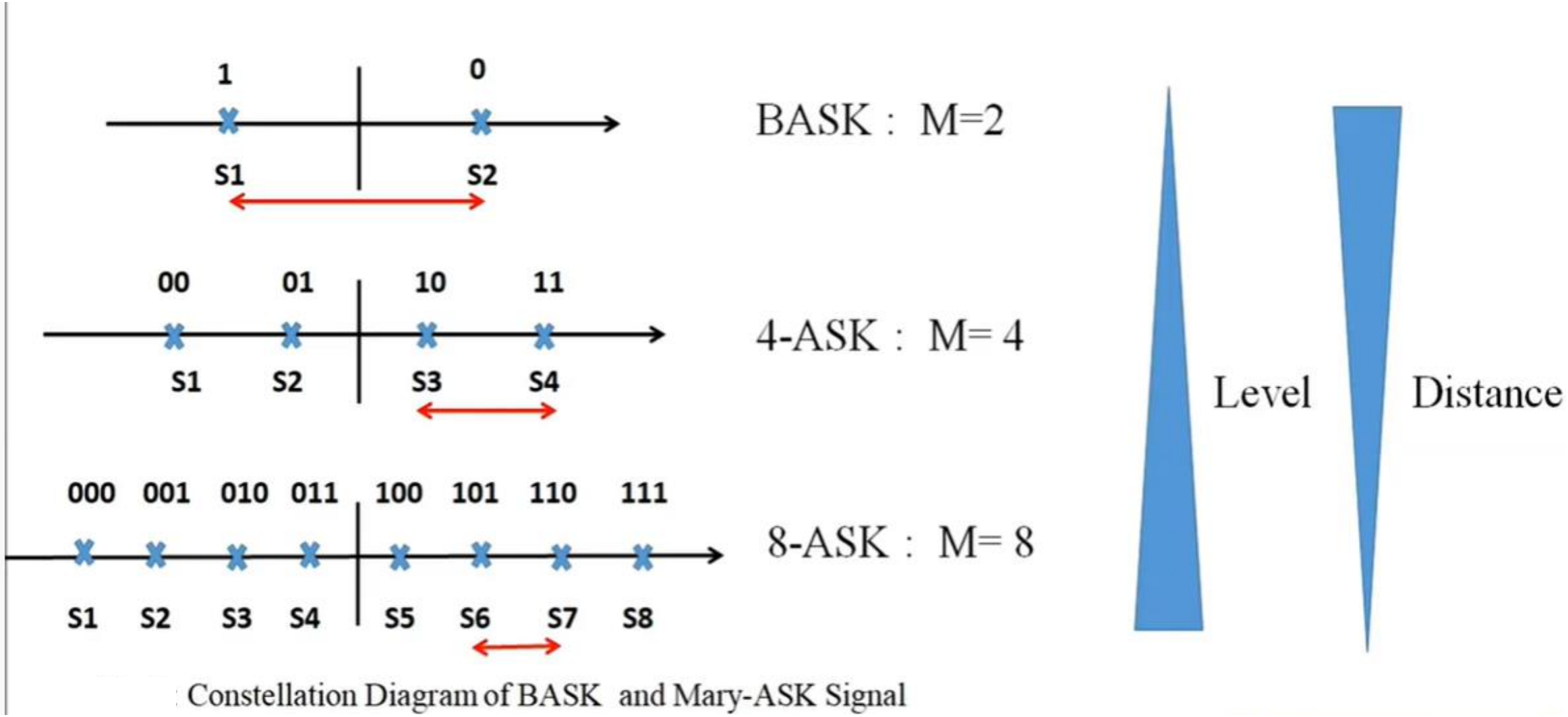
$$s_1(t) = \cos(2\pi f_c t), \quad s_2(t) = -\cos(2\pi f_c t),$$
$$s_3(t) = 3\cos(2\pi f_c t), \quad s_4(t) = -3\cos(2\pi f_c t).$$



BASK : 2 Level
1 bit in 1 symbol duration

M-ASK : 4 Level
2 bit in 1 symbol duration

BASK and Mary ASK Signal Representation



Constellation Diagram of BASK and Mary-ASK Signal

- M-ary ASK is also used in PAM.
- Implementation is simple.
- M-ary ASK is susceptible to noise and distortion.

M-ary FSK Modulation Technique

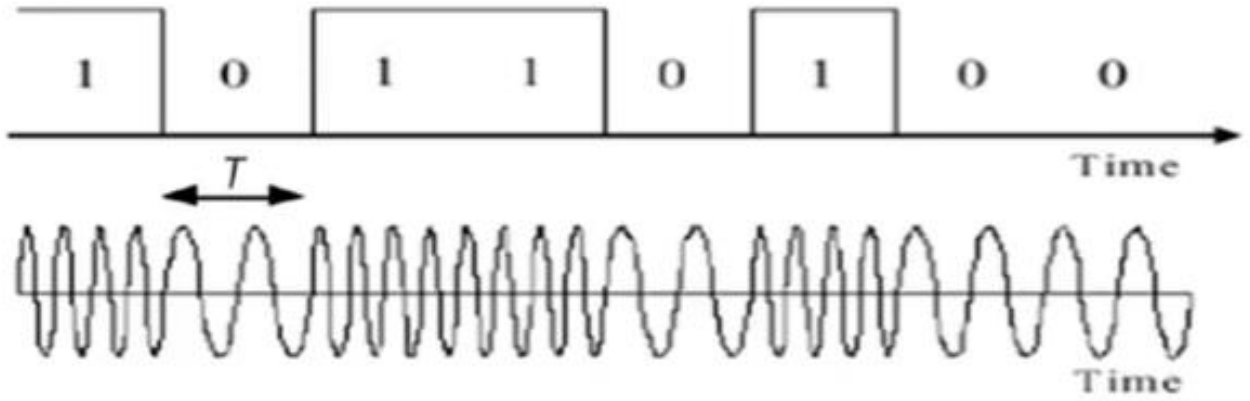
- In M-ary Frequency Shift Keying (M-ary FSK), there are *M* different carrier frequencies are used to represent M symbols.
- The signal is represented by:

$$s_i(t) = A\cos(2\pi f_{ci}t), \quad 0 \le t \le Ts \quad \text{for } i = 1,2,\ldots,M$$

where M different frequencies are $f_{ci} = f_c + \left(i - \frac{M}{2}\right)\Delta f$, for i= 1,2,…M

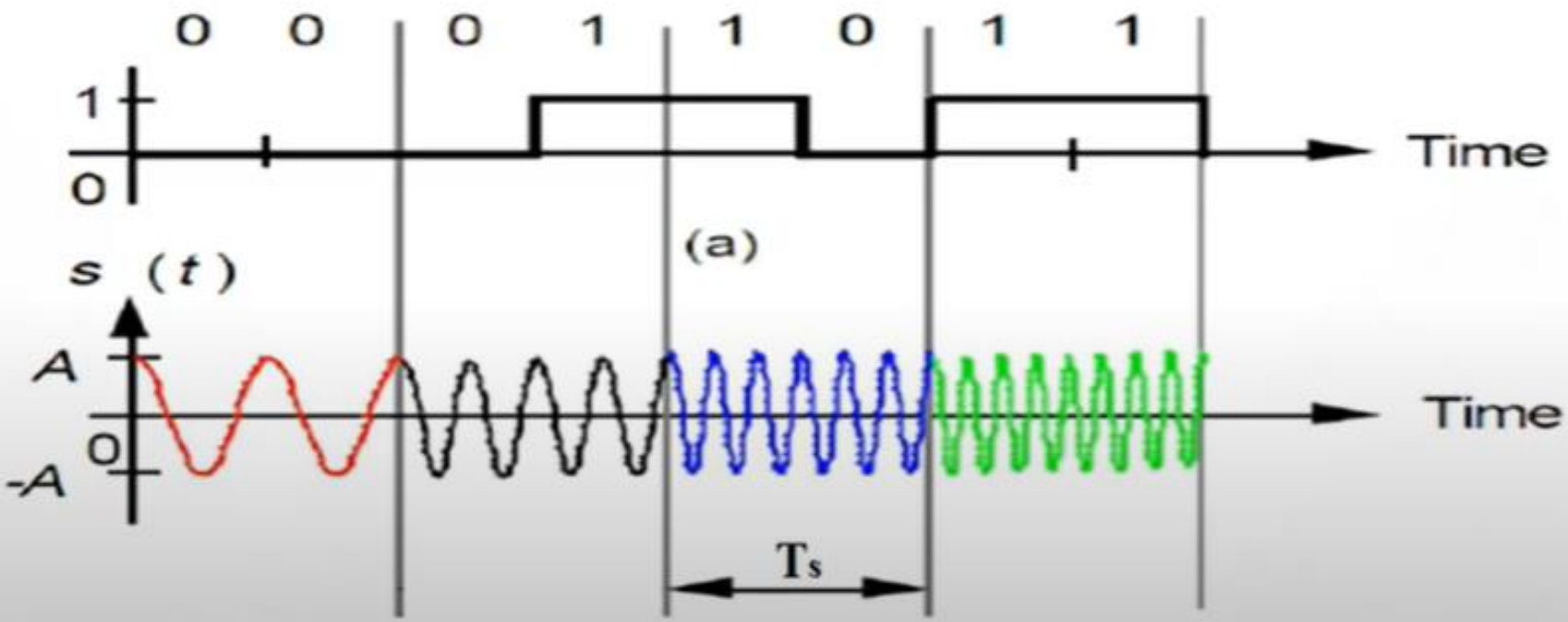
Δf is difference between two adjacent frequencies.

- Frequencies are chosen such that they are orthogonal to each other and there is no interference between adjacent carriers.



BFSK : 2 Level

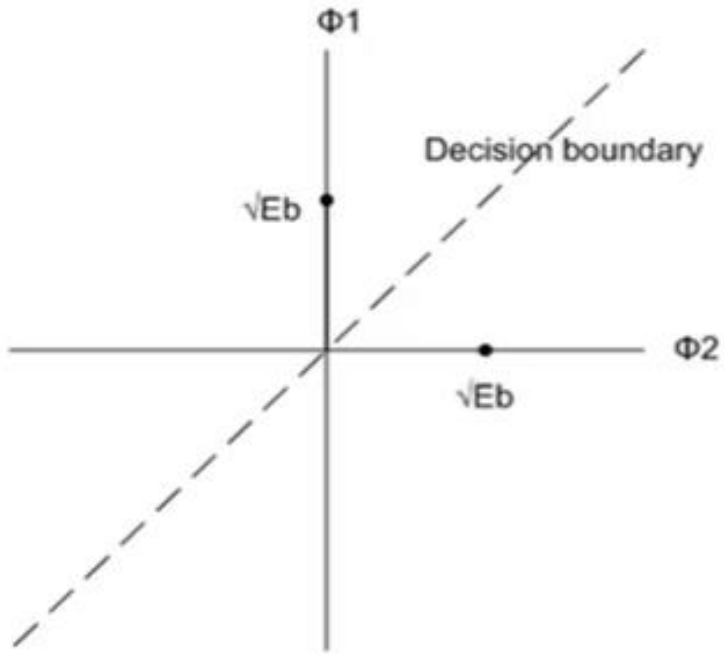
1 bit in 1 symbol duration



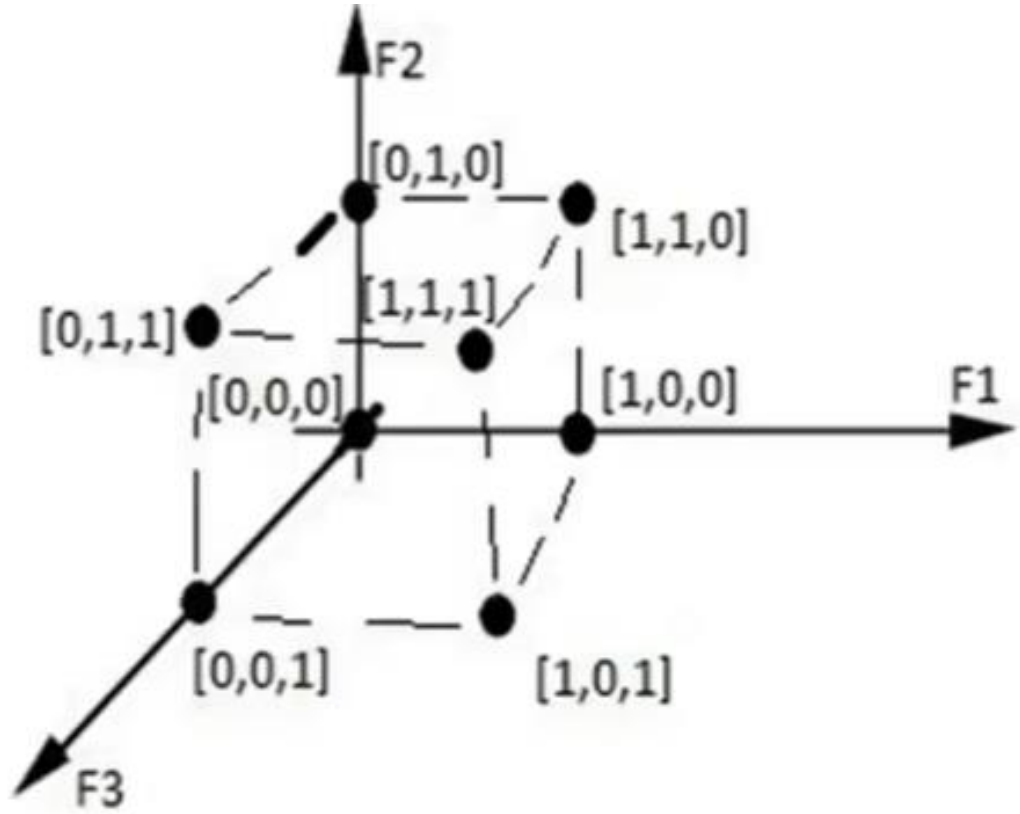
M-FSK : 4 Level

2 bit in 1 symbol duration

BFSK and Mary-FSK Signal Representation



BFSK : M=2



8-FSK : M= 8

Constellation Diagram of BFSK and Mary -FSK Signal

- Better noise immunity than Mary-ASK.
- The transmitted M number of signals are equal in energy and duration.
- The signals are separated such that they are orthogonal to each other and therefore there is no crowding in the signal space.
- The bandwidth efficiency of M-ary FSK decreases and the power efficiency increases with the increase in M.

M-ary PSK Modulation Technique

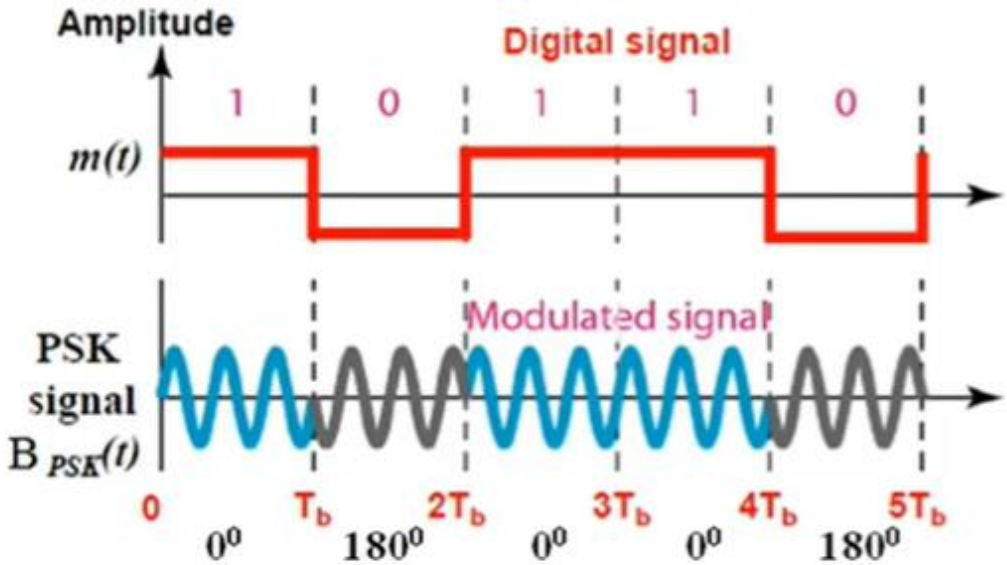
- In M-ary Phase Shift Keying (M-ary PSK), M different phase angles are used to represent M symbols.
- The signal is represented by:

$$s_i(t) = A\cos(2\pi f_c t + \phi_i)\,,\ 0 \le t \le T_s \ \text{for}\ i = 1,2,\dots,M$$

where $\phi_i = \frac{2\pi}{M}(i-1) + \text{constant}\,,$ for $i = 1,2,\dots M$

- $M = 4$ and the constant = 0, then four phases are 0, $\pi/2$, π, $3\pi/2$.

  $M = 4$ and the constant = $\pi/4$, then four phases are $\pi/4$, $3\pi/4$, $5\pi/4$, $7\pi/4$
- Both 4-PSK are called Quadrature Phase Shift Keying or QPSK

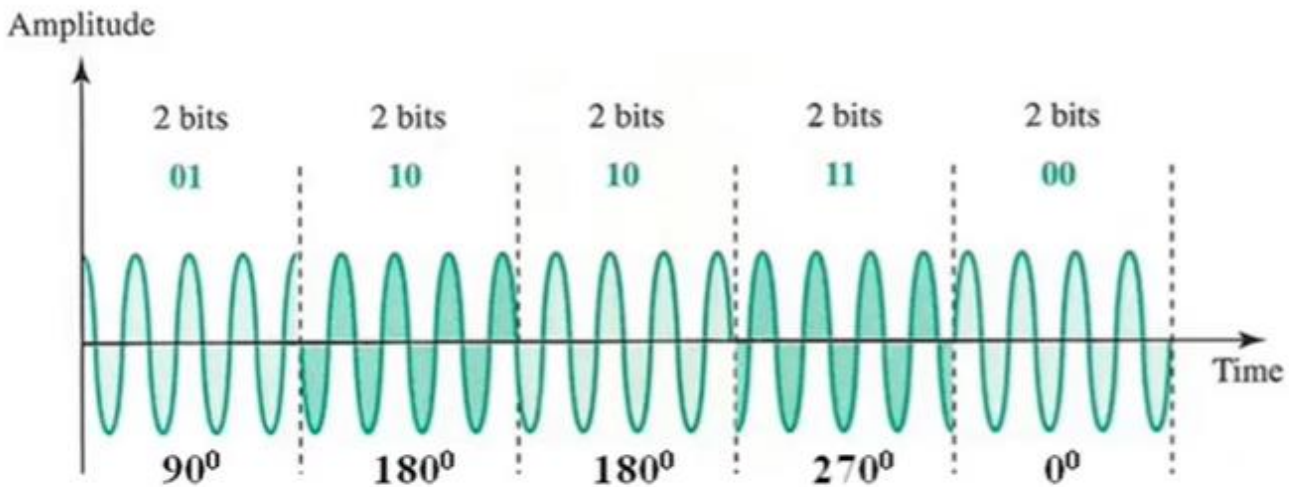


BPSK : 2 Level

1 bit in 1 symbol duration

Bit 1 is represented as 0 degree phase shift

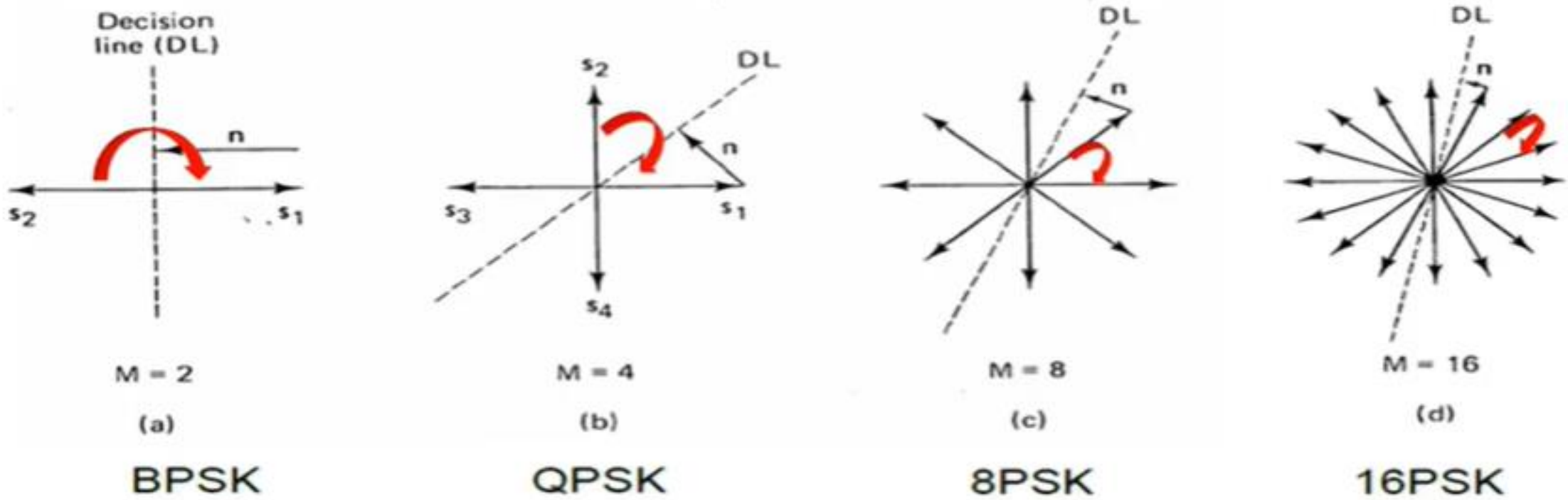
Bit 0 is represented as 180 degree phase shift



M-PSK : 4 Level

2 bit in 1 symbol duration

BPSK and Mary-PSK Signal Representation



Level

Phase Difference

Constellation Diagram of BPSK and Mary-PSK Signal

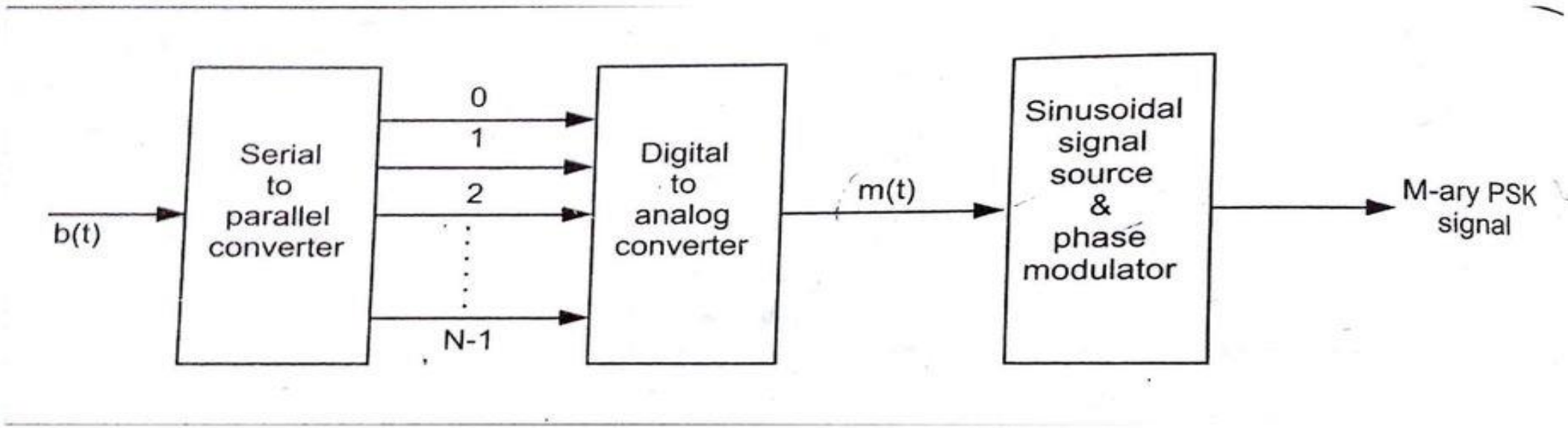
- Bandwidth of MPSK , $B = \frac{2}{Ts}$ Hz

- Bandwidth Efficiency , $\rho = \frac{log_2 M}{2}$ Bits/ s /Hz

As M increases, Bandwidth Efficiency of Mary PSK increases

- Uses constant envelope so immune to noise.
- There is considerable reduction in bandwidth requirement.
- Better performance than ASK and FSK.
- Increase in probability of errors with increase in number of bits per symbol.
- Design of M-ary PSK modulator and demodulator is complex.

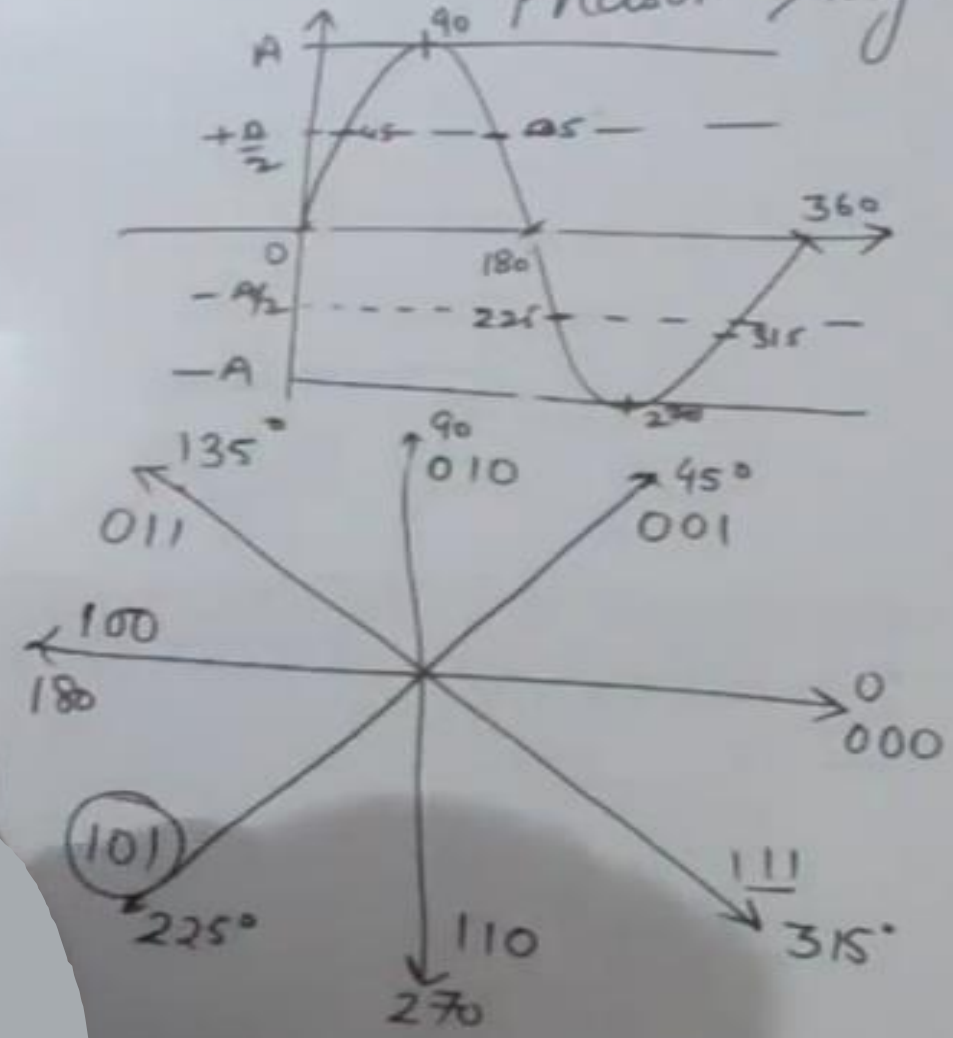
M-ARY PSK TRANSMITTER



Digital Modulation (8-PSK, 8-QAM, 16-QAM)

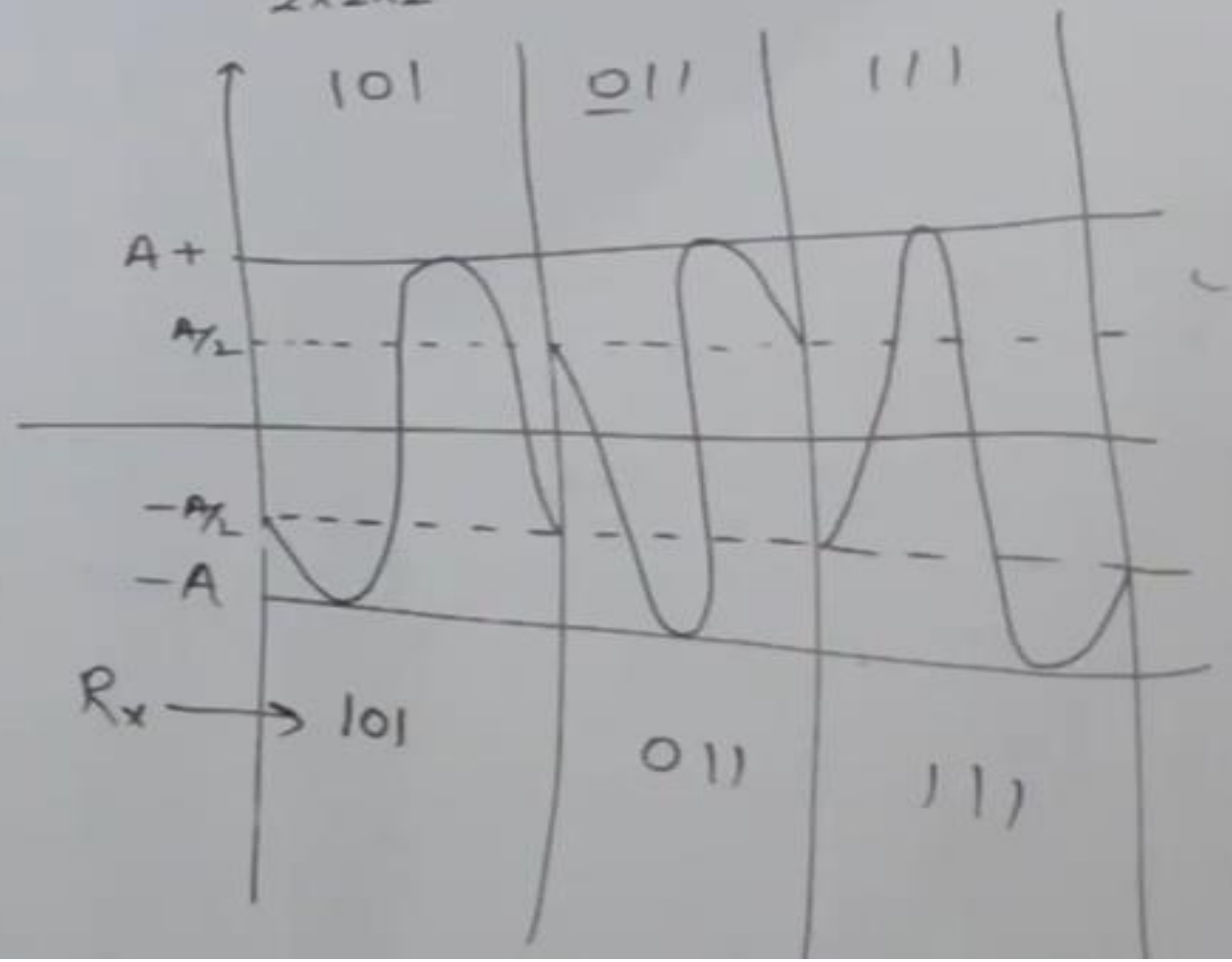
Phase shift Keying

Phasor Diagram



3 bits → $2^3 = 8$

$2 \times 2 \times 2$



QAM

M-ary A P S K

M-ary Amplitude & Phase-shift keying

Quadrature Amplitude Modulation, **QAM** is a signal in which two carriers shifted in phase by 90 degrees (i.e. sine and cosine) are **modulated** and combined. Often one signal is called the In-phase or "I" signal, and the other is the quadrature or "Q" signal.

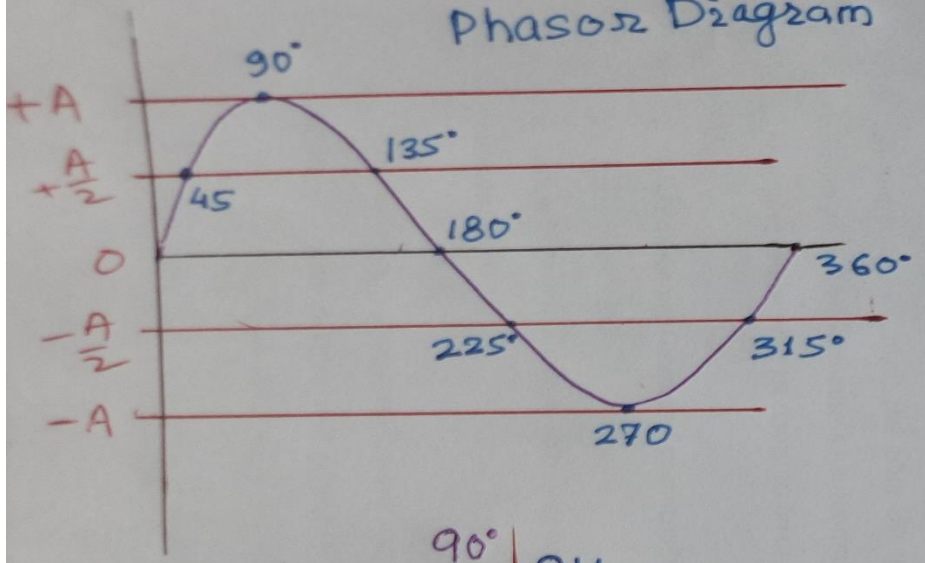
Quadrature Phase Shift Keying (**QPSK**)ASK and PSK can be combined to create **QAM** where both the phase and amplitude are changed. The receiver then receives this modulated signal, detects the shifts and demodulates the signal back into the original data stream.

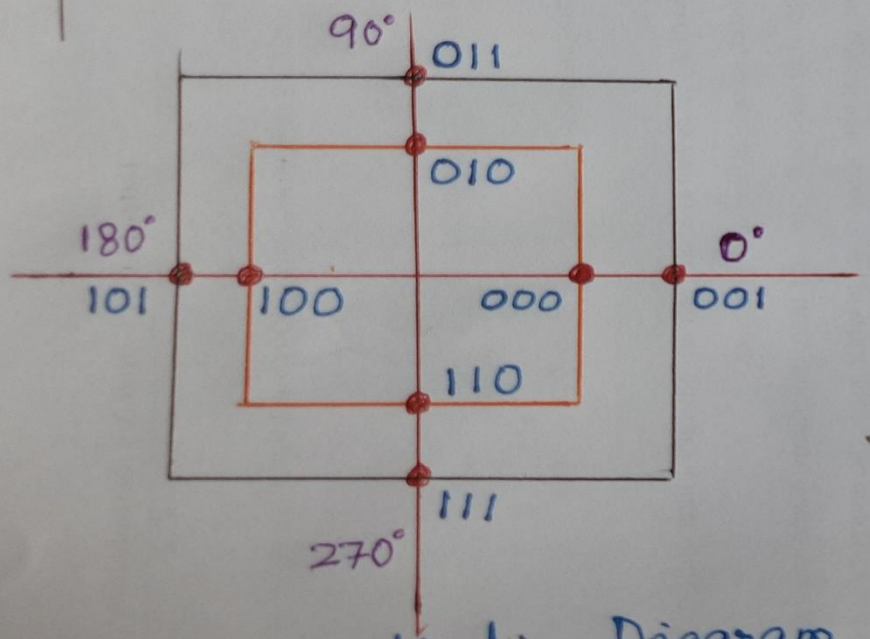
QAM is used extensively as a modulation scheme for digital telecommunication systems, such as in 802.11 Wi-Fi standards. Arbitrarily high spectral efficiencies can be achieved with **QAM** by setting a suitable constellation size, limited only by the noise level and linearity of the communications channel.

8-QAM

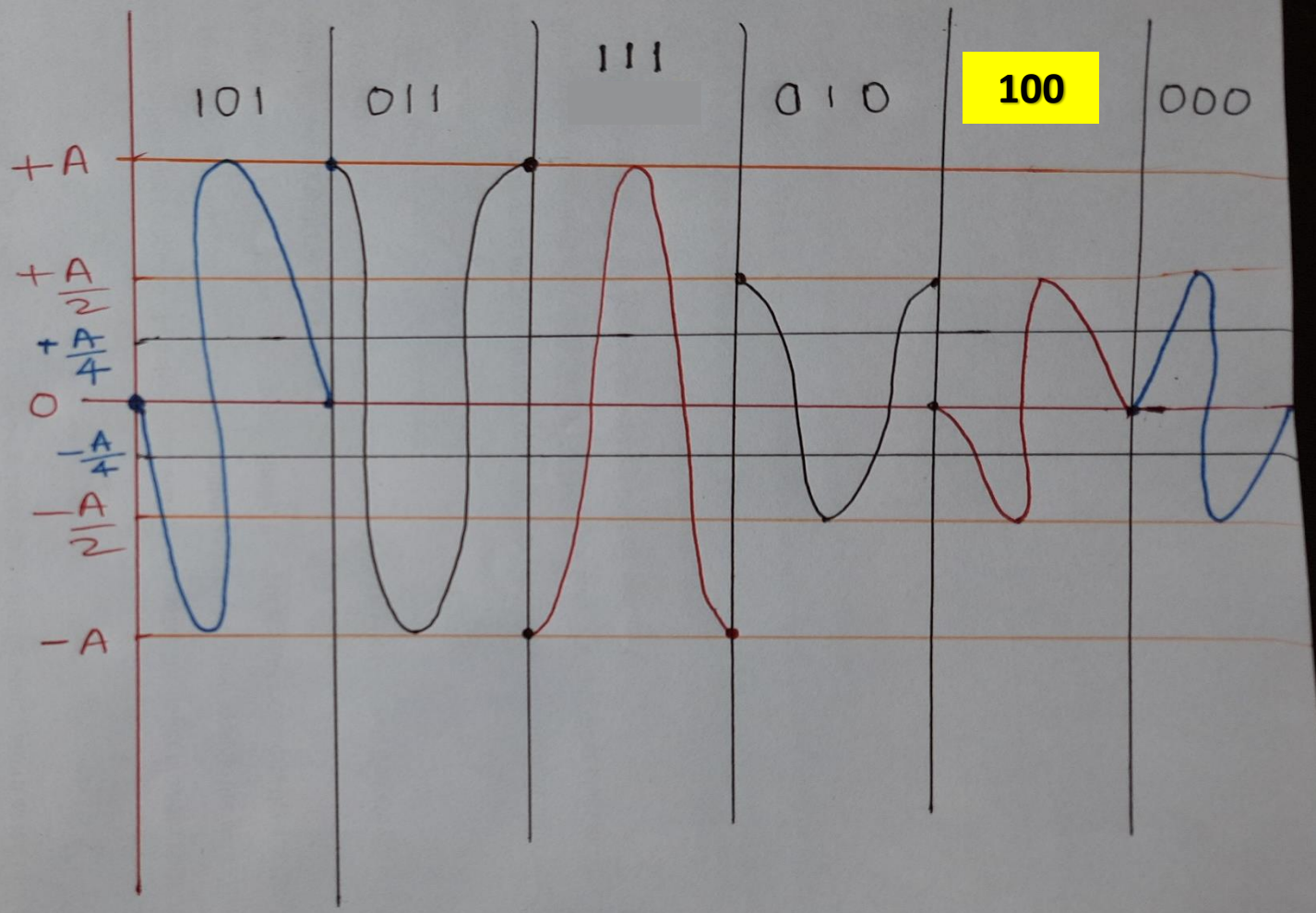
2 Amp × 4 Phase

Phasor Diagram





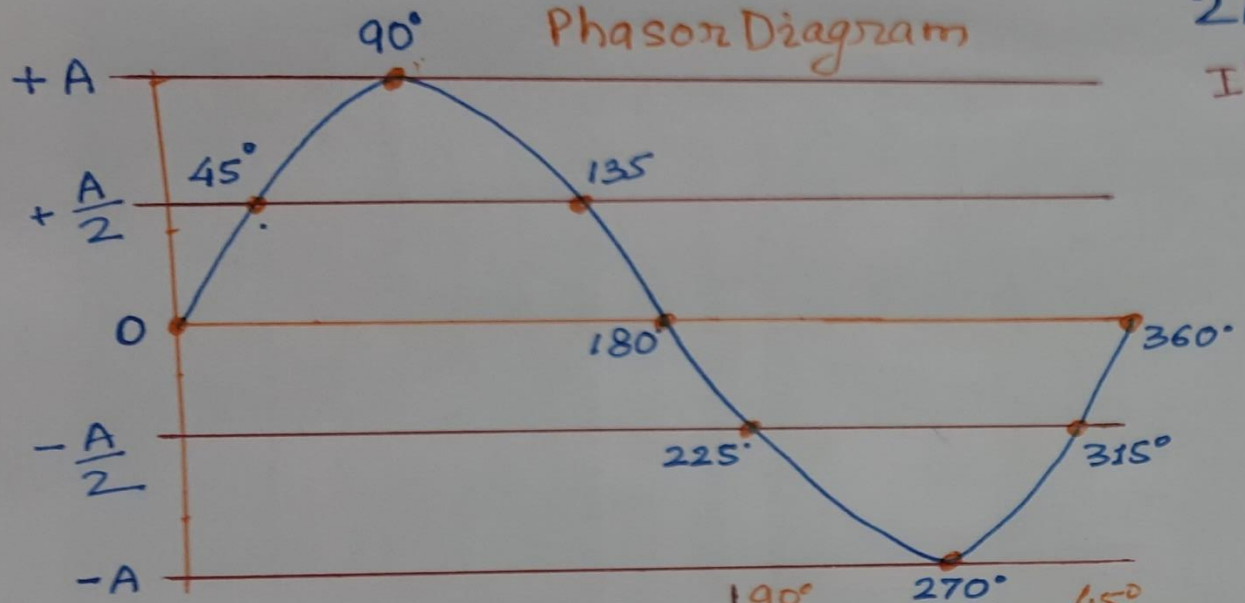
Constellation Diagram

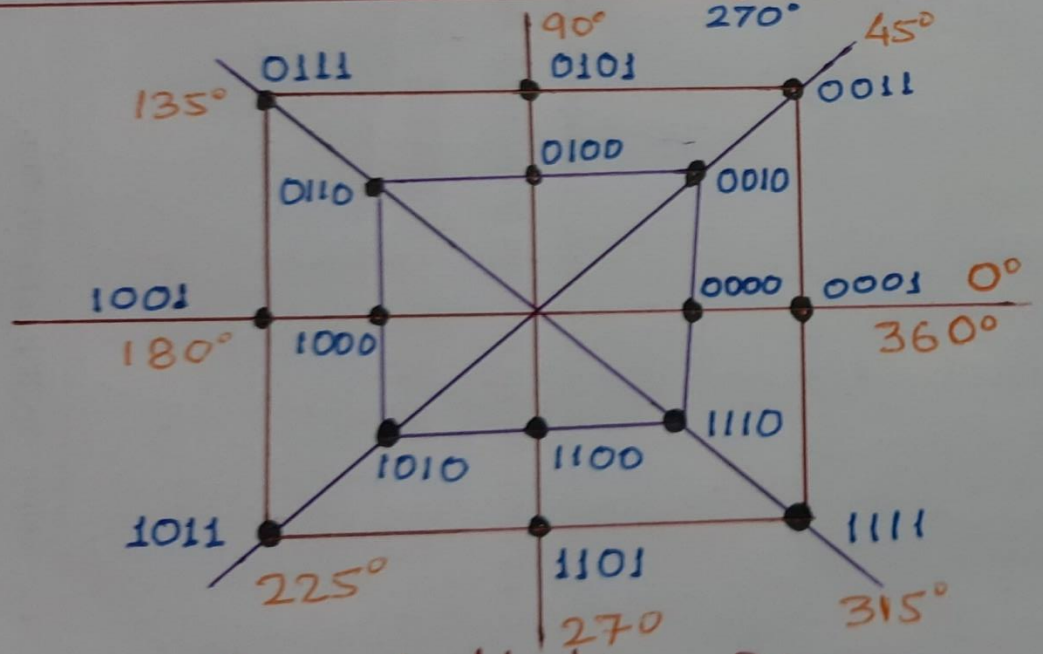


16-QAM

2 Amp X 8 Phase = 16 QAM

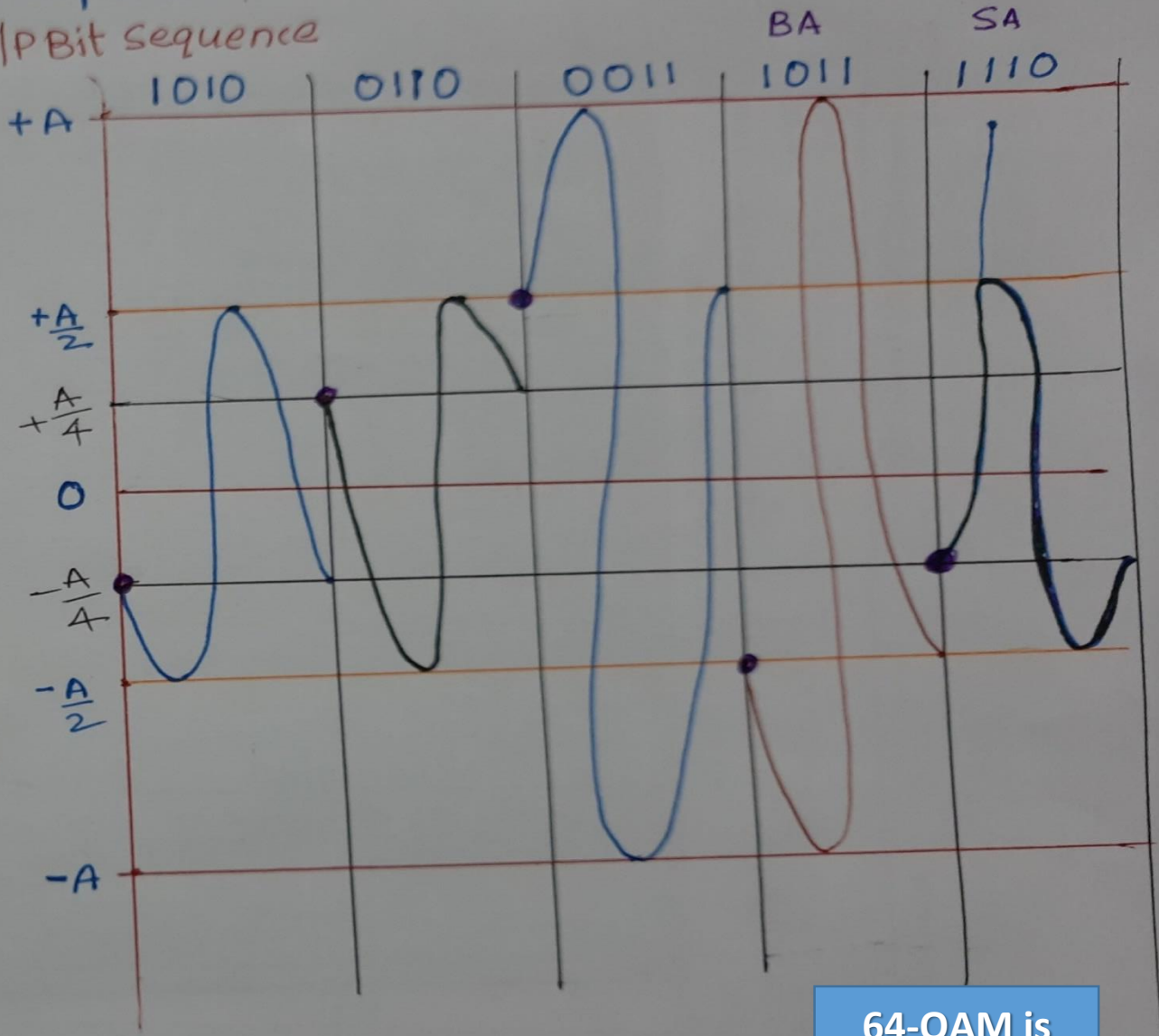
2^4 = 16 (4 bits)





Constellation Diagram

I/P Bit sequence



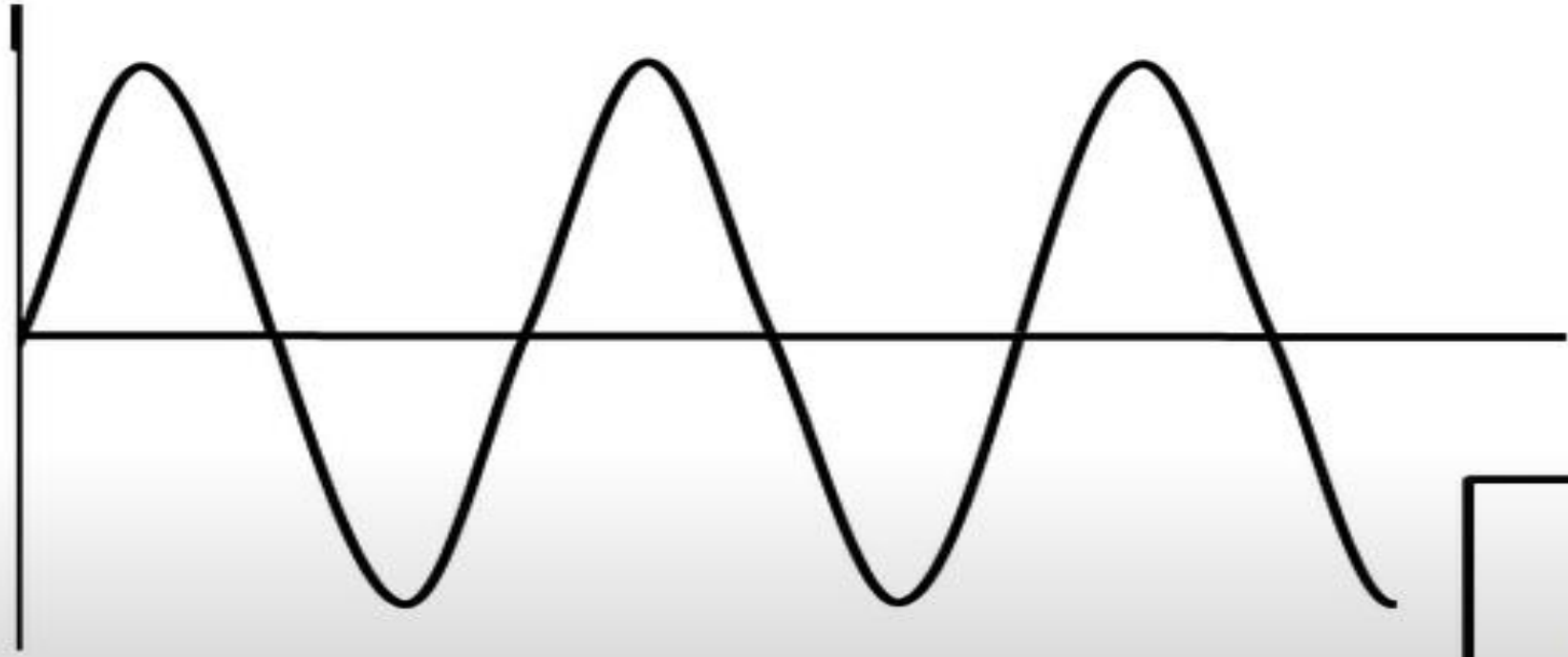
SA = Small Amplitude

BA = Big Amplitude.

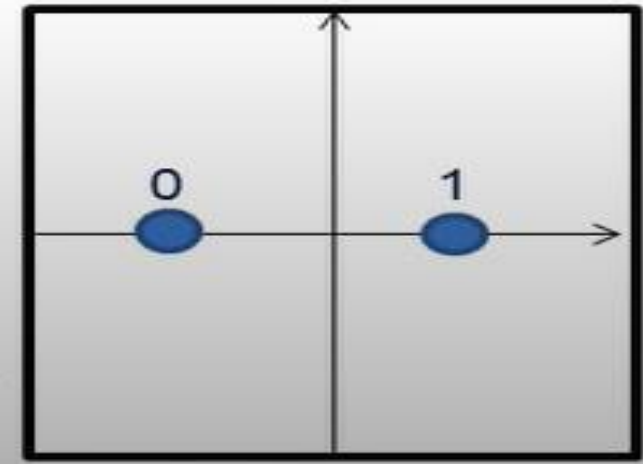
64-QAM is used in 4G

After this slide the content of QAM is for reference and for further understanding but for exam point of view you can skip and jump to OFDM

Modulation Binary Phase Shift Keying (BPSK)



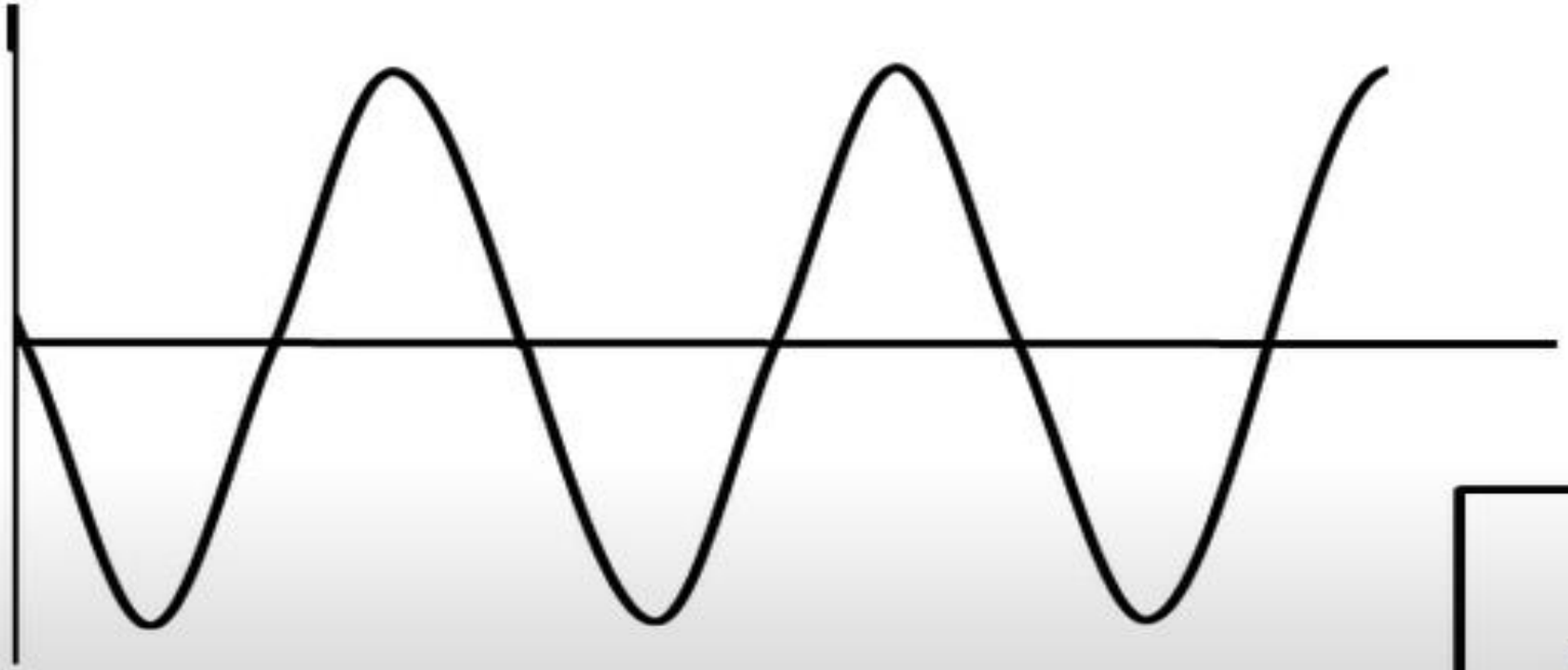
= 1



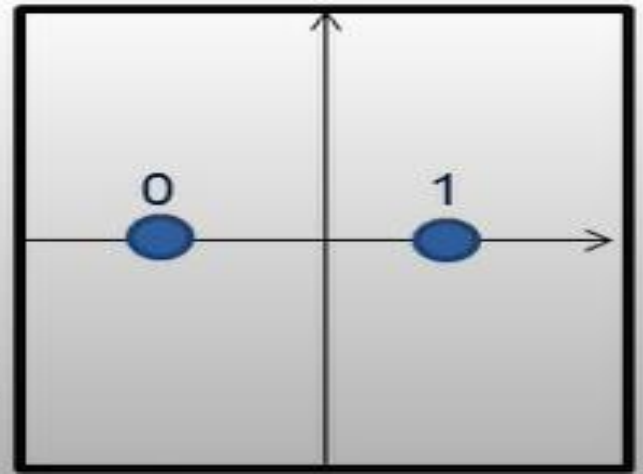
Constellation diagram of BPSK

Modulation Binary Phase Shift Keying (BPSK)

Change The Phase by 180°



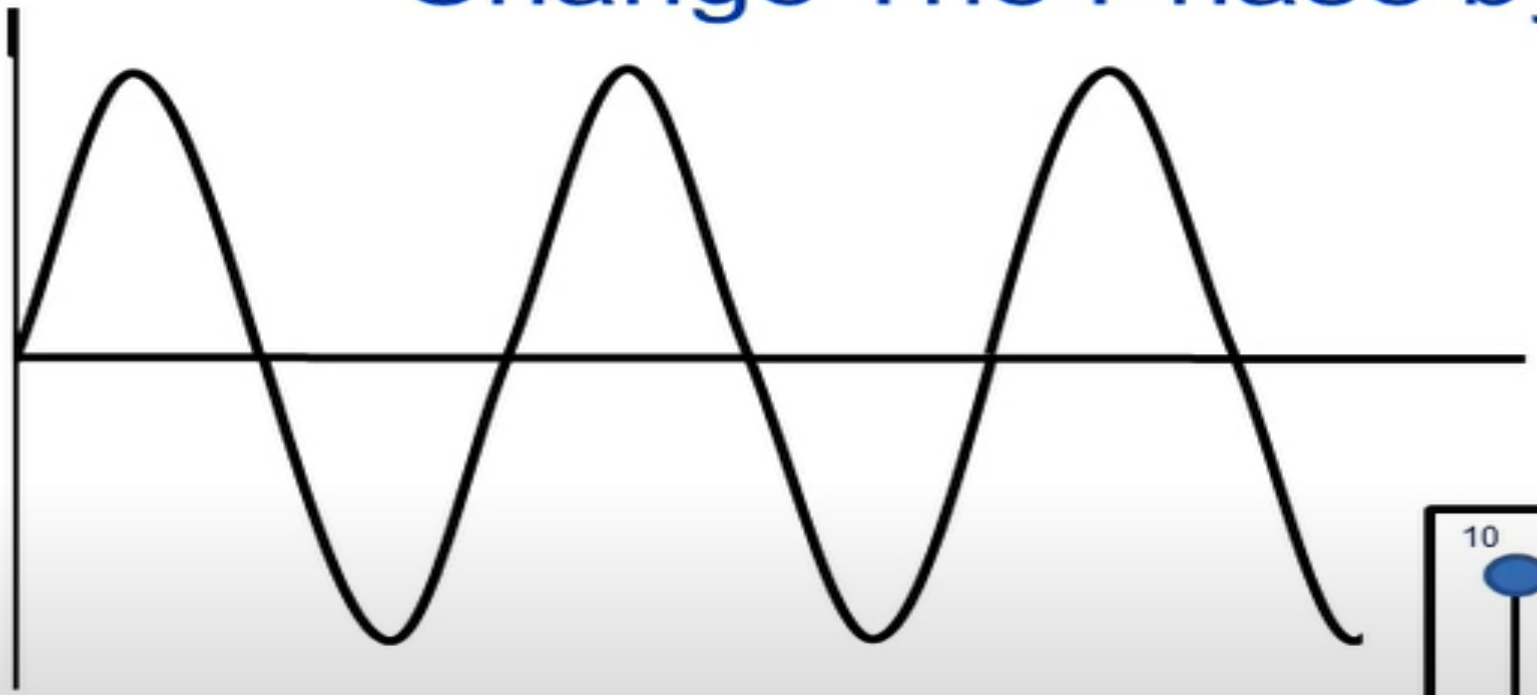
= 0



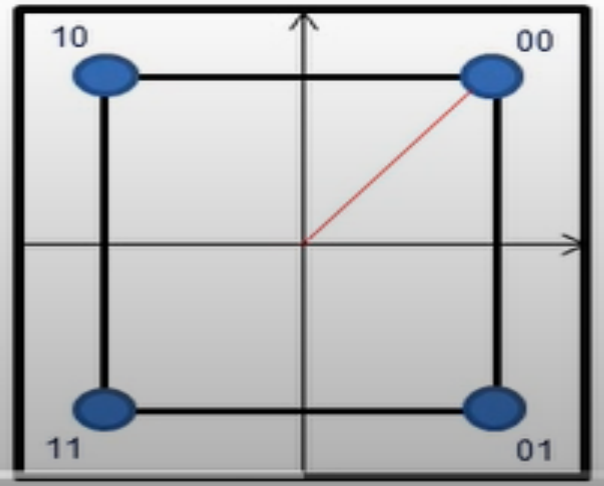
Constellation diagram of BPSK

Quadrature Phase Shift Keying (QPSK)

2 Bit Modulation Symbol
Change The Phase by 90°



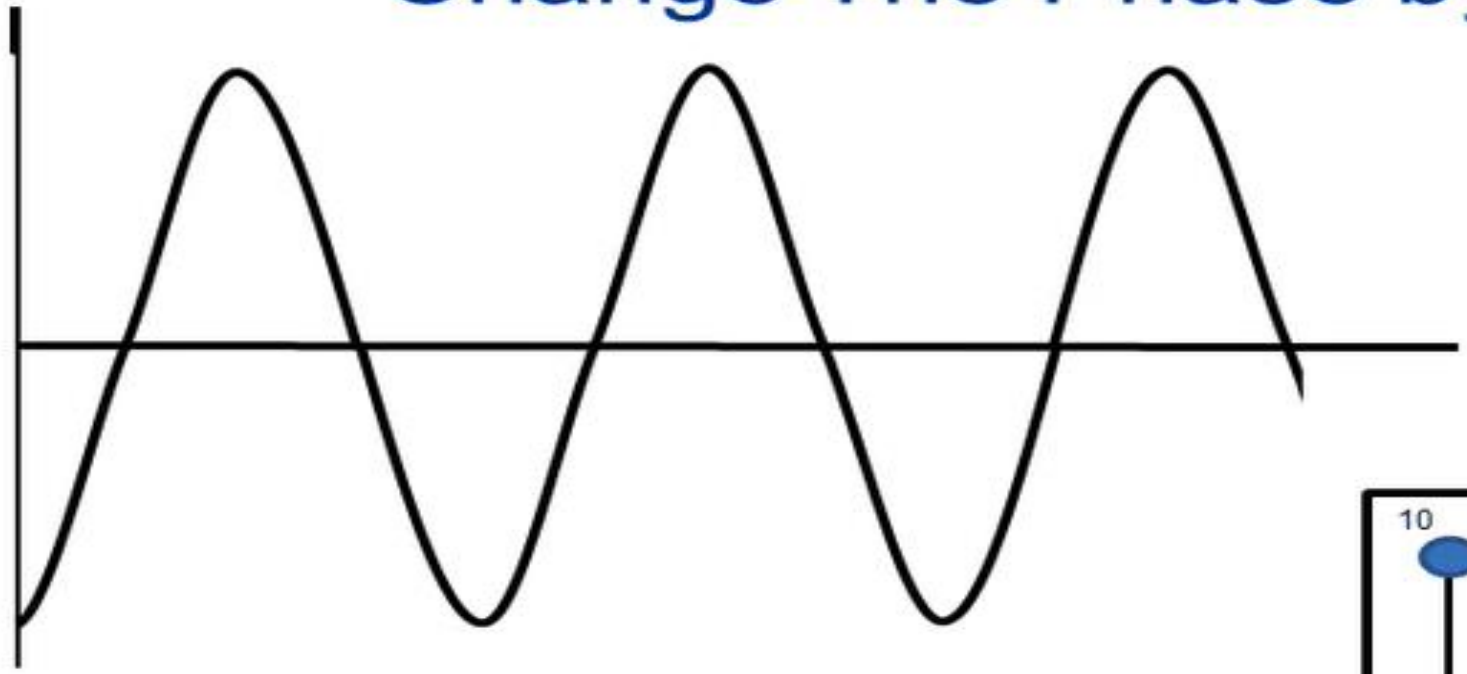
= 00



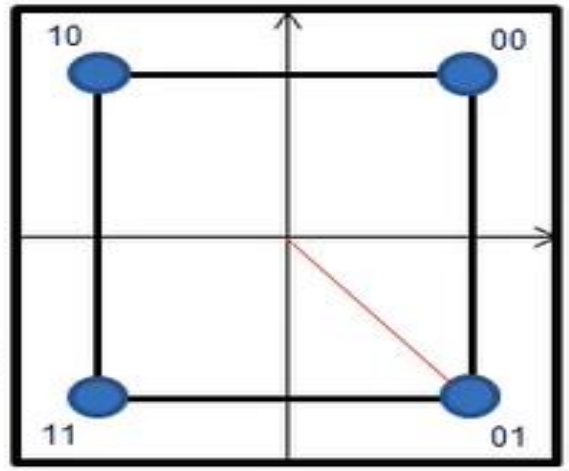
Constellation diagram of QPSK

Quadrature Phase Shift Keying (QPSK)

2 Bit Modulation Symbol
Change The Phase by 90°



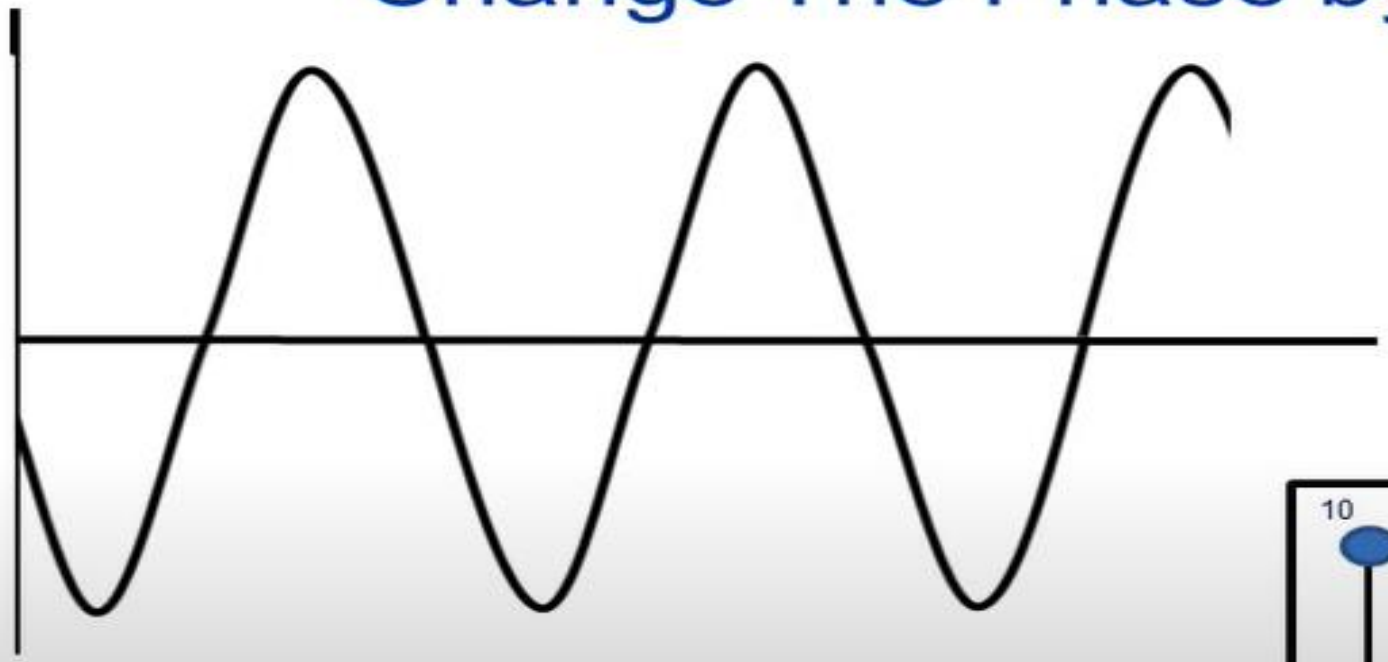
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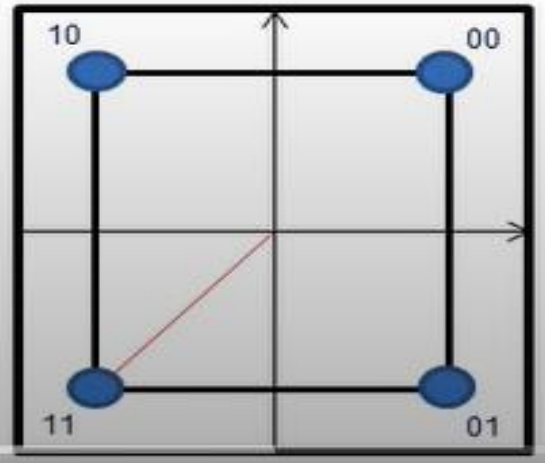
Constellation diagram of QPSK

Quadrature Phase Shift Keying (QPSK)

2 Bit Modulation Symbol
Change The Phase by 90°



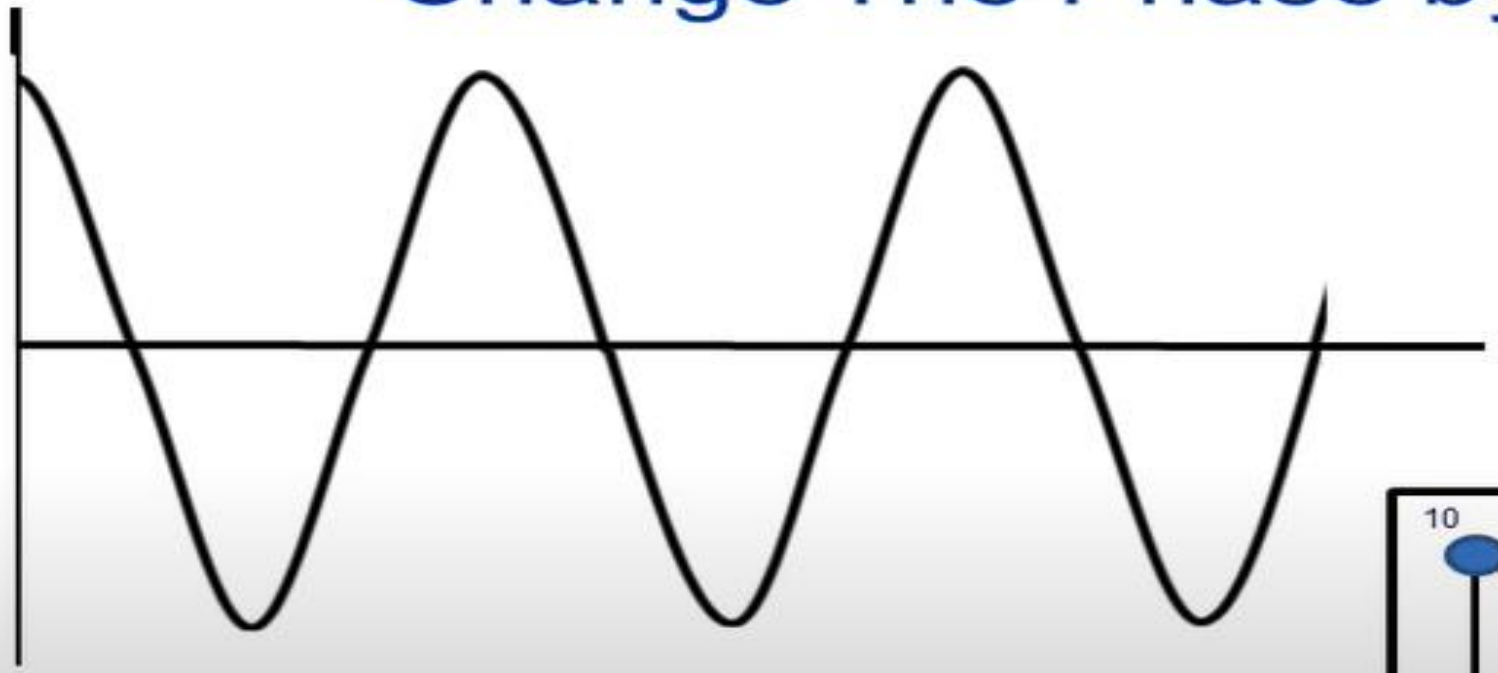
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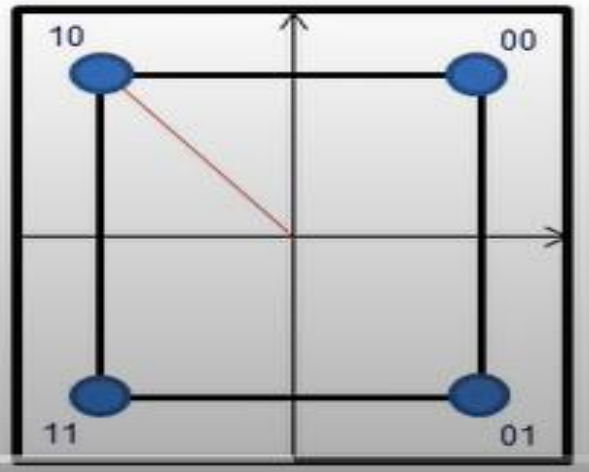
Constellation diagram of QPSK

Quadrature Phase Shift Keying (QPSK)

2 Bit Modulation Symbol
Change The Phase by 90°



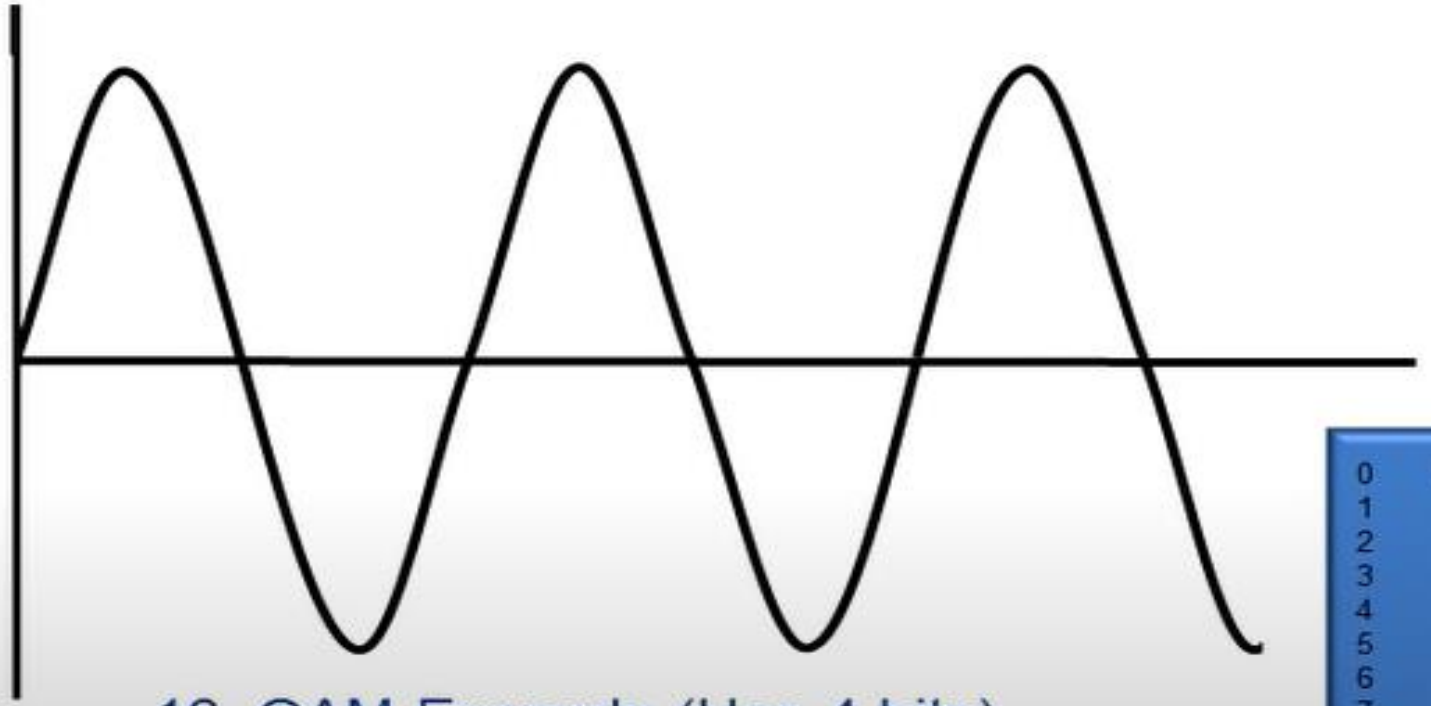
= 10



Constellation diagram of QPSK

Quadrature Amplitude Modulation (QAM)

Change The Phase by 90° & Amplitude



16 QAM Example (Has 4 bits)

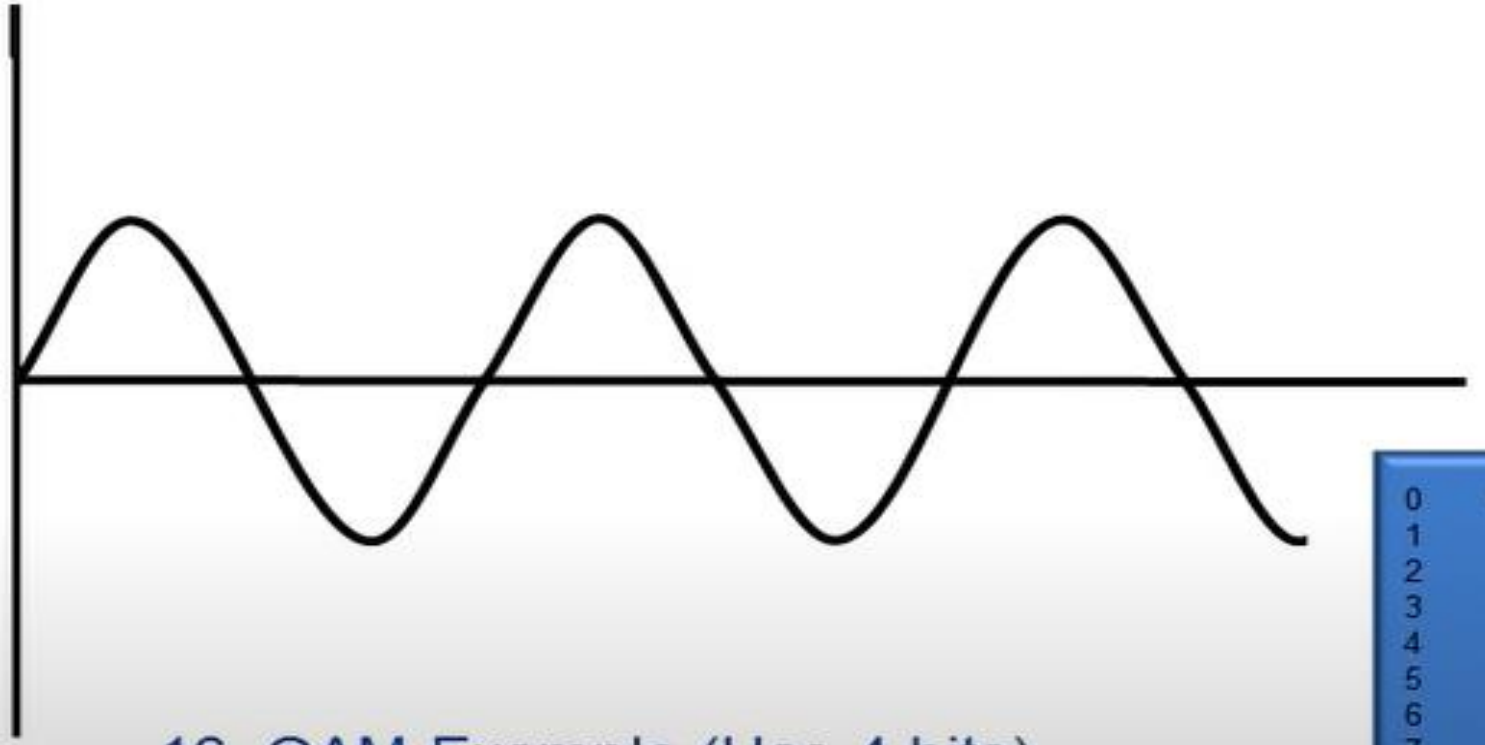
=0000

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

SPIRENT

Quadrature Amplitude Modulation (QAM)

Change The Amplitude



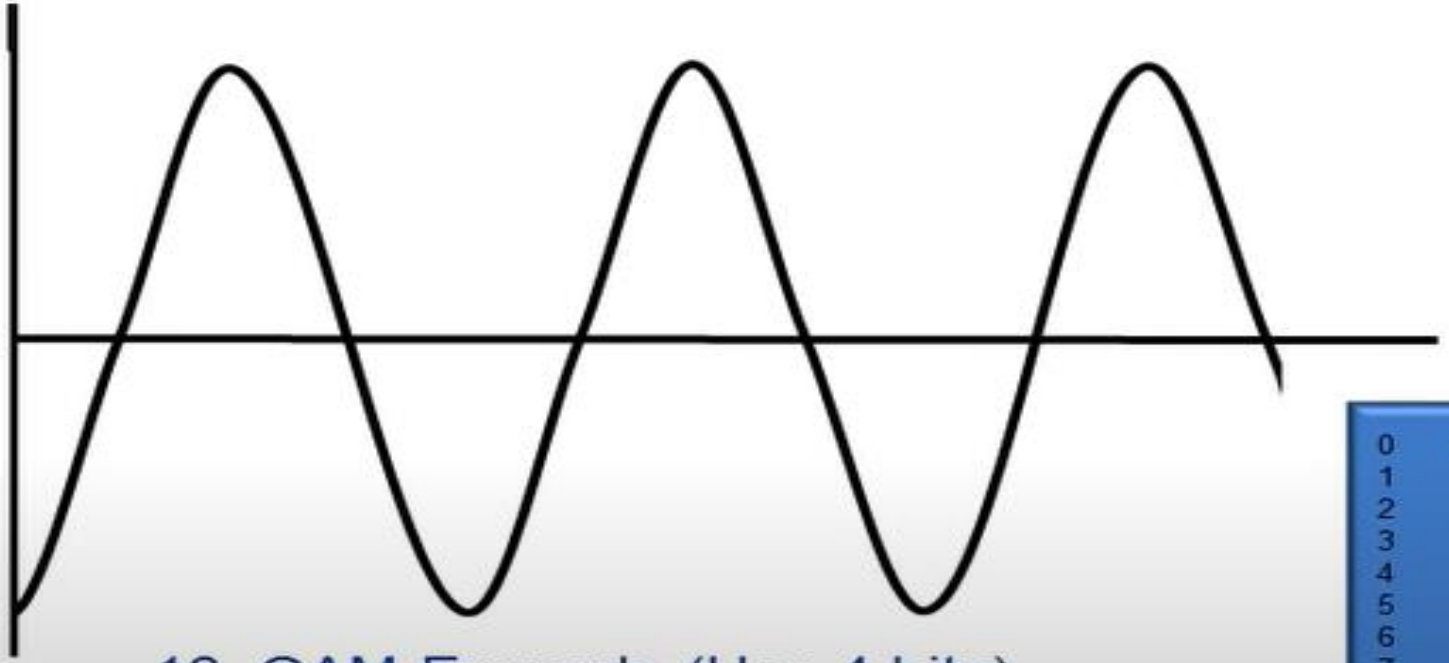
=0001

16 QAM Example (Has 4 bits)

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110

Quadrature Amplitude Modulation (QAM)

Change The Phase by 90°



16 QAM Example (Has 4 bits)

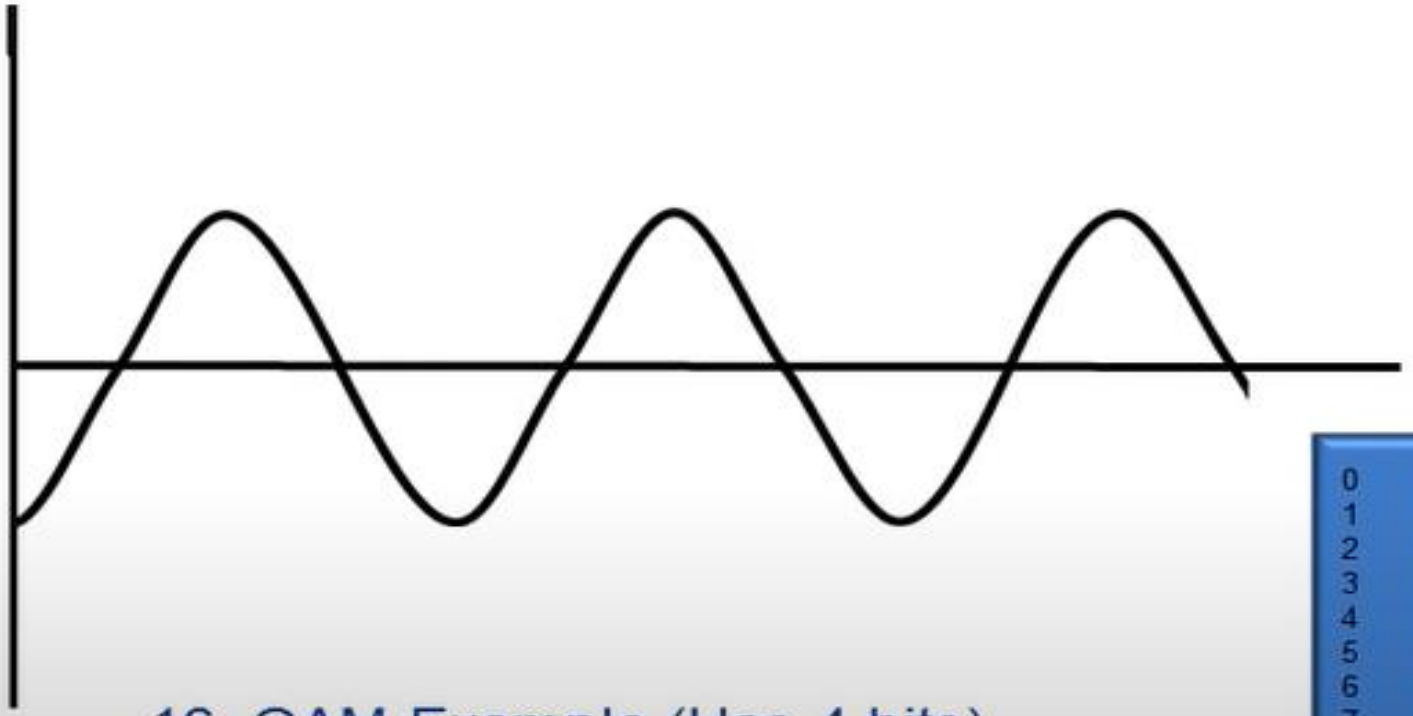
=0010

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Modulation & QAM Basics - YouTube - Google Chrome

Quadrature Amplitude Modulation (QAM)

Change The Amplitude

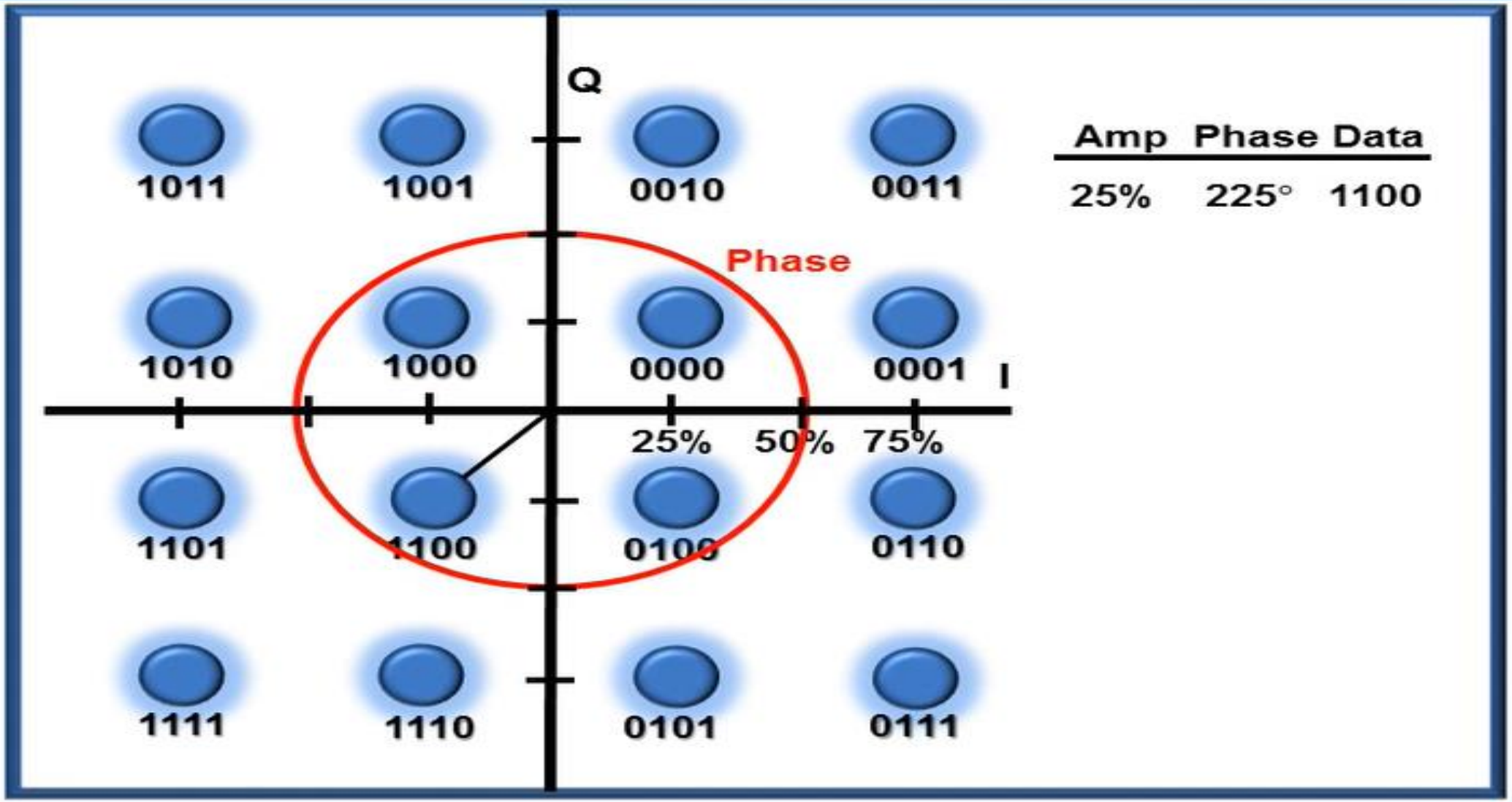


16 QAM Example (Has 4 bits)

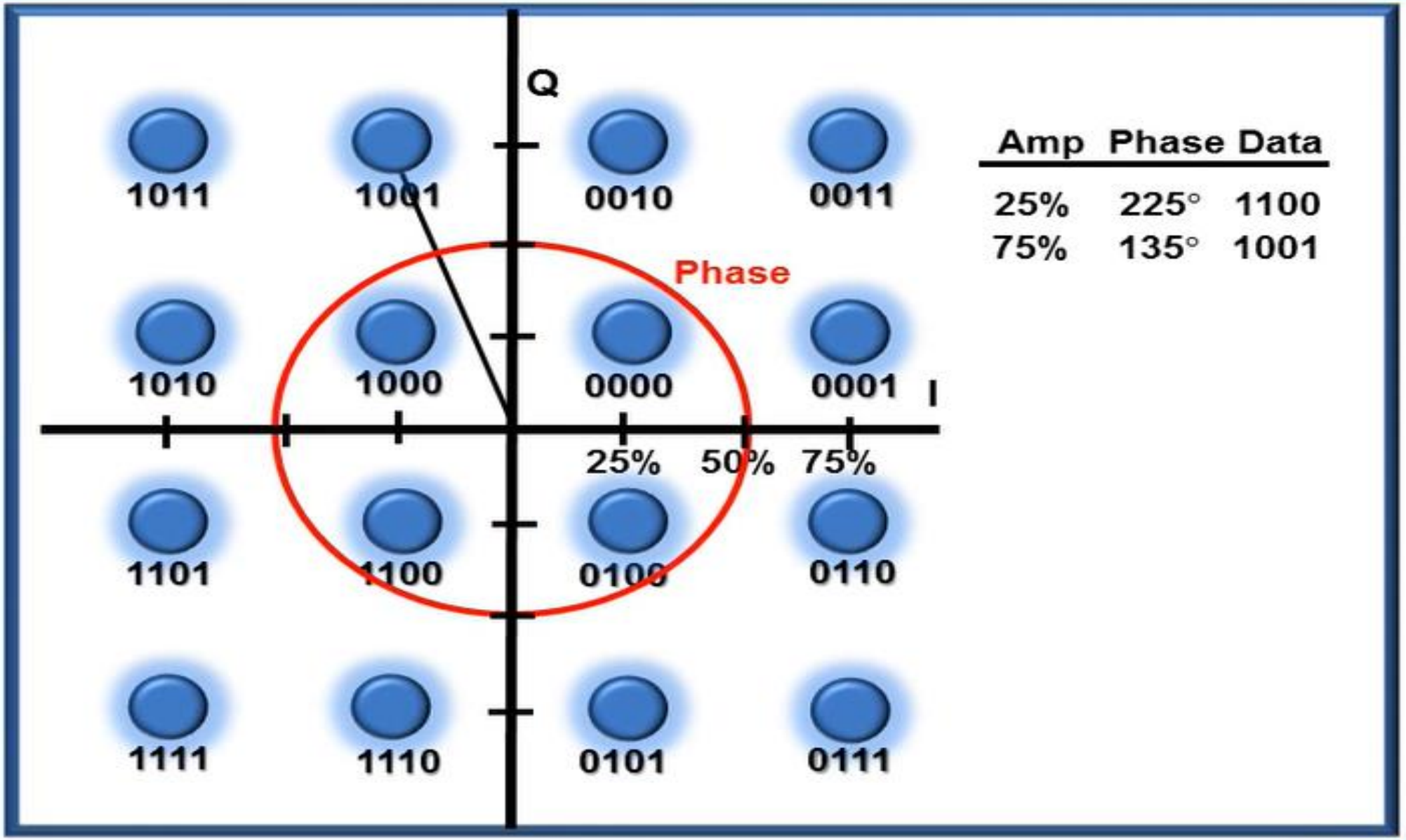
=0011

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110

16 QAM Constellation Diagram

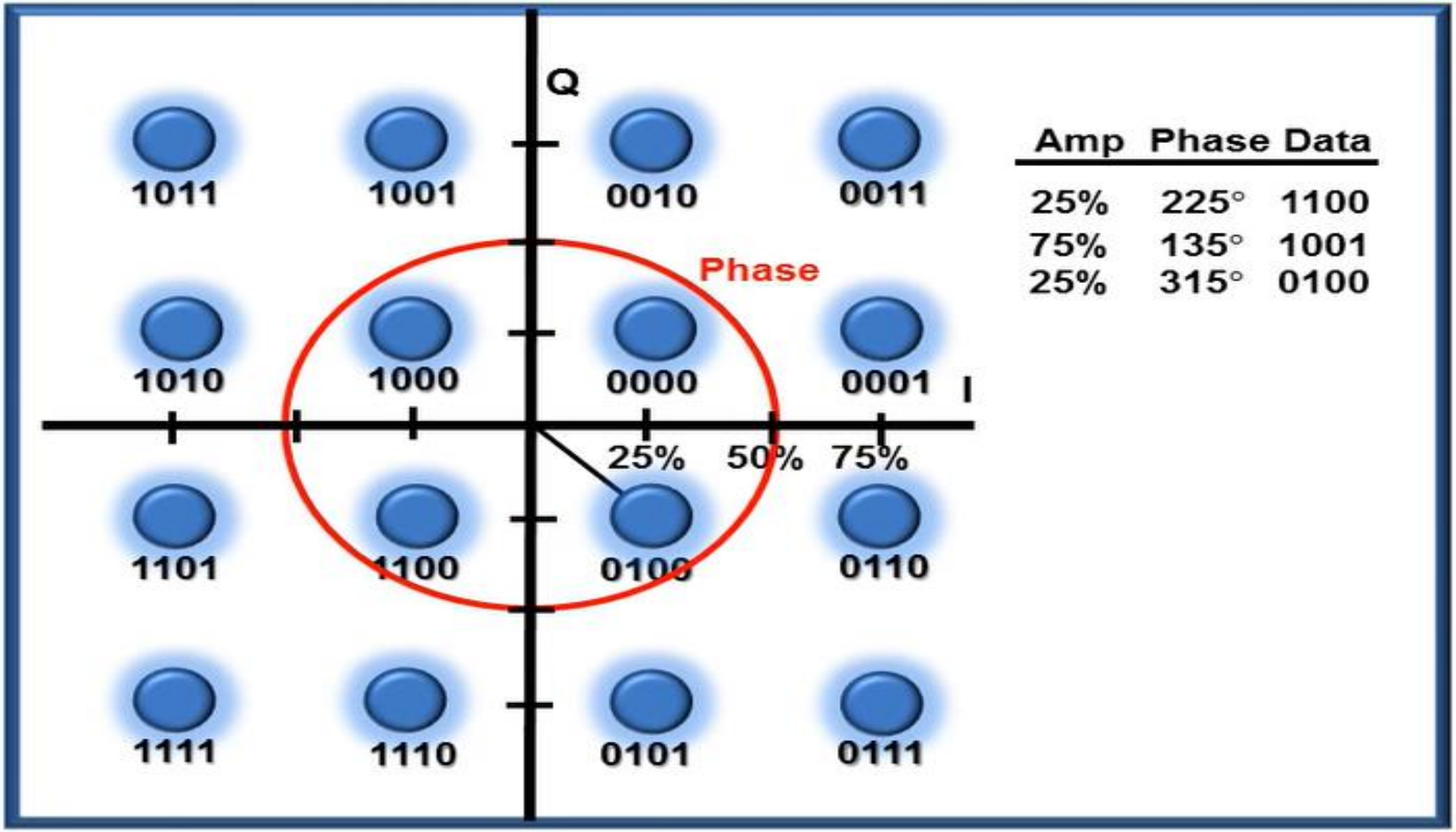


16 QAM Constellation Diagram



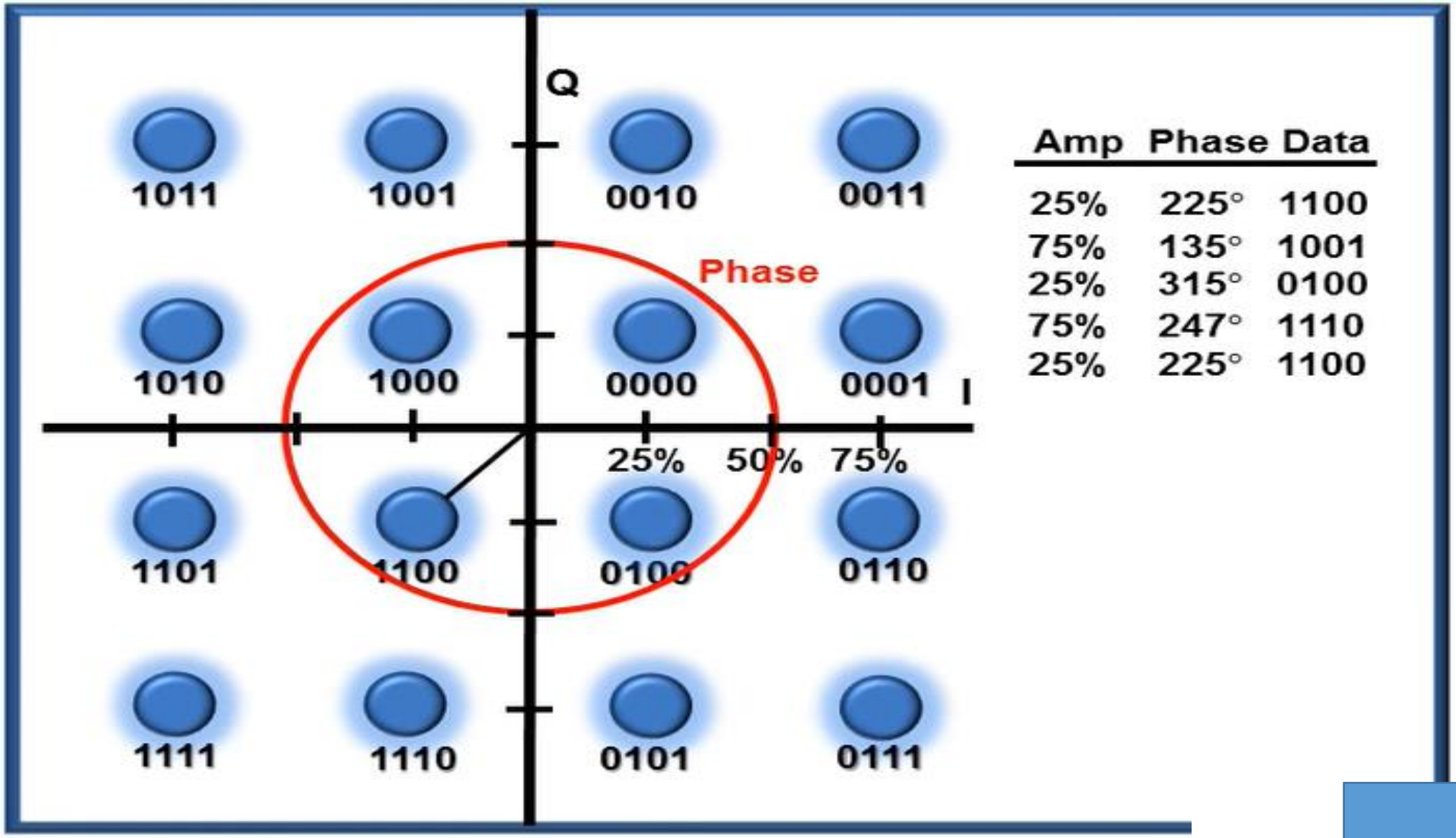
Amp	Phase	Data
25%	225°	1100
75%	135°	1001

16 QAM Constellation Diagram



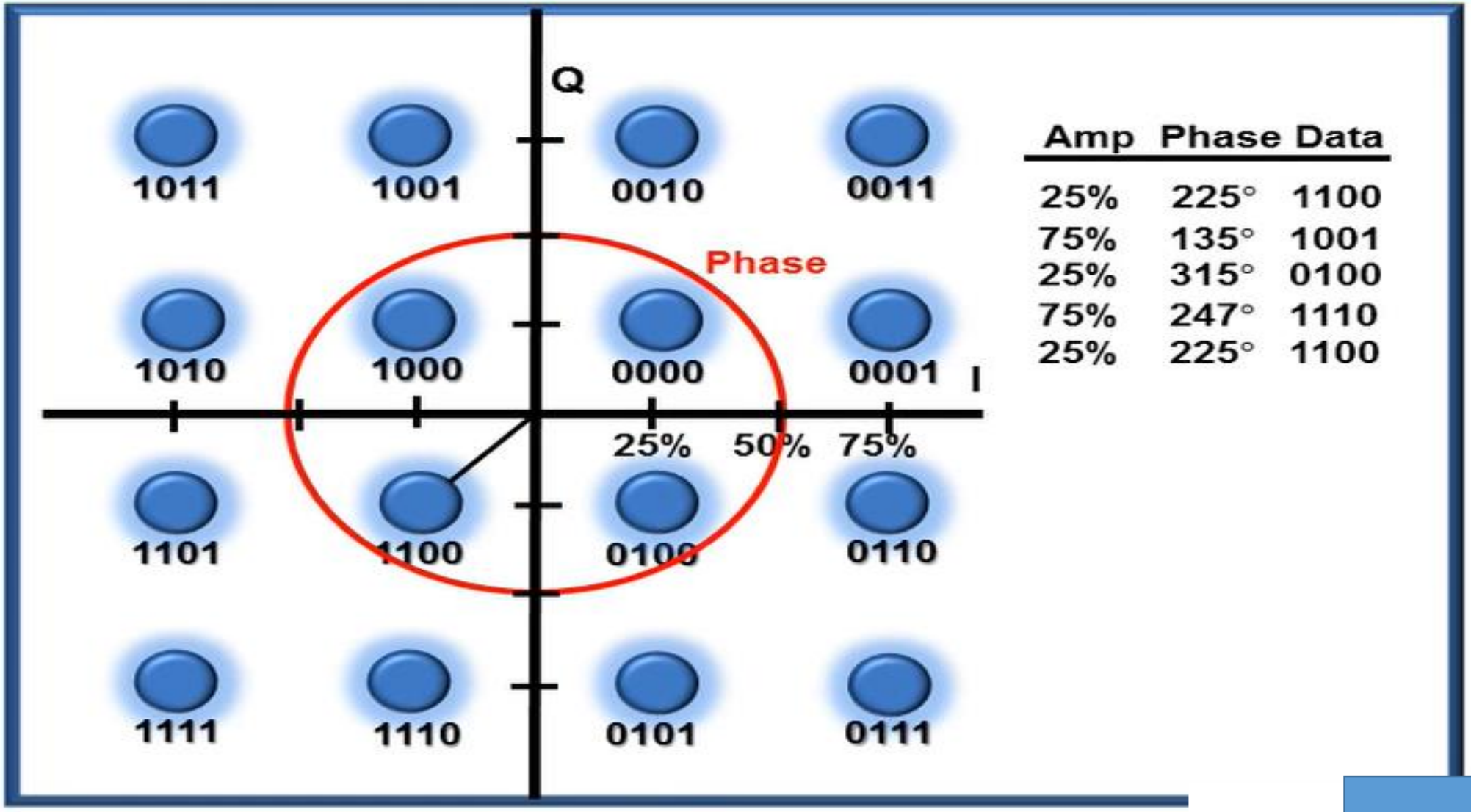
Amp	Phase	Data
25%	225°	1100
75%	135°	1001
25%	315°	0100

16 QAM Constellation Diagram



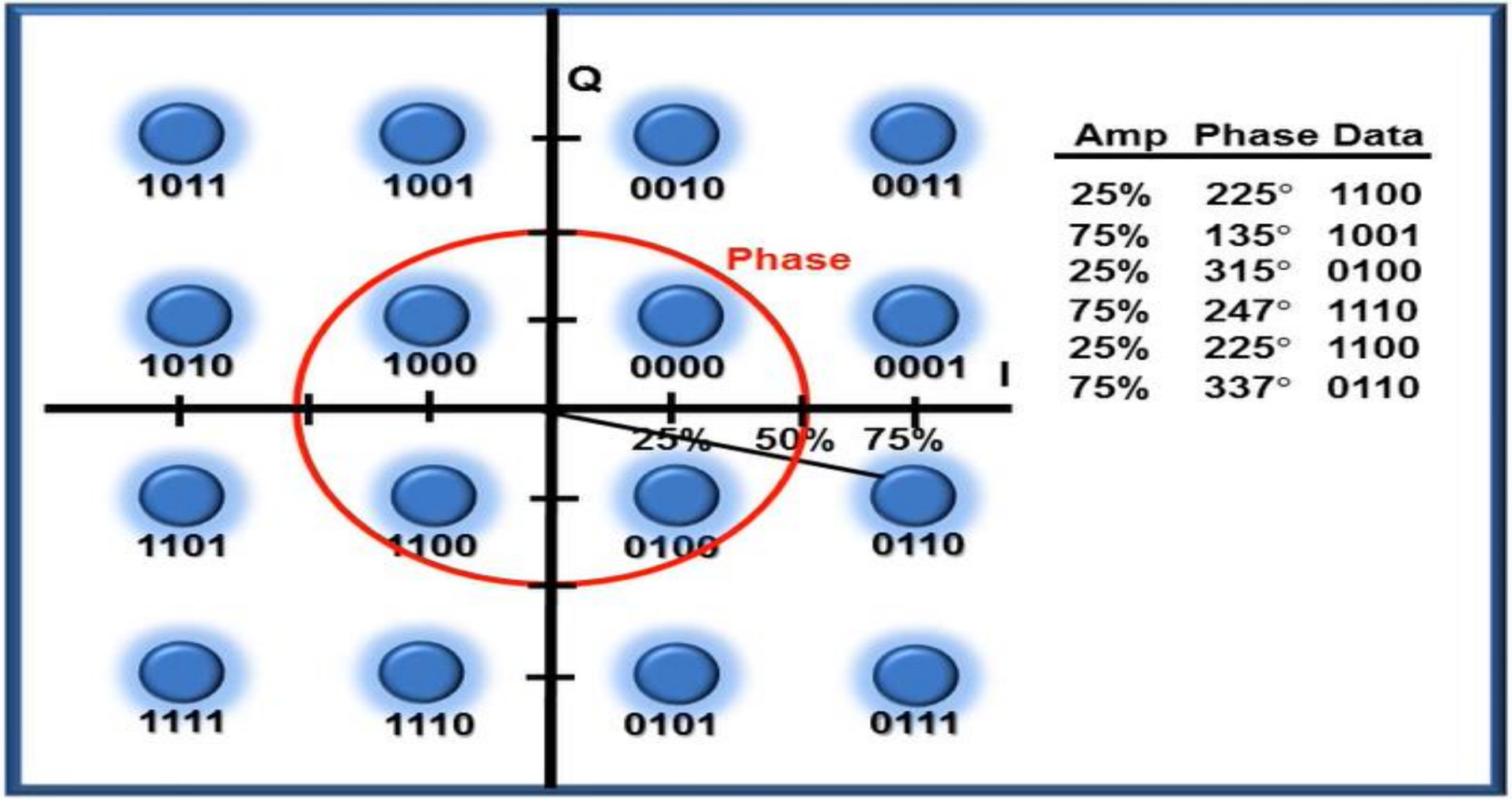
Amp	Phase	Data
25%	225°	1100
75%	135°	1001
25%	315°	0100
75%	247°	1110
25%	225°	1100

16 QAM Constellation Diagram



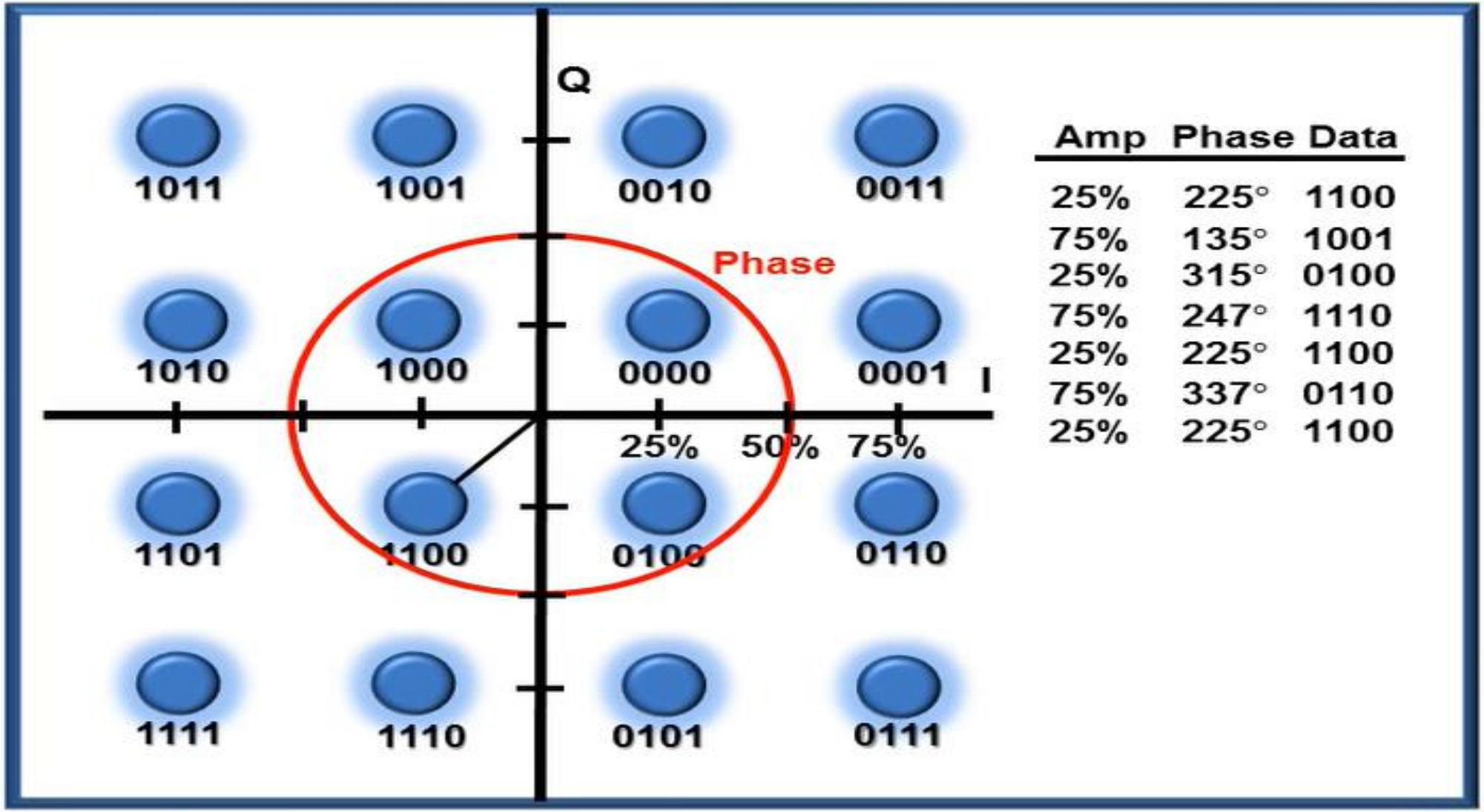
Amp	Phase	Data
25%	225°	1100
75%	135°	1001
25%	315°	0100
75%	247°	1110
25%	225°	1100

16 QAM Constellation Diagram



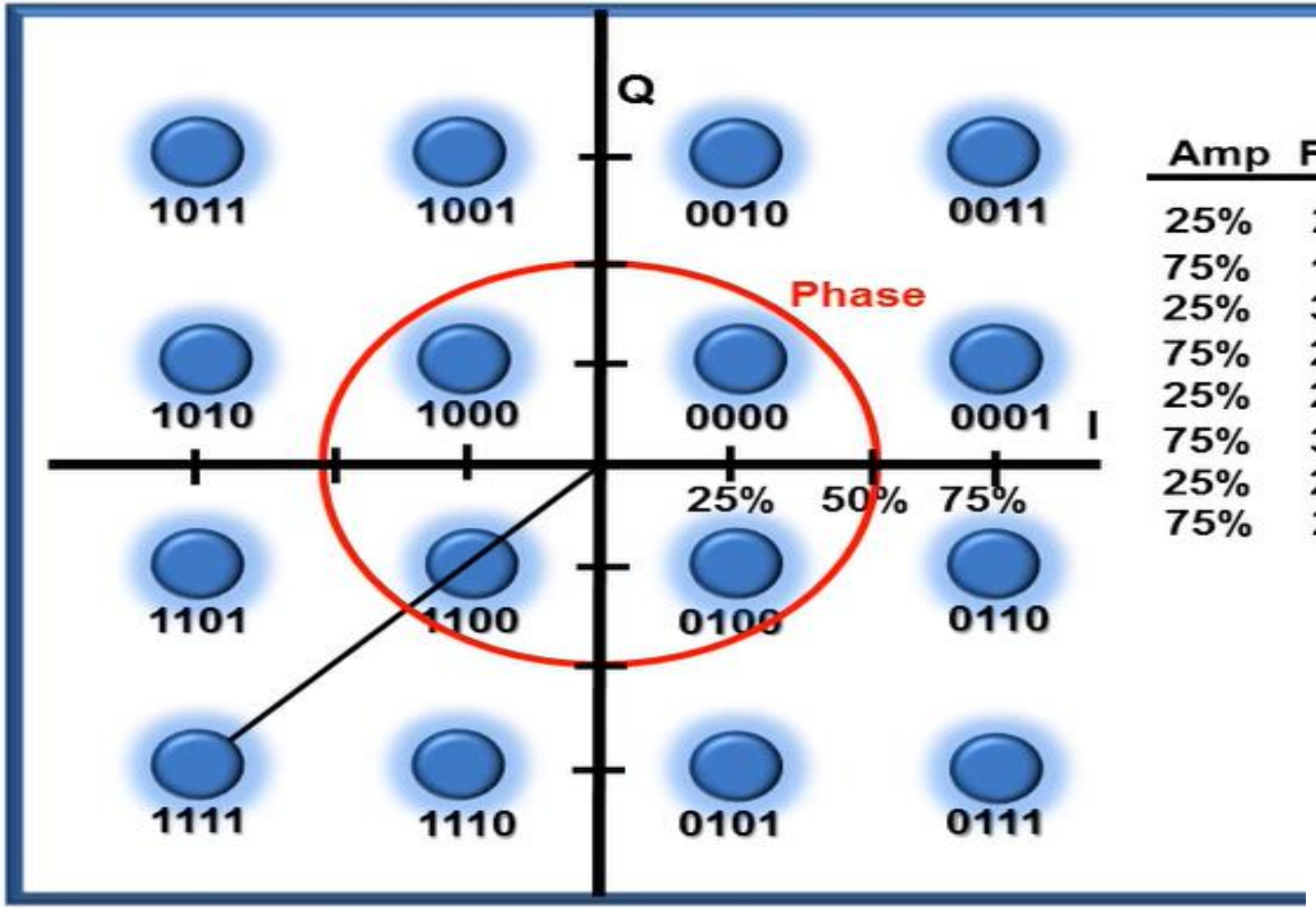
Amp	Phase	Data
25%	225°	1100
75%	135°	1001
25%	315°	0100
75%	247°	1110
25%	225°	1100
75%	337°	0110

16 QAM Constellation Diagram

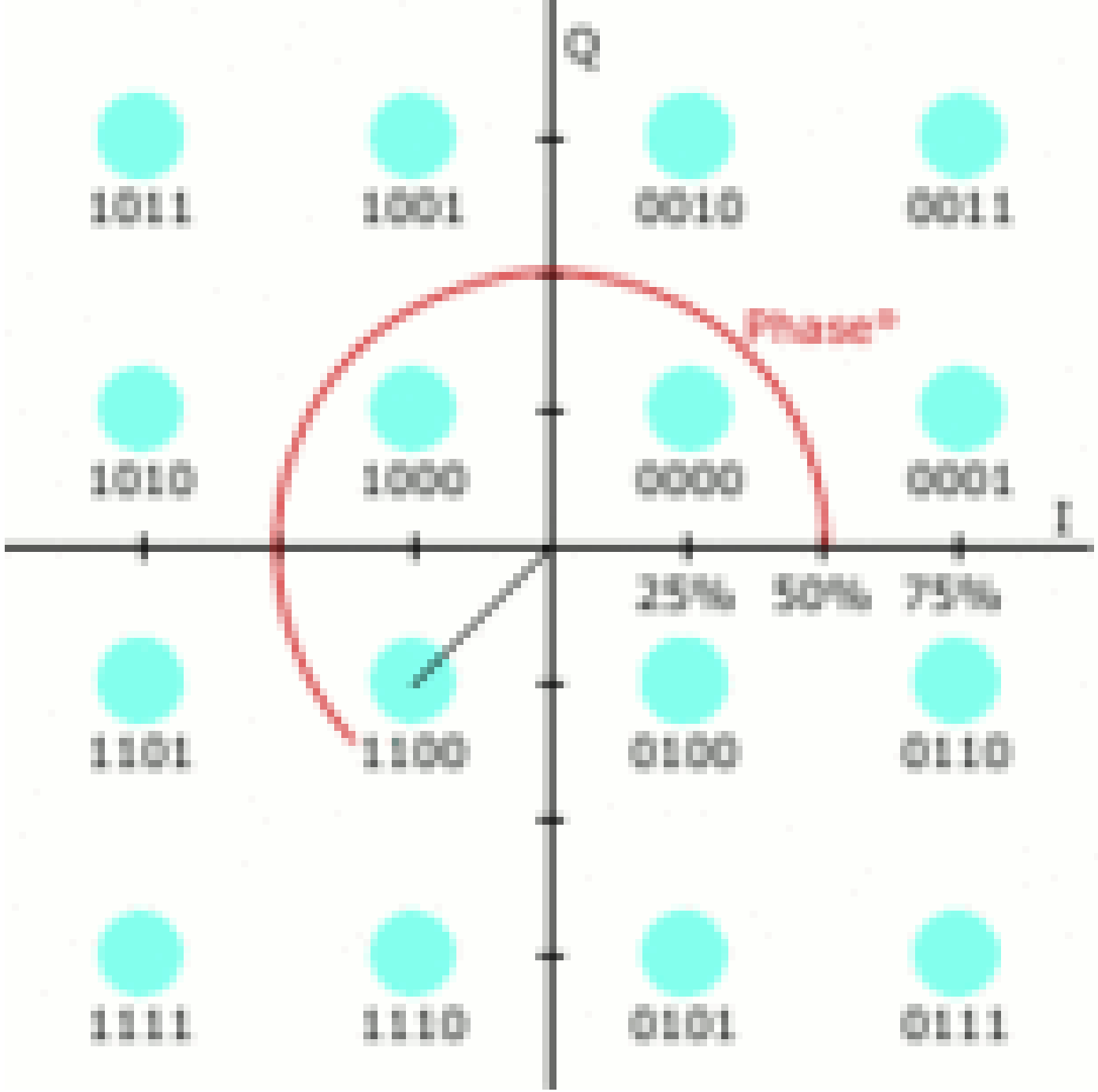


Amp	Phase	Data
25%	225°	1100
75%	135°	1001
25%	315°	0100
75%	247°	1110
25%	225°	1100
75%	337°	0110
25%	225°	1100

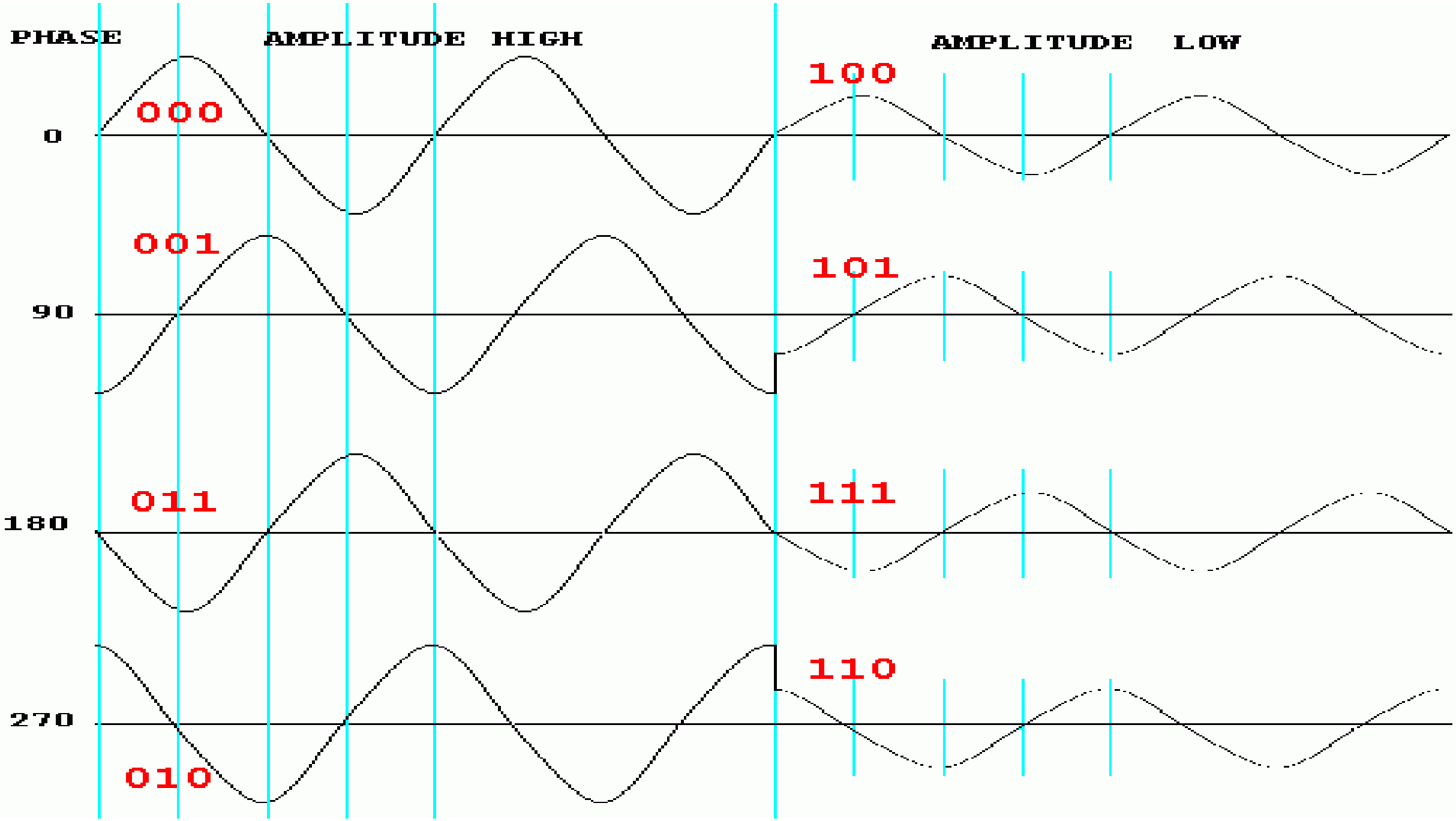
16 QAM Constellation Diagram



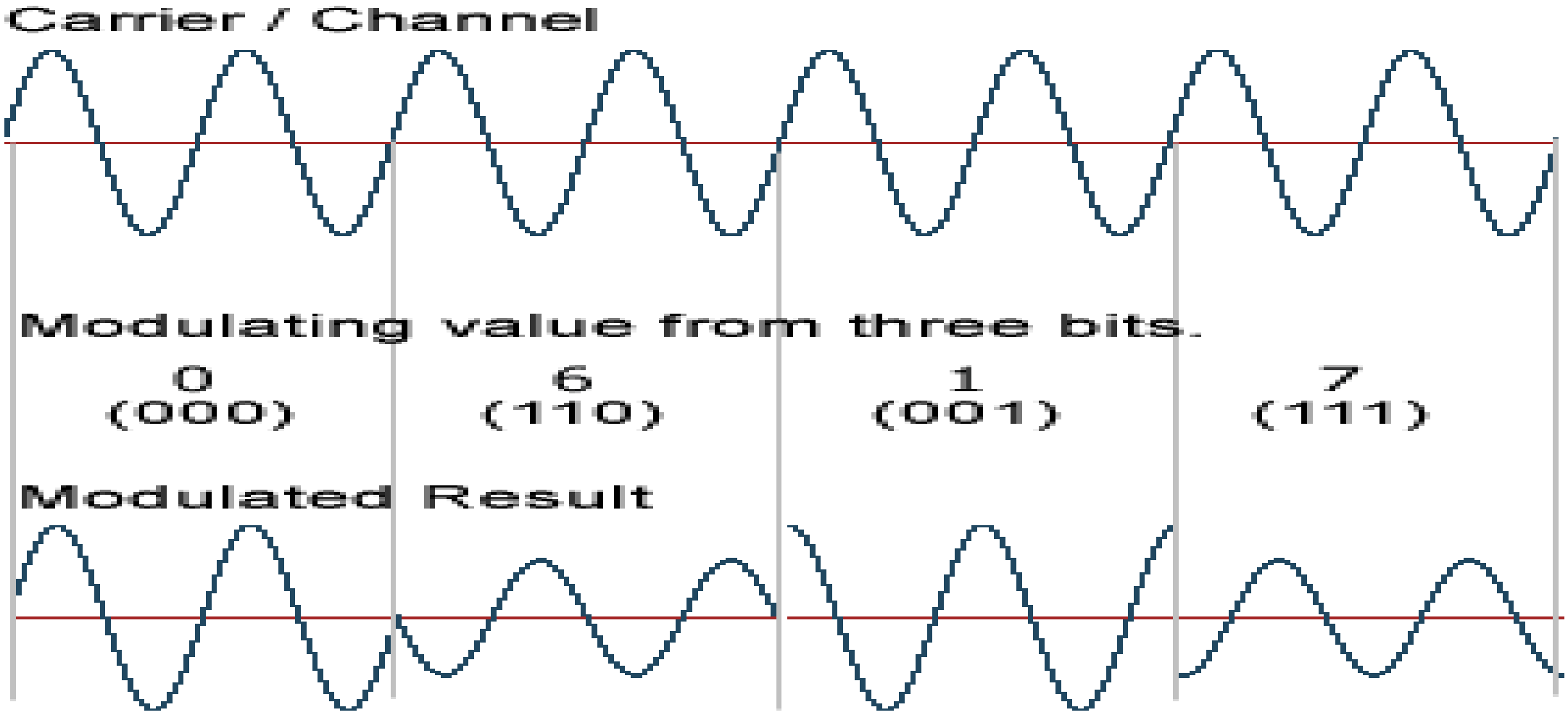
Amp	Phase	Data
25%	225°	1100
75%	135°	1001
25%	315°	0100
75%	247°	1110
25%	225°	1100
75%	337°	0110
25%	225°	1100
75%	225°	1111



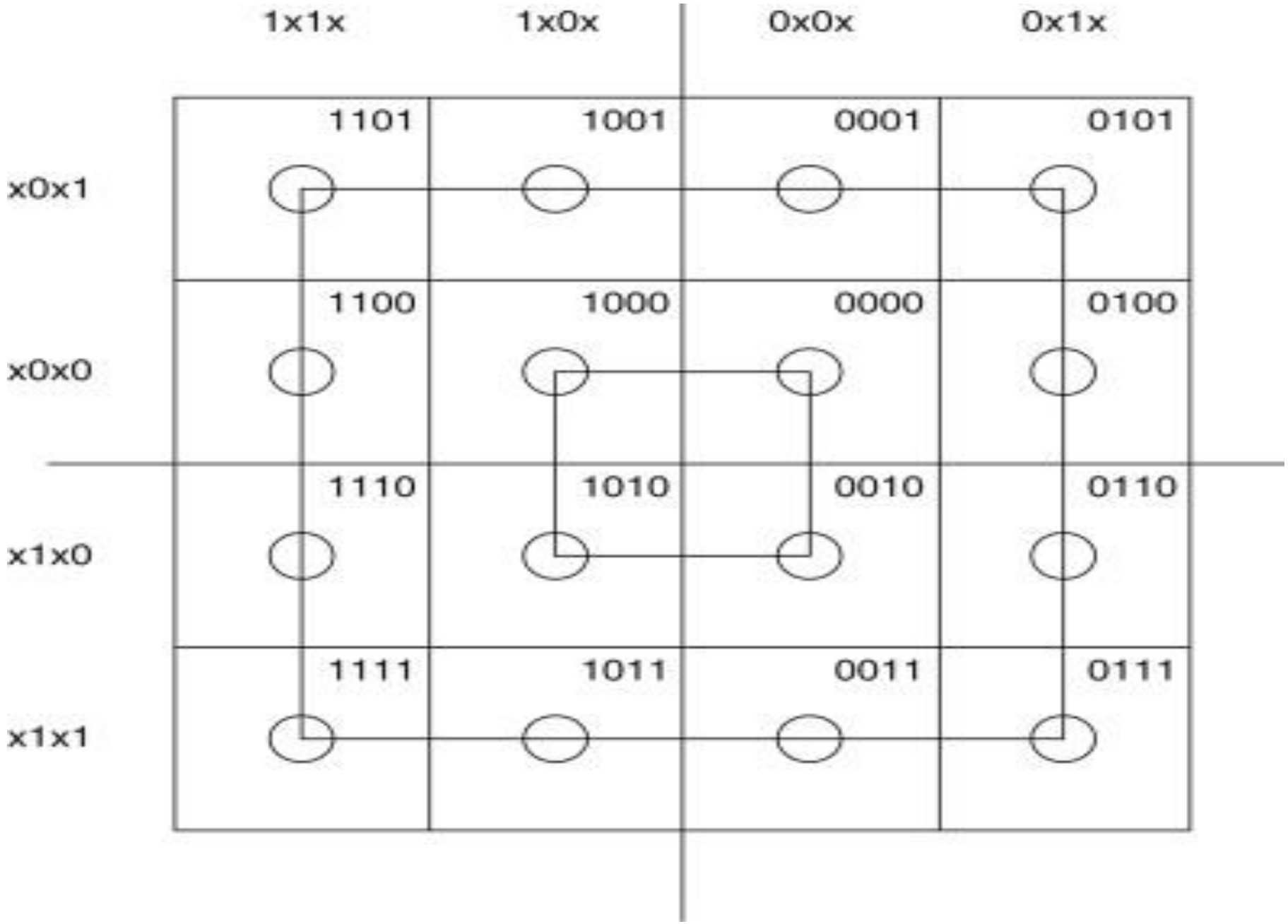
Amp	Phase	Data
25%	225°	1100

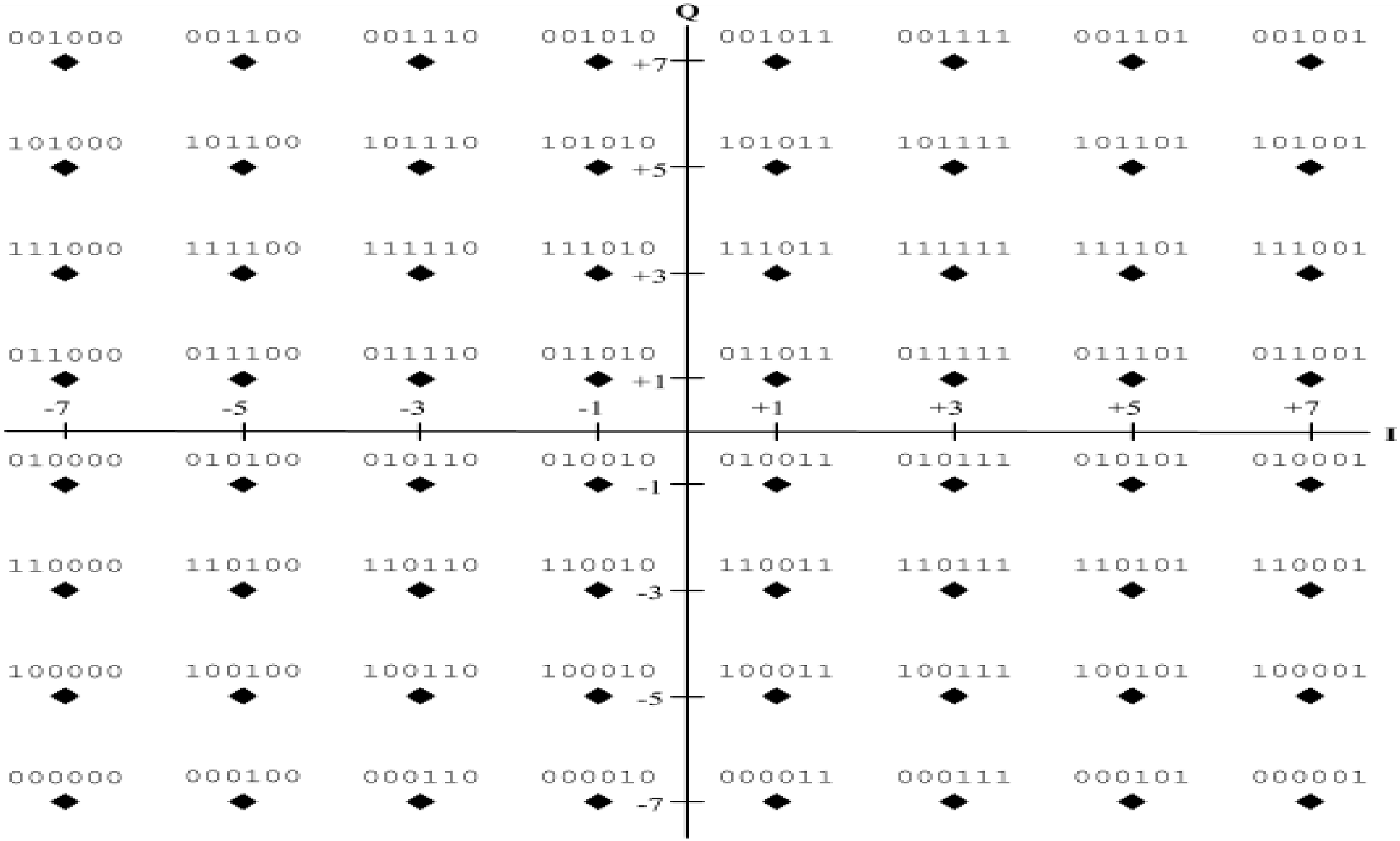

PHASE
AMPLITUDE HIGH
AMPLITUDE LOW
000
100
0
001
101
90
011
111
180
110
270
010

DIGITAL QAM (8QAM)



Note: Only four (0, 6, 1 and 7) out of the eight possible modulation states (0–7) are shown in this illustration.

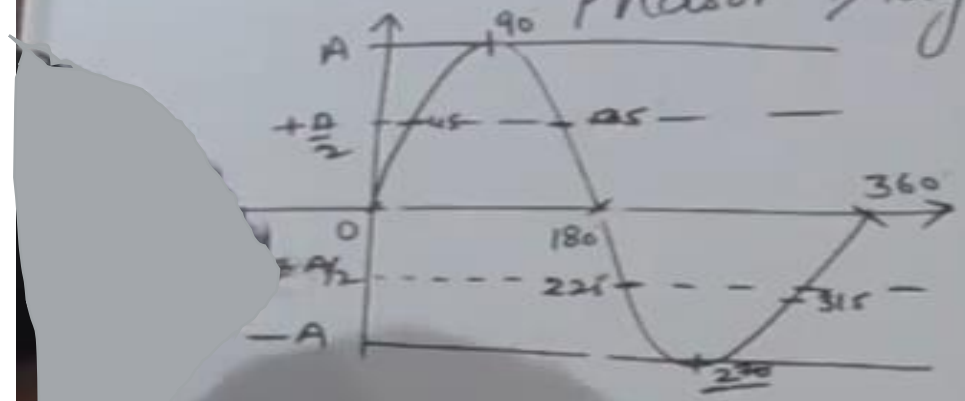

1x1x
1x0x
0x0x
0x1x
x0x1
x0x0
x1x0
x1x1
1101
1001
0001
0101
1100
1000
0000
0100
1110
1010
0010
0110
1111
1011
0011
0111

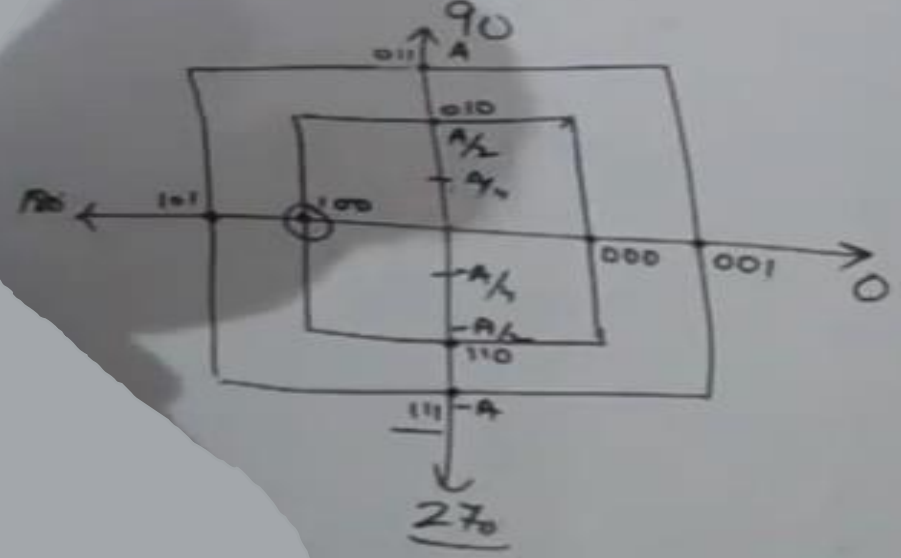

Q
I
001000 001100 001110 001010 001011 001111 001101 001001
101000 101100 101110 101010 101011 101111 101101 101001
111000 111100 111110 111010 111011 111111 111101 111001
011000 011100 011110 011010 011011 011111 011101 011001
010000 010100 010110 010010 010011 010111 010101 010001
110000 110100 110110 110010 110011 110111 110101 110001
100000 100100 100110 100010 100011 100111 100101 100001
000000 000100 000110 000010 000011 000111 000101 000001
+7
+5
+3
+1
-1
-3
-5
-7
-7 -5 -3 -1 +1 +3 +5 +7

QPSK	Gross Thoughput	Net Throughput
1.6 MHz	2.56 Mbps	2.24 Mbps
3.2 MHz	5.12 Mbps	4.49 Mbps
6.4 MHz	10.24 Mbps	8.97 Mbps
8-QAM	Gross	Net
1.6 MHz	3.84 Mbps	3.37 Mbps
3.2 MHz	7.68 Mbps	6.73 Mbps
6.4 MHz	15.36 Mbps	13.46 Mbps
16-QAM	Gross	Net
1.6 MHz	5.12 Mbps	4.49 Mbps
3.2 MHz	10.24 Mbps	8.97 Mbps
6.4 MHz	20.48 Mbps	17.95 Mbps
32-QAM	Gross	Net
1.6 MHz	6.40 Mbps	5.61 Mbps
3.2 MHz	12.80 Mbps	11.22 Mbps
6.4 MHz	25.60 Mbps	22.44 Mbps
64-QAM	Gross	Net
1.6 MHz	7.68 Mbps	6.73 Mbps
3.2 MHz	15.36 Mbps	13.46 Mbps
6.4 MHz	30.72 Mbps	26.92 Mbps

Digital Modulation (8-PSK, 8-QAM, 16-QAM)

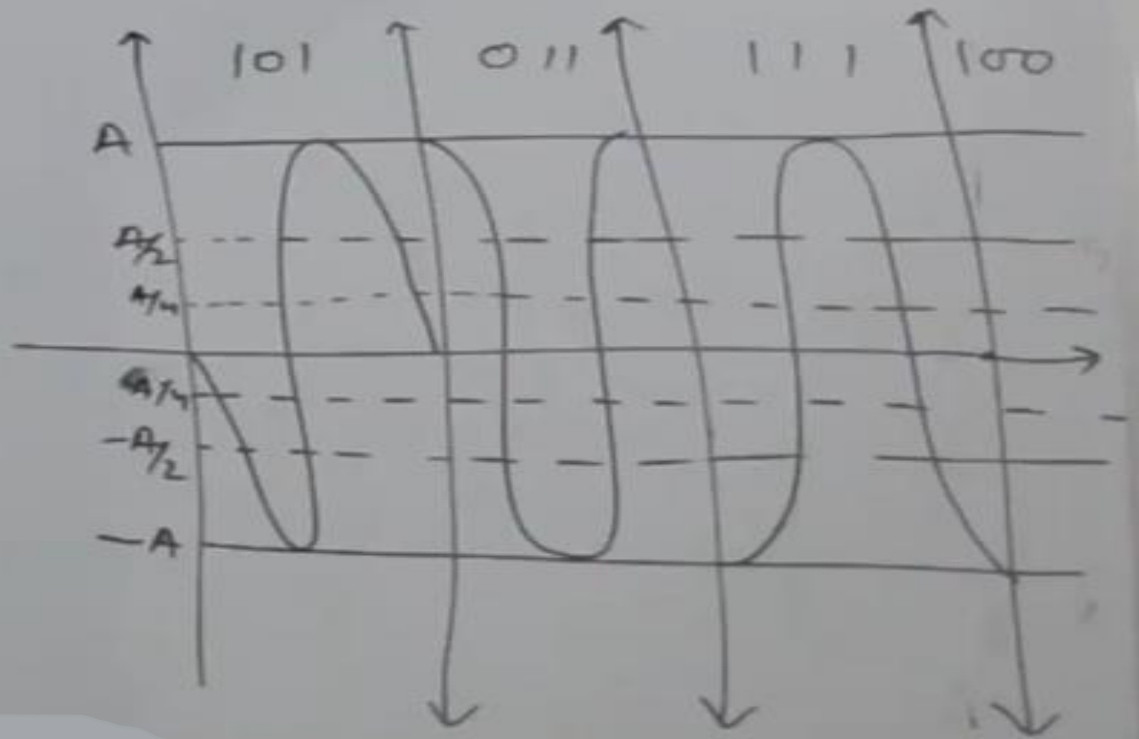
Phasor Diagram





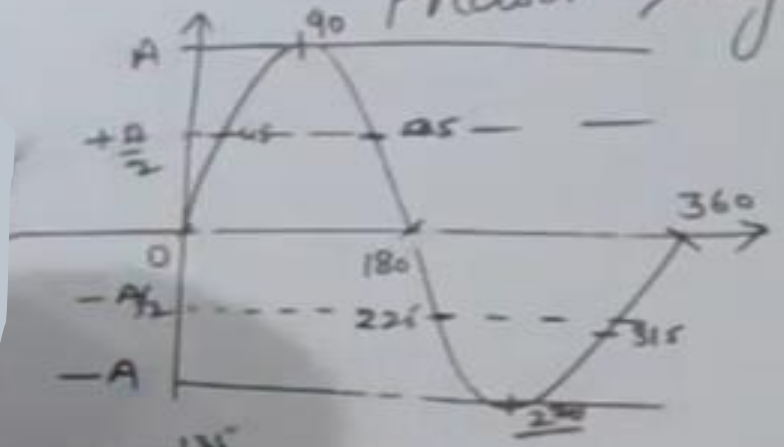
2 Amp X 4 phase
8QAM

Quadrature Amplitude Modulation

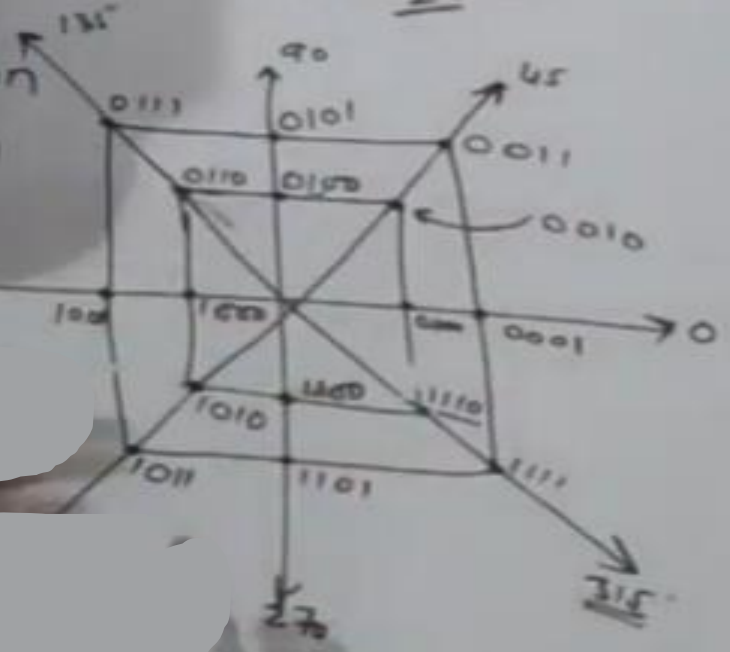


Digital Modulation (8-PSK, 8-QAM, 16-QAM)

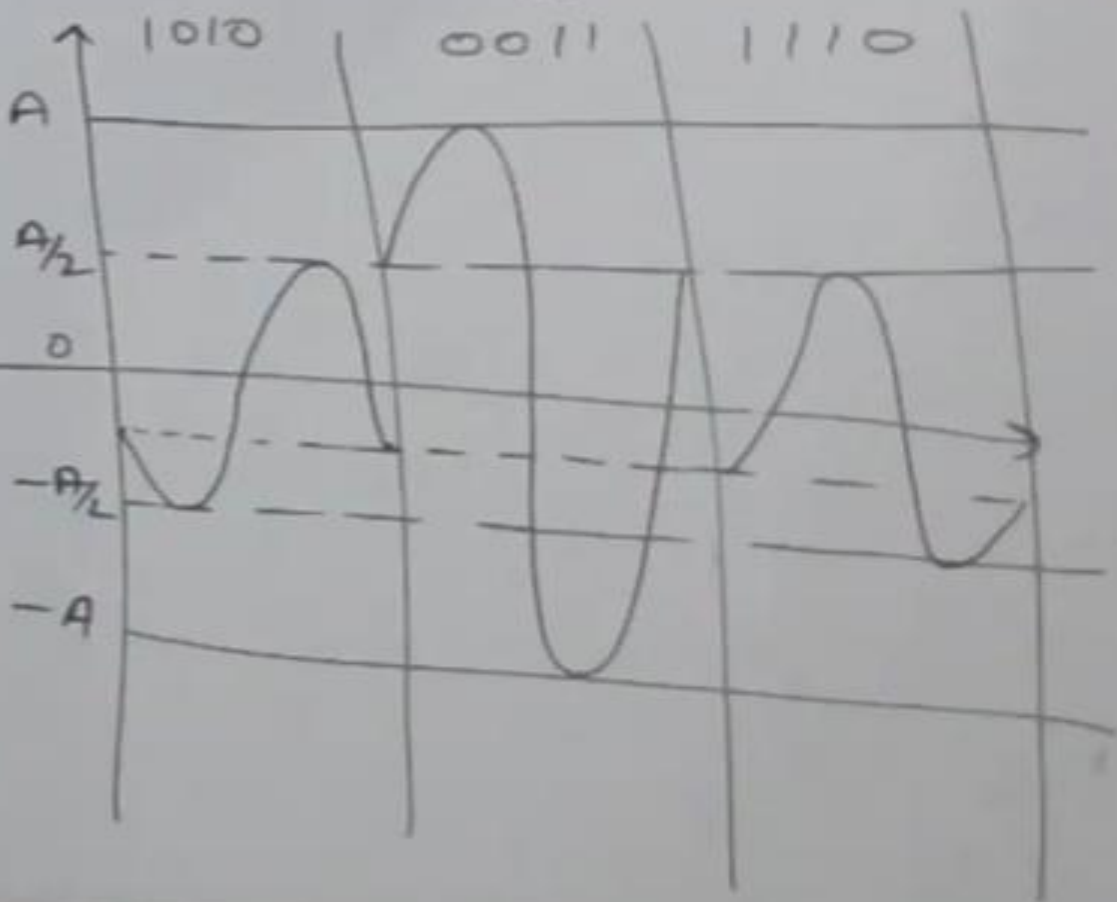
Phasor Diagram



Constellation Diagram



2 Amp × 8 Phase
= 16 QAM
2^4 → 4 bits



OFDM

Orthogonal Frequency-Division Multiplexing

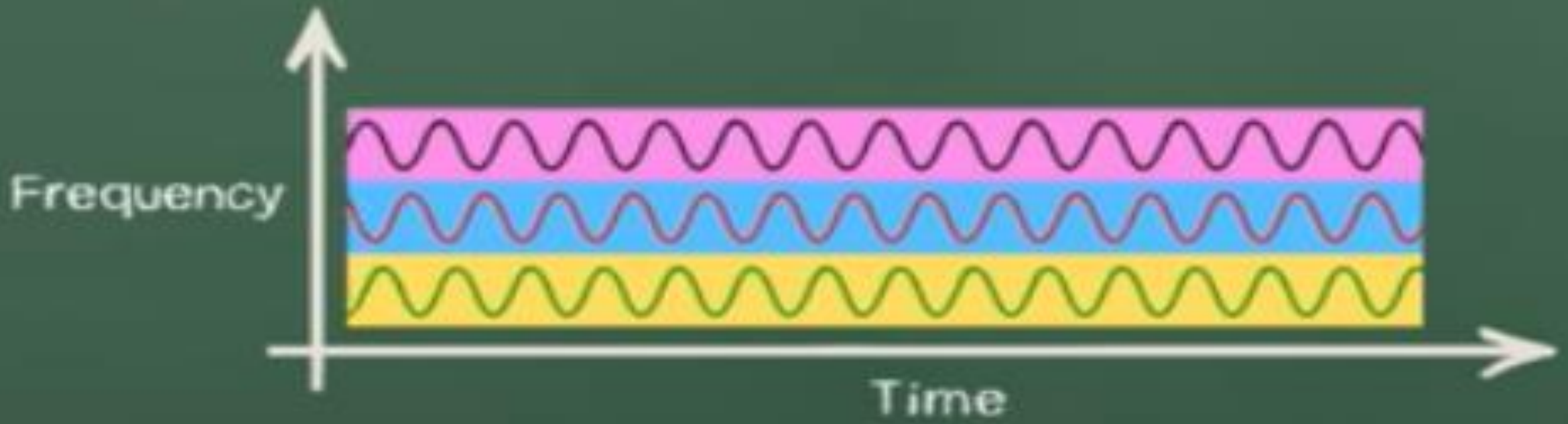
[a variation of FDM]

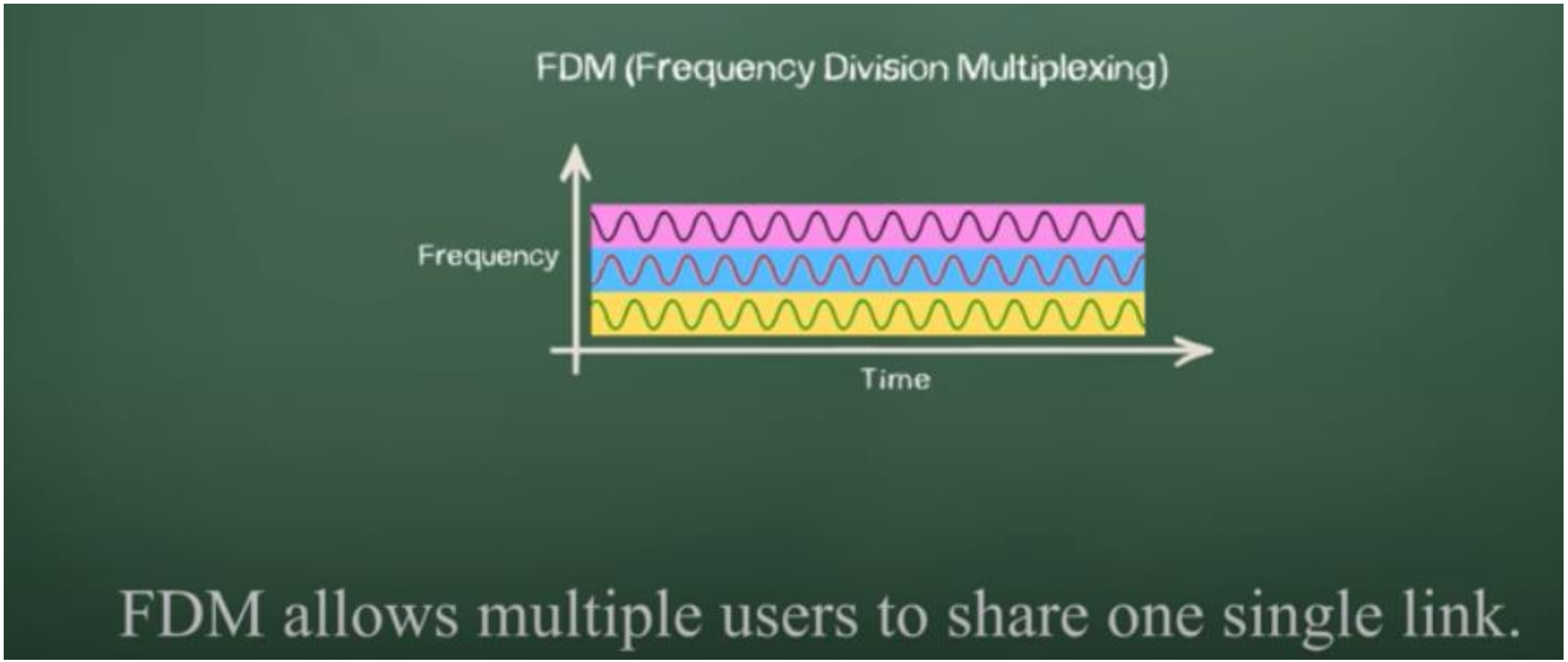
OFDM — Orthogonal Frequency Division Multiplexing.

→ Can efficiently overcome ISI

OFDM is a very popular multiplexing method used for many of the latest wireless and telecommunication standards,

FDM (Frequency Division Multiplexing)



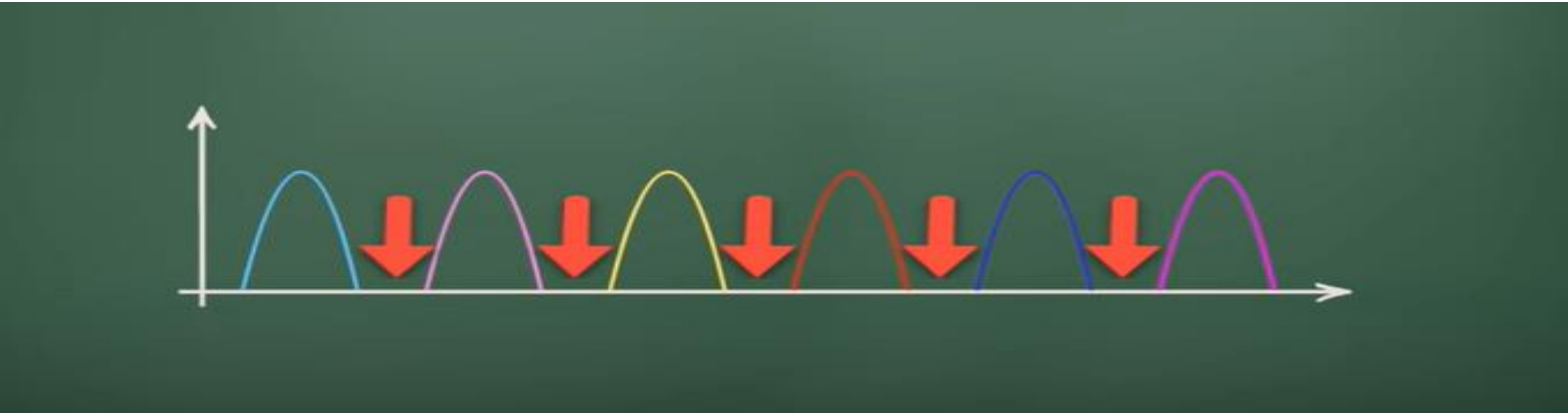

FDM (Frequency Division Multiplexing)
Frequency
Time
FDM allows multiple users to share one single link.

OFDM is a very popular multiplexing method used for many of the latest wireless and telecommunication standards,

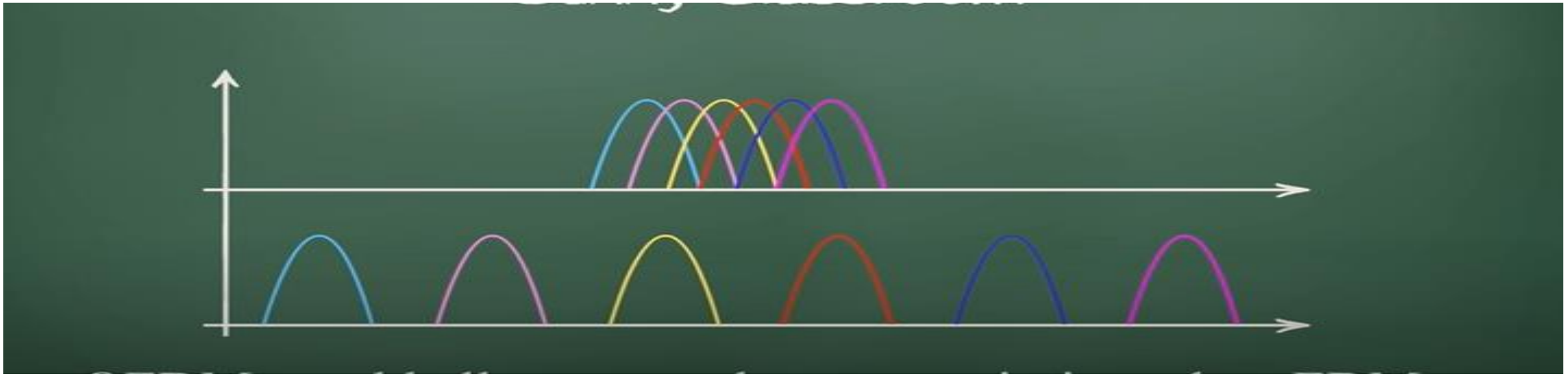
such as Wi-Fi 802.11ac, 4G and 5G cellular phone technologies, WiMAX, satellite and many others.


What is OFDM?
Why is it better than FDM?

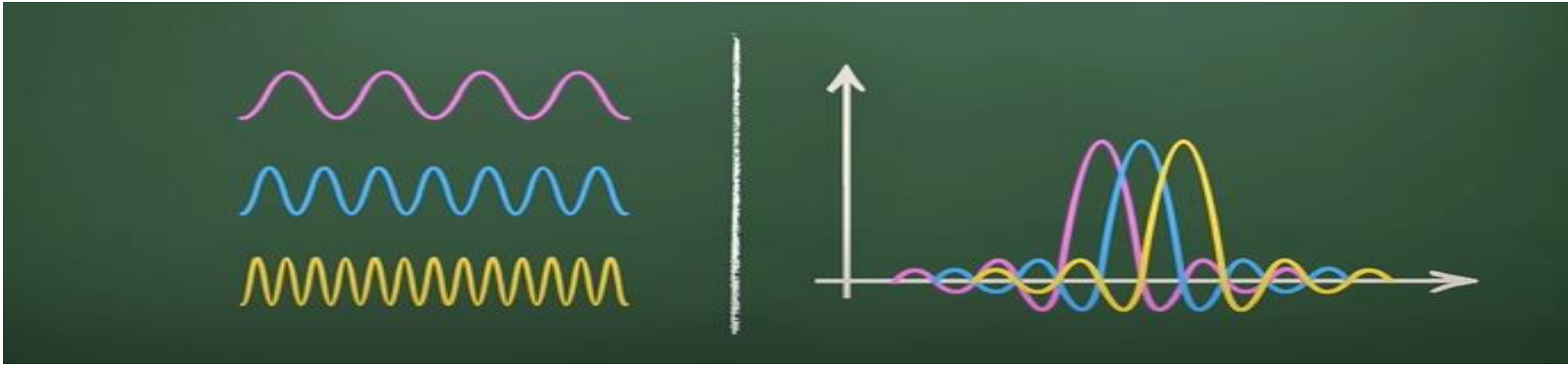
1st let's compare FDM and OFDM in general FDM allows multiple users to share one link by dividing available bandwidth into different non overlapping sub channels a guard band, a narrow frequency range is inserted between adjacent sub channels so that different channels travel separately and simultaneously without interfering with each other



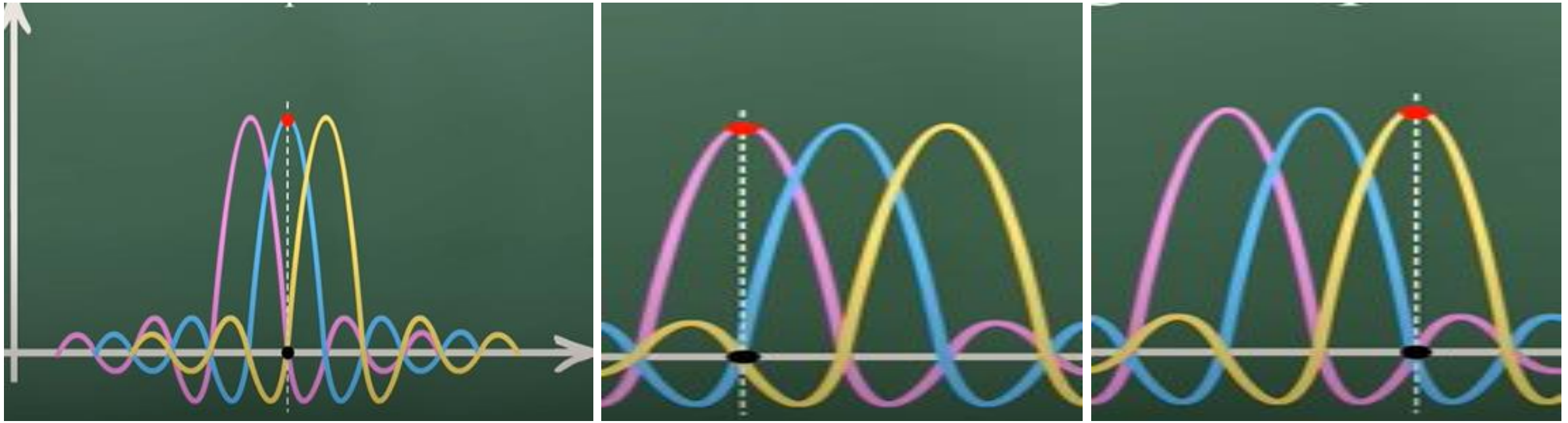
Not only there is no guard band between them but actually there are overlapped we can say that with the same available bandwidth OFDM allow more data transmissions than FDM. But How does OFDM prevents interference while multiple sub-channels overlap each other



Suppose we have three different signals send over one channel . OFDM would combine them closely together in a way they Orthogonal to each other . What is the meaning of the word orthogonal anyway ? The word orthogonal means that two or multiple objects act independently in this case any neighbor signals in OFDM operate without dependence on or interference with one another . Let's explain why?.....

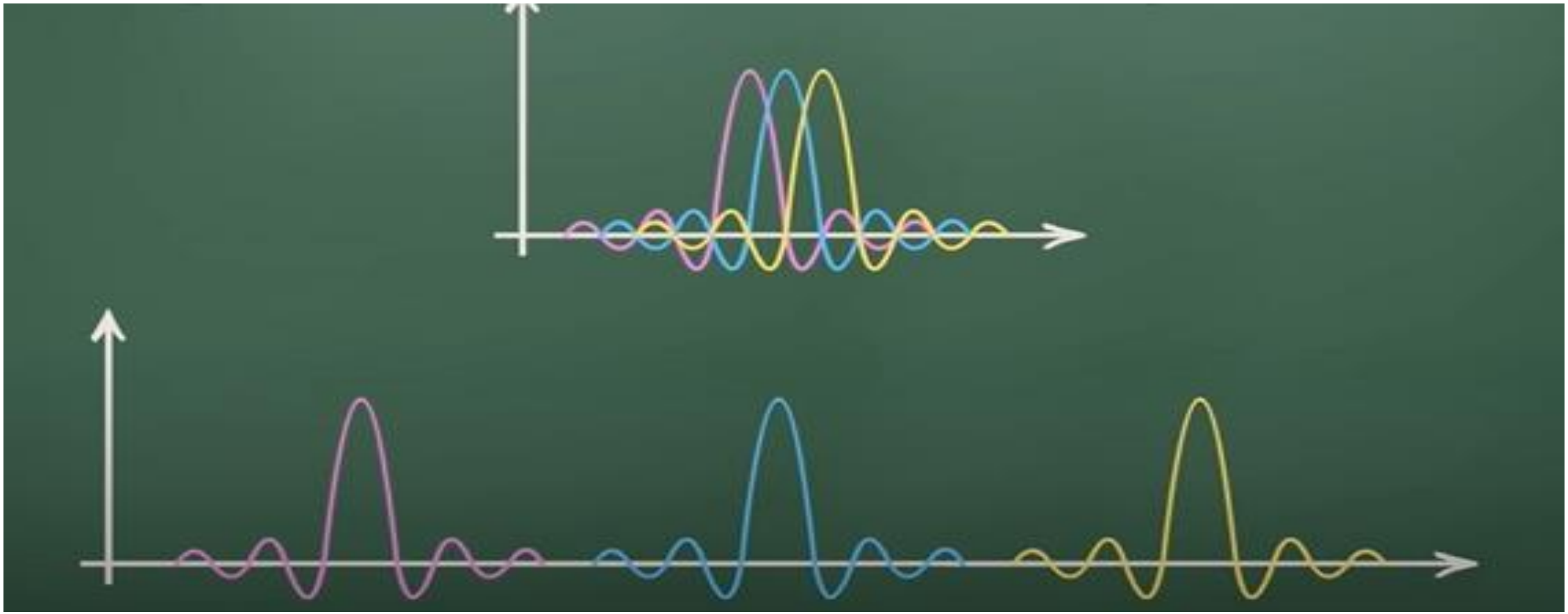


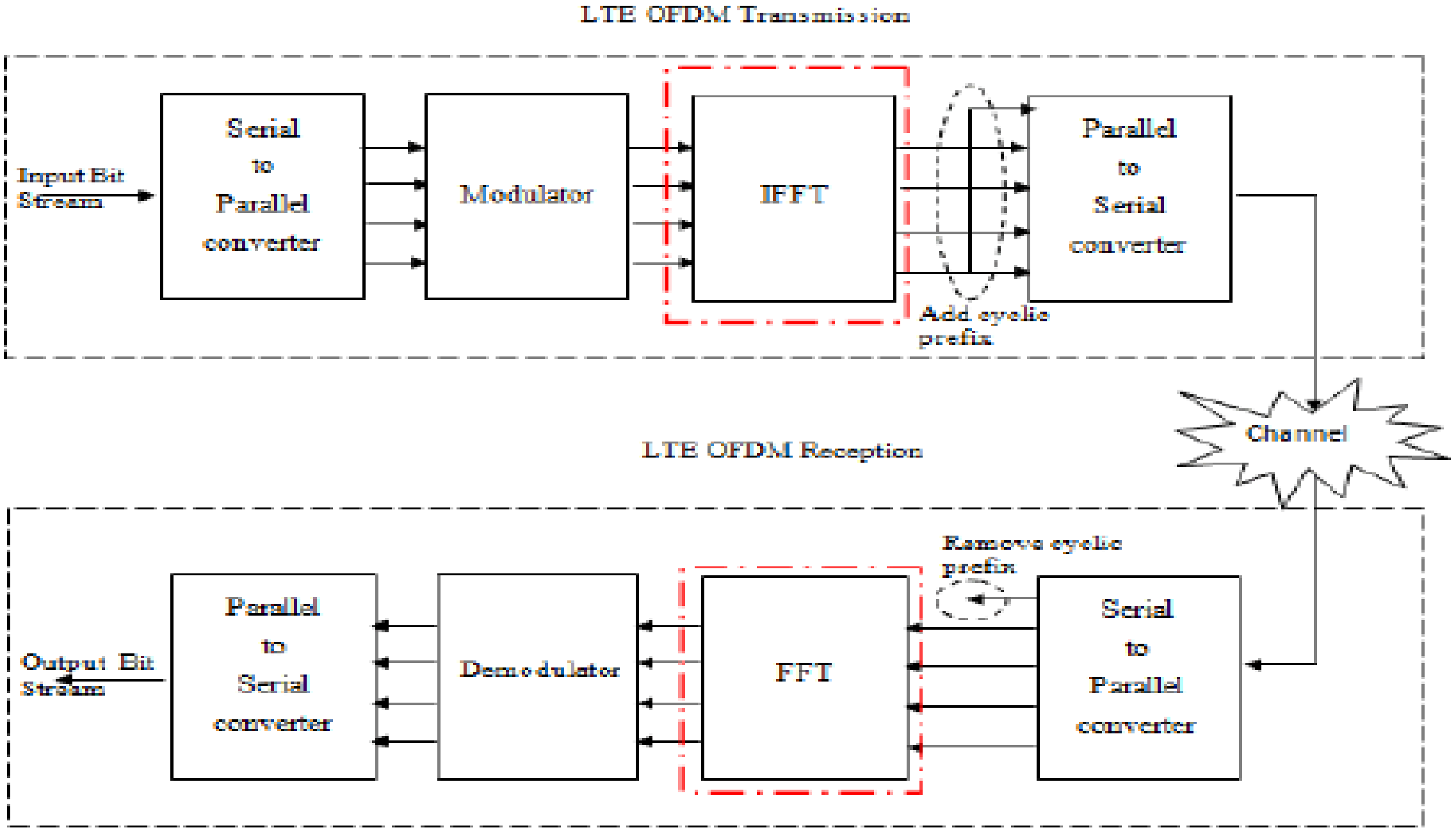
Now look closely at these three signals to see why they are orthogonal o each other. Let's focus on the middle signal in color sky blue/ cyan. When the signal reaches its peak, the highest point its two neighbors, pink signal and yellow signal are at there 0 point or at null. The same is the true for pink signal, the other two signals land at zero point with same pattern, yellow signal's peak corresponds with 0 point of other two signals. Therefore , Orthogonal means signals are multiplexed in a way that the peak of one signal occurs at null of the other neighbor signals.



At the receiver end , the demultiplexer would separate them based on this orthogonal feature.

OFDM would better utilize the available bandwidth thus offering higher data transmission rate than FDM




LTE OFDM Transmission
Input Bit Stream
Serial to Parallel converter
Modulator
IFFT
Add cyclic prefix
Parallel to Serial converter
Channel
LTE OFDM Reception
Remove cyclic prefix
Parallel to Serial converter
Demodulator
FFT
Serial to Parallel converter
Output Bit Stream

In telecommunications, orthogonal frequency-division multiplexing (OFDM) is a type of digital transmission and a method of encoding digital data on multiple carrier frequencies.

What is OFDM and how it works?
OFDM is a form of multicarrier modulation. An OFDM signal consists of a number of closely spaced modulated carriers. When modulation of any form - voice, data, etc. is applied to a carrier, then sidebands spread out either side. ... To see how OFDM works, it is necessary to look at the receiver.

What is OFDM used for?
Orthogonal frequency division multiplexing (OFDM) is a modulation technique that is used in several applications ranging from cellular systems (3GLTE, WiMAX), wireless local area networks (LANs), digital audio radio, underwater communications, and even optical light modulation.

What are the main strengths of OFDM?

Immunity to selective fading: One of the **main advantages of OFDM** is that is more resistant to frequency selective fading than single carrier systems because it divides the overall channel into multiple narrowband signals that are affected individually as flat fading sub-channels.

How OFDM signal is generated?

OFDM is a combination of **modulation** and multiplexing. Multiplexing generally refers to independent **signals**, those **produced** by different sources. ... In **OFDM** the **signal** itself is first split into independent channels, modulated by data and then re-multiplexed to create the **OFDM** carrier.

Introduction to
Spread Spectrum
PN Sequence
DS-SS & FH-SS
Performance of DS-SS and FH-SS

A collective class of signaling techniques are employed before transmitting a signal to provide a secure communication, known as the **Spread Spectrum Modulation**. **The main advantage of spread spectrum communication technique is to prevent "interference" whether it is intentional or unintentional.**

The signals modulated with these techniques are hard to interfere and cannot be jammed. An intruder with no official access is never allowed to crack them. Hence, these techniques are used for military purposes. These spread spectrum signals transmit at low power density and has a wide spread of signals.

Pseudo-Noise Sequence

A coded sequence of **1s** and **0s** with certain auto-correlation properties, called as **Pseudo-Noise coding sequence** is used in spread spectrum techniques. It is a maximum-length sequence, which is a type of cyclic code.

Narrow-band and Spread-spectrum Signals

Both the Narrow band and Spread spectrum signals can be understood easily by observing their frequency spectrum as shown in the following figures.

Narrow-band Signals

The Narrow-band signals have the signal strength concentrated as shown in the following frequency spectrum figure.

Following are some of its features –

- **Band of signals occupy a narrow range of frequencies.**
- **Power density is high.**
- **Spread of energy is low and concentrated.**
- **Though the features are good, these signals are prone to interference.**

Spread Spectrum Signals

The spread spectrum signals have the signal strength distributed as shown in the following frequency spectrum figure.

Following are some of its features −

- Band of signals occupy a wide range of frequencies.
- Power density is very low.
- Energy is wide spread.

With these features, the spread spectrum signals are highly resistant to interference or jamming. Since multiple users can share the same spread spectrum bandwidth without interfering with one another, these can be called as **multiple access techniques**

FHSS and DSSS / CDMA

Spread spectrum multiple access techniques uses signals which have a transmission bandwidth of a magnitude greater than the minimum required RF bandwidth.

These are of two types.

- Frequency Hopped Spread Spectrum FHSS
- Direct Sequence Spread Spectrum DSSS

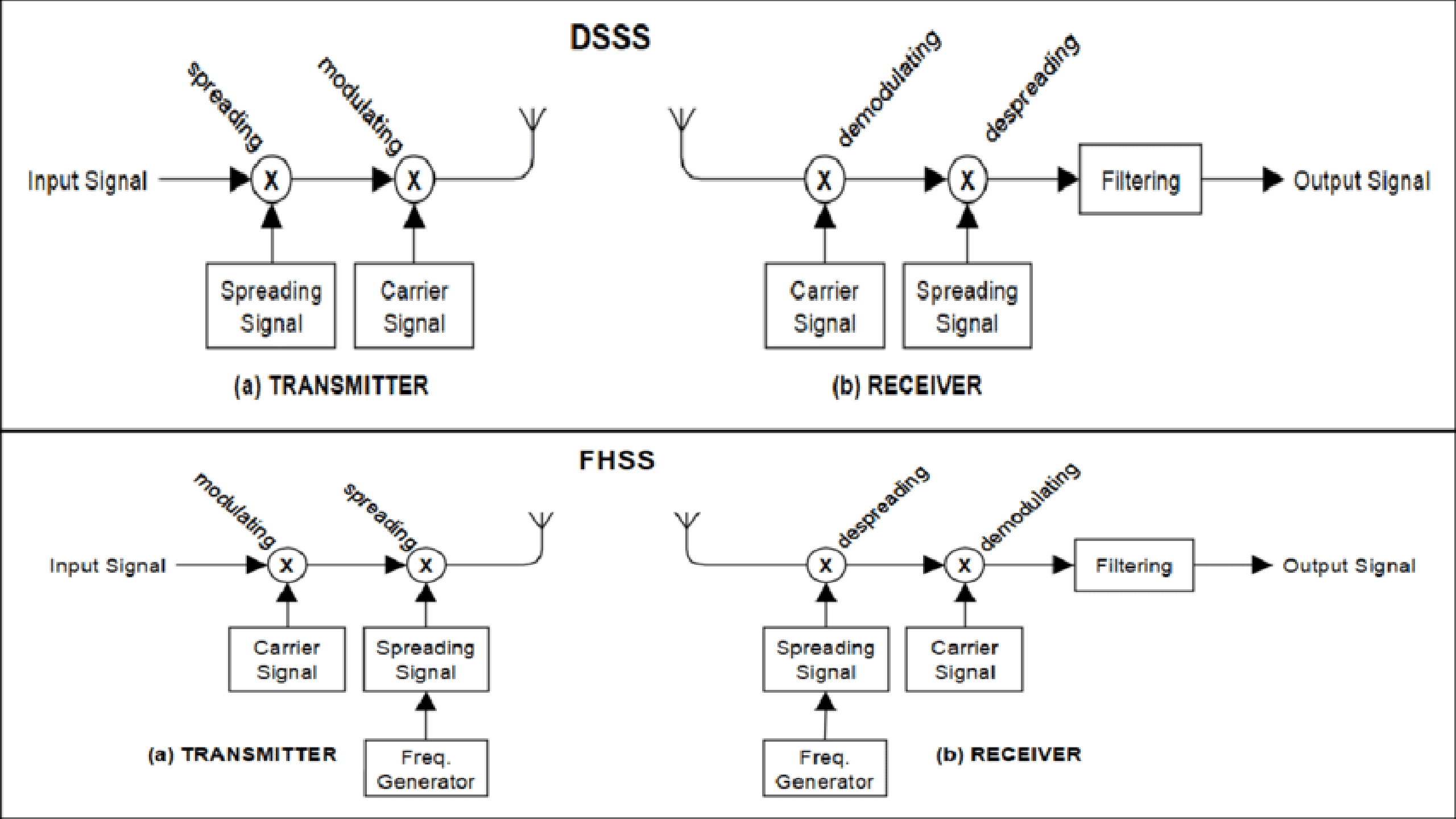
Frequency Hopped Spread Spectrum [FH SS]

This is frequency hopping technique, where the users are made to change the frequencies of usage, from one to another in a specified time interval, hence called as frequency hopping. For example, a frequency was allotted to sender **1** for a particular period of time. Now, after a while, sender **1** hops to the other frequency and sender **2** uses the first frequency, which was previously used by sender **1**. This is called as **frequency reuse**.

The frequencies of the data are hopped from one to another in order to provide a secure transmission. The amount of time spent on each frequency hop is called as Dwell time.

Direct Sequence Spread Spectrum [DS SS]

Whenever a user wants to send data using this DSSS technique, each and every bit of the user data is multiplied by a **secret code**, called as **chipping code**. This **chipping code** is nothing but the **spreading code** which is multiplied with the original message and transmitted. The receiver uses the same code to retrieve the original message


DSSS
spreading
modulating
Input Signal
X
X
Spreading Signal
Carrier Signal
(a) TRANSMITTER
demodulating
despreading
X
X
Filtering
Output Signal
Carrier Signal
Spreading Signal
(b) RECEIVER
FHSS
modulating
spreading
Input Signal
X
X
Carrier Signal
Spreading Signal
Freq. Generator
(a) TRANSMITTER
despreading
demodulating
X
X
Filtering
Output Signal
Spreading Signal
Carrier Signal
Freq. Generator
(b) RECEIVER

Comparison between FH SS and DS SS / C D M A

Both the spread spectrum techniques are popular for their characteristics. To have a clear understanding, let us take a look at their comparisons

FHSS	DSSS / CDMA
Multiple frequencies are used	Single frequency is used
Hard to find the user's frequency at any instant of time	User frequency, once allotted is always the same
Frequency reuse is allowed	Frequency reuse is not allowed
Sender need not wait	Sender has to wait if the spectrum is busy
Power strength of the signal is high	Power strength of the signal is low
Stronger and penetrates through the obstacles	It is weaker compared to FHSS
It is never affected by interference	It can be affected by interference
It is cheaper	It is expensive
This is the commonly used technique	This technique is not frequently used

Advantages of Spread Spectrum

Following are the advantages of spread spectrum −

- Cross-talk elimination
- Better output with data integrity
- Reduced effect of multipath fading
- Better security
- Reduction in noise
- Co-existence with other systems
- Longer operative distances
- Hard to detect
- Not easy to demodulate/decode
- Difficult to jam the signals

Although spread spectrum techniques were originally designed for military uses, they are now being used widely for commercial purpose.